

Time Series Econometrics (TSE)

Henri Nyberg

October 08, 2026

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Preface

i Foreword

This material is compiled from various sources. It is largely based on the Aikasarja-analyysi (Time Series Analysis) course taught at the University of Turku from 2016 to 2025. That course material was, in turn, heavily based on the numerous time series analysis courses taught by Professor Emeritus Pentti Saikkonen at the University of Helsinki. My/our sincere thanks to him for providing access to his course materials, particularly for the course “Stationaariset aikasarjat” (2015). Thanks also to Professor Mika Meitz, whose lecture notes, “Lecture Notes on Time Series Econometrics” were valuable reference for the English translation and terminology.

Special thanks to Professor Markku Lanne for sharing his lecture notes on linear regressions with time series variables, originally part of the “Econometrics 2” course at the University of Helsinki. I also wish to thank MSocSc and doctoral researcher Roope Rihtamo for setting up the Quarto and GitHub documents of this material.

All remaining errors are solely the responsibility of the author.

Other sources and potential references include the following:

- Brockwell, P.J. & R.A. Davis (1996 or 2002). Introduction to Time Series and Forecasting. Springer (2002, 2nd ed.).
- Franses, P.H. & D. van Dijk (2000). Non-linear time series models in empirical finance. Cambridge University Press.
- Hamilton, J. (1994). Time Series Analysis. Princeton University Press.
- Lütkepohl, H. & Krätzig, M. (2004). Applied Time Series Econometrics. Cambridge University Press.
- Tsay, R.S. (2010). Analysis of Financial Time Series. Wiley
- Verbeek, M. (2008). A Guide To Modern Econometrics. John Wiley & Sons. Third Edition

This lecture manuscript primarily focus on univariate time series models. However, we will also briefly cover key aspects of multiple time series analysis, particularly vector autoregressive (VAR) models and cointegration.

In addition, we will introduce some fundamental concepts of machine learning techniques relevant to time series econometrics.

The empirical examples, implemented using RStudio, serve to illustrate to theoretical concepts. While most examples are drawn from (macro)economic and financial applications, the methods are broadly applicable across various fields.

Some of the empirical examples and materials are based on well-known journal articles:

- Hall, R.E. (1978). Stochastic implications of the life cycle-permanent income hypothesis: Theory and evidence. *Journal of Political Economy*, 86(6), 971–987.
- Stock, J.H. & M.W. Watson (1988). Variable trends in economic time series. *Journal of Economic Perspectives*, 2(3), 147–174
- Stock, J.H. & M.W. Watson (2001). Vector autoregressions. *Journal of Economic Perspectives*, 15(4), 101–115.

More background information (journal articles and their references)

- Engle, R.F., Lilien, D.M. & Robins, R.P. (1987). Estimating time varying risk premia in the term structure: The ARCH-M model. *Econometrica*, 55, 391–407.
- Glosten, L.R., Jagannathan, R. & Runkle, D.E. (1993). On the relation between the expected value of the volatility of the nominal excess return on stocks. *Journal of Finance*, 48, 1779–1801.
- Lanne, M. & Saikkonen, P. (2006). Why is it so difficult to uncover the risk-return tradeoff in stock returns? *Economics Letters*, 92, 118–125.
- Newey, W.K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Nyberg, H. (2012). Risk-return tradeoff in U.S. stock returns over the business cycle. *Journal of Financial and Quantitative Analysis*, 47, 137–158.
- Scruggs, J. (1998). Resolving the puzzling intertemporal relation between the market risk premium and conditional market variance: A two-factor approach. *Journal of Finance*, 53, 575–603.

Datasets and data sources used in this lecture manuscript:

- “Shiller data” refers to Robert Shiller’s updated data on Irrational Exuberance: <http://www.econ.yale.edu/~shiller/data.htm>.
 - See Shiller, R. (2005). *Irrational Exuberance*. 2nd Edition. Currency Books.
- FRED data (Federal Reserve Bank of St. Louis): <https://fred.stlouisfed.org/>.
 - Hereafter “FRED” (“FRED data”) refers to the Federal Reserve Economic Data (FRED) database, maintained by the Federal Reserve Bank of St. Louis.

- quantmod R package: <https://cran.r-project.org/web/packages/quantmod/index.html>.
 - Data retrieved via the quantmod R package is seemingly often from Yahoo Finance (<https://finance.yahoo.com>)
 - This package provides tools for quantitative financial modeling and data retrieval. Note that when using financial data in more “serious analyses”, like in your thesis work, ensure always that you have the appropriate license and permission to access and use the data sources, such as possibly integrated via quantmod.

i Notation

On this course, we follow, for example, the following key notational selections:

- We will use the same notation for random quantities and their realized values. The context should make the difference clear.
- In vector and matrix calculations, vectors are interpreted as column vectors and we might also denote, for example, $\mathbf{y} = [y_1 \cdots y_T]'$, where the superscripted comma denotes transpose of a vector (here $T \times 1$ vector), or elsewhere also of a matrix. Vectors (column vectors) are also at times denoted as $\mathbf{y} = (y_1, \dots, y_T)'$, indicating the same vector as above (a change made in 2026 vs. 2025 notation).

Concerning different (stochastic/time series) processes, we use mainly notations y_t , x_t and z_t .

- z_t is typically a zero mean process and denoting often a demeaned process. That is $z_t = y_t - E(y_t)$, where y_t is the “original” time series.
- x_t , or $\mathbf{x}_t$ in the case of multiple time series, is used for a general time series or a vector of time series for different notational purposes, but most often they are in the role of explanatory (predictive) variables.

The R code of the course (the R Lab sections at the end of the chapters) is included only in the online version of this material.

Chapter 1

Introduction

1.1 Time series data

A **time series** is a collection of data where observations correspond to consecutive time periods. A common notation for time series data is y_t , $t = 1, \dots, T$, where t denotes the time period and T the total number of observations. The interval between observations y_{t-1} and y_t , or y_t and y_{t+1} , represents a unit of time, for example, an hour, a month, a quarter or one year. In many cases, it is natural to use more concrete labels for the time periods, such as $t = 2000, \dots, 2024$ for yearly data, or $t = 2000 : 1, 2000 : 2, \dots, 2024 : 12$ for monthly observations. Therefore, occasionally T may also refer to the final time label, such as 2024:12 (December 2024).

- Time series data are analyzed in various fields, including economics and finance, as well as social, natural, engineering, and medical sciences.
- In (macro)economics and finance, data sets are typically time series, which has led to the specialized subfields of **macroeconometrics** and **financial econometrics**.

In this course, we assume that the observations are recorded at equally spaced time intervals (or can reasonably be treated as such). Irregularly spaced data require different methods and are not covered here. If the time series observations y_t , $t = 1, \dots, T$, are scalar valued, we refer to them as a **univariate time series**. Conversely, if $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})'$ is vector-valued (indicated by the bold symbol), we refer to it as a **multiple time series** or **vector-valued time series**. In these latter cases, relationships between two or more time series are of interest.

We restrict ourselves to the common case where y_t is a **real-valued** time series, or is **at least treated as such**, meaning that its realizations are real numbers. In discrete and limited dependent time series, y_t takes only discrete values, such as binary values ($y_t = 1$ or $y_t = 0$) or values where the range is limited (such

as the case where y_t is nonnegative). These types of time series require specific nonlinear models, which we will not cover in this course, except for the models used in volatility modelling, where positive-valued variables are relevant.

When analyzing time series data, a useful and natural first step is to plot the data and inspect its main characteristics visually. Below, we plot the monthly Consumer Price Index (CPI) for the United States over the time period 1990:1–2025:6 (January 1990–June 2025). In this figure, as is almost always the case when plotting time series data, the line connects observation points to highlight the sequential nature of the data and the dependency between observations.

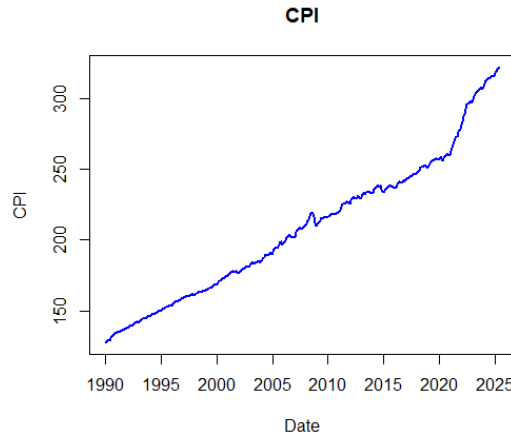


Figure: Monthly U.S. CPI index (levels) (1990:1–2025:6).

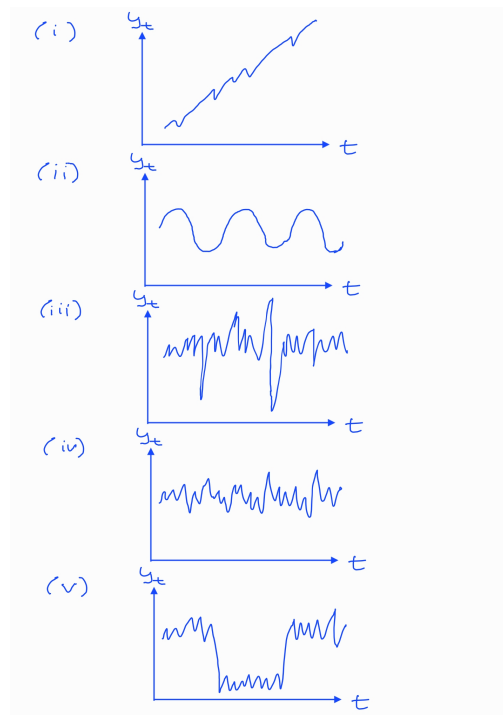
The aim of time series analysis is often to build a **statistical model** that represents the observed time series and its fluctuation over time. This typically involves examining the dependence structure among consecutive observations. **Autocorrelation** (serial correlation) between observations is a key property of time series and must be accounted for in statistical and econometric modelling.

- This violates the no-autocorrelation assumption typically imposed in linear regression models for cross-sectional datasets, as introduced in basic econometrics and statistics courses.
- With cross-sectional data, random sampling generally ensures that the error terms of the linear regression model are mutually independent (i.e., not autocorrelated), so autocorrelation is usually not an issue.

As noted above, **plotting the series and inspecting its main characteristics visually** is a natural first step, to be done alongside any underlying theory or background knowledge of the phenomenon the time series represents. This provides at least a preliminary understanding of the dependence structures

behind the **data generating process (DGP)**. Some important questions to consider when examining a time series graph include:

- (i) Is there a **trend** (that is, do the values of the time series tend to increase or decrease over time)?
- (ii) Are there **seasonal effects** (a repeating regular pattern of highs and lows related to, e.g., seasons, quarters, months or days of the week)?
- (iii) Are there major **outliers** (observations far away from other data points)?
- (iv) Is the **variability** (variance) relatively **constant over time**?
- (v) Are there any **abrupt changes**, such as **breaks** or **regime switches** (such as in figure (v)), in either the level or variance of the series?



As an example, in the U.S. CPI time series shown above, a clear feature is the persistent trend over time. However, rather than focusing on the level of the CPI, we are primarily interested in its changes as a measure of inflation, as will be discussed below.

An upward trend is also evident in the monthly S&P 500 stock market index. Stock prices have risen substantially between 1990 and 2025, and a pronounced upward surge is clearly visible. At the same time, several significant market corrections, commonly referred to as “bear markets”, occurred during this

period. It is important to note that this trending behavior differs markedly from that observed in the CPI series. The characteristics underlying these types of stochastic trends, particularly those seen in the S&P 500 index, will be explored later in this course when we discuss nonstationary time series.

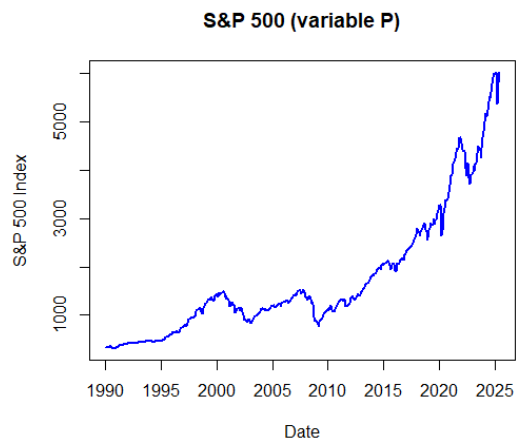


Figure: S&P 500 stock market index between January 1990 and June 2025 (source: Shiller data).

The next figure depicts the monthly number of accidental deaths in the U.S. between 1973:1–1978:12. In this series, there is no apparent upward or downward trend. Instead, the series exhibits a clear **seasonal variation**, with the yearly maximum consistently occurring in July and the minimum in February.

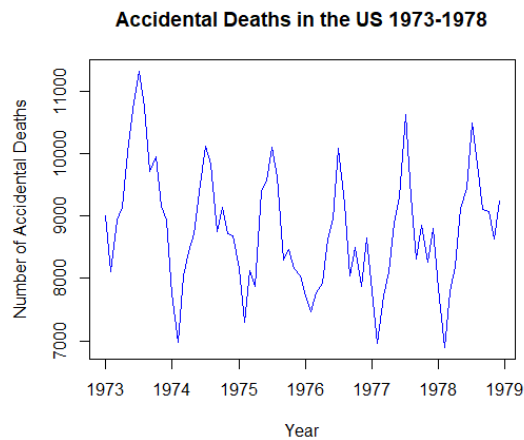


Figure: Monthly accidental deaths in the U.S. between January 1973 and December 1978 (source: Brockwell and Davis, 1996).

The following figure shows the yearly number of sunspots for the period 1770–1870. This series has no trend or seasonal variation, although the **cyclical pattern** could easily be mistaken for seasonality. However, the pattern is not seasonal because the nature of the cycle changes over time. In particular, the distance between consecutive peaks is not constant, as it would be in the case of seasonal variation. Another apparent feature is the positive correlation between consecutive observations: a large observation is typically followed by another large one, and a small observation by another small one.

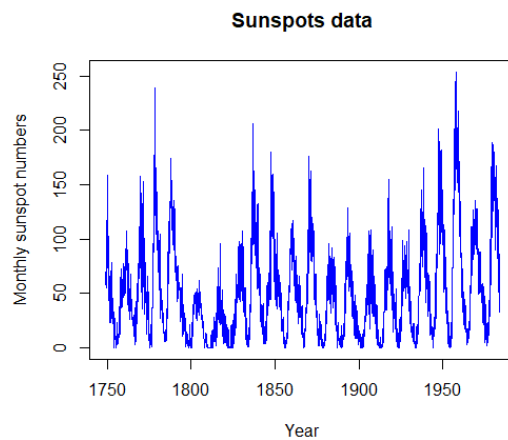


Figure: The number of sunspots between the years 1770–1870 (source: Brockwell and Davis, 1996).

Overall, these visual findings may already suggest which type of **time series model** could be appropriate. Once a model has been selected, as we will consider in more detail in the following sections, one can then:

- Estimate the unknown parameters of the model,
- Assess whether the estimated model is compatible with the observed data,
- Test hypotheses concerning the model parameters, and
- Use the model for its intended end purpose.

The ultimate purpose of the statistical model depends on the context and the field of application.

- One common use is to provide a **summarized description** of the observed dataset, which can help in understanding the underlying data-generating process.
- Another central use of a time series model is to **predict (forecast)** future values of one or more series. In the multivariate case, a series of interest can be predicted using its own past values but also information contained in other series as explanatory or predictive variables.
- A further objective is to explore potential **temporal and contemporaneous dependencies** between time series, such as the relationship between inflation and stock market returns (after applying relevant transformations, as discussed next). For example, one may investigate whether higher inflation tends to occur simultaneously with rising stock prices.

1.2 Transformations

As described above, in time series modelling a good first step is always to plot the time series under investigation. This provides an initial view of its main characteristics, such as trends, seasonal patterns, and any sudden breaks or outliers. In the following sections, we assume that such patterns are absent. If the observed series does exhibit these features, appropriate transformations should be applied to remove or mitigate them.

In time series analysis, several transformations and adjustments (“manipulations”) are commonly used. These are natural and often essential components of empirical analysis in many applications.

- **The first lag** of the time series y_t is denoted by y_{t-1} . Generally, the k **th lag** is y_{t-k} .
- Perhaps the most commonly used transformation for time series data is **differencing**. The **first difference** of the series, Δy_t , is the **change** between y_t and y_{t-1} . In other words,

$$\Delta y_t = y_t - y_{t-1}.$$

- Often differencing is combined with taking natural **logarithms** (when assuming $y_t > 0, \forall t$). Taking logarithms is common especially when the series exhibits exponential growth or when the variation in the series increases as the level increases.
 - Recall that if $y_t = \exp(\delta t)$, then $\log(y_t) = \delta t$.

The **logarithmic change** is

$$\Delta \log(y_t) = \log(y_t) - \log(y_{t-1}).$$

Importantly, the logarithmic changes can be interpreted as relative changes when the changes are small:

$$\Delta \log(y_t) \approx (y_t - y_{t-1})/y_{t-1}.$$

- Why so? Recall that $\log(1 + x) \approx x$ when x is small. Therefore,

$$\Delta \log(y_t) = \log\left(\frac{y_t}{y_{t-1}}\right) = \log\left(1 + \frac{y_t}{y_{t-1}} - 1\right) \approx (y_t - y_{t-1})/y_{t-1}.$$

- The **percentage of change** between periods t and $t - 1$ can be approximated as $100\Delta \log(y_t)$. For quarterly data y_t , **annualized growth rates** are obtained by multiplying by 400, that is, $400\Delta \log(y_t)$. Similarly, for monthly data, annualized growth rates are computed as $1200\Delta \log(y_t)$.

As an example, consider the monthly U.S. CPI time series. Taking log-differences (first applying the logarithm and then differencing) and multiplying the resulting series by 1200, we obtain a proxy for the annualized inflation rate. Because the CPI typically exhibits an upward trend, log-differencing removes this trend, resulting in a series without a clear deterministic trend. This detrending implies that, instead of an increasing level, the transformed series has a positive mean, which reflects the average inflation rate over the sample period.

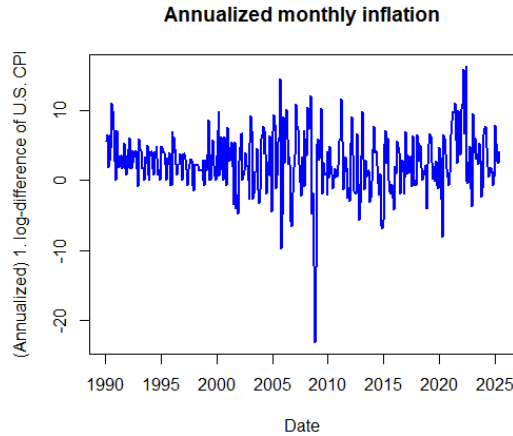


Figure: Annualized U.S. inflation (CPI inflation) rate from February 1990 to June 2025 (1990:2–2025:6).

More on differencing. Differencing may also happen with a longer lag than the first lag: $y_t - y_{t-s}$, ($s \geq 1$).

- As discussed above, the case $s = 1$ is the most common (representing the change from one period to another).

- In the case of seasonal variation, s is often selected to match the length of the seasonal pattern. That is, for example, $s = 4$ for quarterly data and $s = 12$ for monthly series.

If we temporarily denote by y_t the log CPI, taking the seasonal difference $100(y_t - y_{t-12})$ yields another proxy for U.S. inflation (cf. the time series above).

- An example case of seasonal log-differencing for quarterly data: $\log(y_t) - \log(y_{t-4}) = \Delta \log(y_t) + \Delta \log(y_{t-1}) + \Delta \log(y_{t-2}) + \Delta \log(y_{t-3})$.

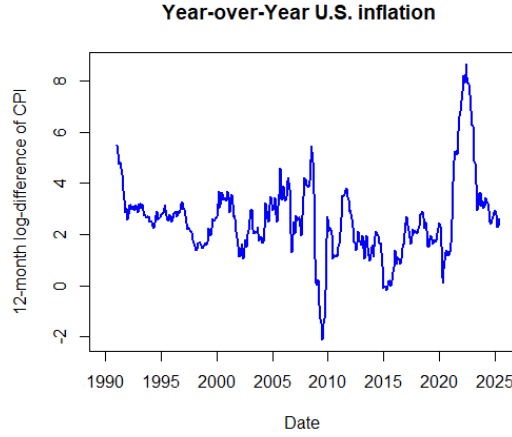


Figure: U.S. inflation (CPI inflation) obtained as year-to-year changes in the CPI (1991:1–2025:6).

Another use of differencing, especially in financial econometrics and empirical finance, is to obtain the returns that different assets yield for investors. Let P_t be the price of an asset at time t . Holding the asset for one period from $t - 1$ to t yields the simple return

$$R_t = \frac{P_t}{P_{t-1}} - 1 = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

In addition, the so-called continuously compounded returns (or log returns) are obtained as

$$r_t = \log(1 + R_t) = \log\left(\frac{P_t}{P_{t-1}}\right) = p_t - p_{t-1},$$

where $p_t = \log(P_t)$. Log returns have several advantages over simple returns, such as time-additivity and better behavior under compounding. In both cases, percentage returns are obtained by multiplying r_t or R_t by 100.

- Importantly, in many financial applications, dividend (and similar) payments must be considered when computing returns. If an asset pays

dividends periodically, the definition of return needs to be adjusted accordingly. We will not explore these issues in detail in this course; they are left for more advanced studies in finance.

Below, we depict the log-differences of the monthly S&P 500 index, multiplied by 100 to express them as percentage returns. These log-differences approximate monthly percentage returns for a broad U.S. market index. The shape of the return time series is clearly very different from the S&P 500 index level data shown above.

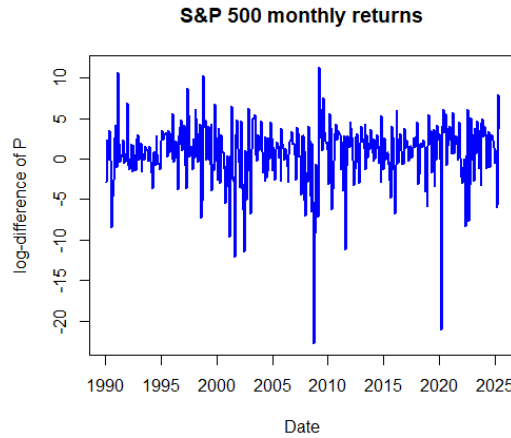


Figure: Monthly percentage changes (log returns) in the S&P 500 index.

Detrending. The trend and/or seasonal patterns in the original time series are often removed through transformations (if necessary), which may require estimating certain parameters. For example, in the case of a linear trend, one can fit a trend line using least squares regression and then consider the resulting residual series using time series methods. In that case, the observed time series y_t would be replaced by the (residual) time series

$$\hat{x}_t = y_t - \hat{\alpha} - \hat{\beta}t, \quad t = 1, \dots, T,$$

where the estimates $\hat{\alpha}$ and $\hat{\beta}$ are obtained by regressing y_t on a constant and t . More generally, instead of the linear function of time, one can use higher-order polynomials or other functional forms to capture more complex trends.

- To model seasonal variation, one could use trigonometric functions of time.
- An often used alternative to remove seasonal variation (if not using already existing seasonally adjusted data) is the so-called seasonal indicators. As an example of this, for quarterly data, one would obtain seasonally-adjusted

series by

$$\hat{x}_t = y_t - \hat{\beta}_1 d_{1t} - \cdots - \hat{\beta}_4 d_{4t}, \quad t = 1, \dots, T,$$

where the i th seasonal indicator d_{it} takes value one when the observation in question corresponds to quarter i and zero otherwise ($i = 1, \dots, 4$).

As for additional (extra) information, there is a substantial body of literature on advanced detrending techniques beyond simple differencing and seasonal dummy variables in econometrics.

1.3 Assumptions and limitations of this course

In summary, we consider time series models/processes under the assumption that the observed time series do not exhibit clear seasonal patterns. If necessary, such patterns have been removed through an appropriate transformation or a **seasonal adjustment method**, so that the modelling is applied to the transformed series.

- A common approach, especially with macroeconomic data, is to use **seasonally adjusted datasets** provided by official statistical agencies, such as Statistics Finland. Therefore, we can reasonably assume that seasonal effects have already been accounted for to a satisfactory degree.

Concerning trends, we mainly focus on **stationary time series processes** and their realizations (i.e. observed time series) in Sections 2–11. These models assume that the time series does not exhibit clear trends or very persistent/trending behaviour.

- That is, the time series considered here resemble the U.S. inflation series and stock market returns (as depicted above), rather than the CPI level series or the S&P 500 price series.
- For now, it is sufficient to understand that non-stationary time series can usually (though not always) be transformed through differencing so that the stationarity assumption holds reasonably well. This allows us to model the resulting series as a stationary process, such as an ARMA(p, q) process, which will be introduced later.
- An exception to this focus on stationary time series occurs in the latter part of the material, where trending behaviour is examined in more detail. The basics of non-stationary processes will be discussed in Sections 12–13, where modelling deterministic and, especially, stochastic trends becomes important in several applications.

Finally, we begin with univariate time series, meaning we analyze a single series at a time in Sections 2–9. Later, in Sections 10–11 and 13, we will introduce the basics of multivariate time series analysis, where multiple time series are modeled jointly.

Chapter 2

A primer on time series models

2.1 Starting point

Relation to linear regression for cross-sectional datasets. In standard econometrics or statistics courses, linear regression models are typically introduced using cross-sectional data. A key assumption in these models is the absence of autocorrelation in the error term. With time series data, however, autocorrelation is a fundamental characteristic and is almost always present.

- While linear regression can be applied to time series variables, the notation changes to reflect the temporal ordering. Variables are indexed by time, such as y_t and x_t (instead of y_i and x_i for cross-sectional observations).

We will return to multivariate regression models with time series variables later. First, we will explore a **pure time series approach**, as characterized by Verbeek (2008).

- In this approach, the current value of a univariate time series, y_t , is modeled as a function of its past values, either directly or indirectly.
- The emphasis is entirely on exploiting the information contained in y_t and its own history for modeling and forecasting purposes.

2.2 Notes on deterministic components

In time series analysis, we often decompose a process y_t into deterministic and stochastic components. This separation allows us to distinguish deterministic

elements, such as the mean or the trend, from random fluctuations, which are typically the primary focus in stochastic models like ARMA or VAR models to be introduced later in this course.

The general decomposition and notation used throughout this material is

$$y_t = \mu_t + z_t,$$

where

- μ_t is the **deterministic component**, representing features such as the mean, a trend or seasonality.
- z_t is the **stochastic component**, capturing random fluctuations around the deterministic part. We assume $E(z_t) = 0$, so z_t represents the de-meaned and detrended process. Importantly, z_t typically exhibits its own autocorrelation structure and is therefore not merely pure random noise.

By construction, the expected value of y_t is then simply its deterministic component:

$$E(y_t) = E(\mu_t + z_t) = \mu_t + E(z_t) = \mu_t.$$

Common forms of the deterministic component. The functional form of μ_t depends on the underlying characteristics of the data and the application. Two of the most common specifications are:

- **Constant mean:** If the time series fluctuates around a fixed, time-invariant level, the deterministic component is simply a constant

$$\mu_t = \mu.$$

Here, μ (sometimes denoted μ_0) represents the unconditional mean of y_t , that is, $E(y_t) = \mu$. In practice, almost all real-world time series have a nonzero mean.

- **Constant and linear time trend:** If the series exhibits a persistent upward or downward movement over time, the deterministic component includes both a constant and a linear trend

$$\mu_t = \mu_0 + \mu_1 t.$$

Here, μ_0 is the constant or intercept (the value of μ_t at $t = 0$), and μ_1 is the trend coefficient, representing the average change in y_t from one period to the next.

In practice, we may first estimate the parameters of the deterministic component μ_t , for example, using Ordinary Least Squares (OLS), and then work with the residual series $\hat{z}_t = y_t - \hat{\mu}_t$. This approach corresponds to the simple detrending technique described in Section 1. Alternatively, and more commonly, the deterministic and stochastic components are modelled simultaneously.

Chapter 3

Stationary processes

3.1 Basic concepts

The aim is to build a statistical model for the observed time series $y_1, \dots, y_T$. At this stage, and in Sections 3–9 and 12, we assume that we have one, univariate, time series. Multiple time series and multivariate models will be briefly considered later in Sections 10–11 and 13.

On a rather general level, the observed time series can be interpreted as the observed value of the random vector $\mathbf{y} = (y_1, \dots, y_T)'$ or, in other words, the **realization** of this random vector.

- Recall the notation: Here the random vector interpretation emphasizes the fact that y_t will be interpreted as a random variable. Notice that we do not make a distinction between random variables and their realizations; the difference should be clear from the context.

When building a statistical model, we would attempt to specify the probability distribution of the T -dimensional random vector $\mathbf{y} = (y_1, \dots, y_T)'$ that produced the observed time series.

- Note and recall that in a time series, the observations are (typically) not independent as in the conventional linear regression, and therefore it is not enough to specify the marginal distributions of the components y_t ($t = 1, \dots, T$).
- Instead, we indeed need to specify the T -dimensional joint distribution.

A concrete and often used way to specify the joint distribution of $\mathbf{y}$ in time series analysis is to:

- Specify a model equation that characterises the dependence structure of consecutive observations, and

- combine this with a suitable distributional assumption.

Before we proceed to such concrete model equations, it is useful to first briefly study some basic concepts of probability theory that form the foundation of time series analysis.

3.2 Stochastic processes

In statistical and econometric modelling of time series, one concentrates on the observed time series. That is, the realization of a random vector $\mathbf{y} = (y_1, \dots, y_T)'$.

From a mathematical point of view, however, it is more convenient to consider an “extension” of this vector to all time indices when defining time series models. This leads us to the concept of (a discrete time) **stochastic process** (or just simply **process**), which is a collection of random variables $\{y_t; t = 0, \pm 1, \pm 2, \dots\}$.

- Occasionally, the cases $t = 0, 1, 2, \dots$ or $t = 1, 2, \dots$ are also considered.
- Because the set of values the time index takes is usually known, it does not need to be emphasized, and often we can simply denote the stochastic process by y_t , or $\{y_t\}$ if one wants to emphasize the distinction to a single component of the process.
- The terms “**process**” and “**model**” are often used interchangeably. In this material, however, “process” refers specifically to the underlying theoretical mechanism—namely, the stochastic process—that we aim to represent (learn) when specifying an empirical time series “model” for the observed empirical data.

From a practical point of view, it is **important to understand that we have only a single realization available**, the observations $(y_1, \dots, y_T)$, of the process $\{y_t\}$.

- In other words, **we have only one realization from the process and that is only partial**, that is, observations are available only for the time periods $t = 1, \dots, T$.

These above-mentioned remarks imply that in order to investigate the properties of y_t by making use of the observed data $y_1, \dots, y_T$ only, one must make some restrictive assumptions about y_t for this to be possible.

3.3 Weak stationarity

To illustrate the final remarks in the previous section, consider the **expected value function** of the process y_t (assume here $E(y_t^2) < \infty$ for all t)

$$\mu_t = E(y_t), \quad t = 0, \pm 1, \pm 2, \dots$$

as well as the **covariance function** of y_t

$$\gamma_{s,t} = \text{Cov}(y_s, y_t) = E[(y_s - \mu_s)(y_t - \mu_t)], \quad s, t = 0, \pm 1, \pm 2, \dots$$

Estimating these functions by making use of one realization $y_1, \dots, y_T$ only is of course impossible, unless we somehow restrict/limit their dependence on time. **The situation simplifies if it is assumed that these quantities do not depend on time.** Indeed, it is common to assume that such a dependence on time does not (in a particular sense) exist.

A process y_t is called **weakly stationary**, or **covariance stationary**, if

$$E(y_t) = \mu < \infty \quad \text{for all } t = 0, \pm 1, \pm 2, \dots$$

and

$$\text{Cov}(y_t, y_{t+h}) = \gamma_{t,t+h} = \gamma_{0,h} < \infty \quad \text{for all } h, t = 0, \pm 1, \pm 2, \dots$$

In other words, **a form of time invariance holds**: The expected value function μ_t is constant over time, and the covariance function $\gamma_{s,t}$ does not depend on the time indices s and t but only on their distance $t - s$.

- For brevity, denote $\gamma_{0,h} \equiv \gamma_h$.
- When h is fixed, γ_h is called the **autocovariance (coefficient)** of y_t at lag h . As a function of h , γ_h is called the **autocovariance function** of y_t .
- The parameter μ is called the **expected value** or the **mean**.

To summarize, a characteristic feature of a weakly stationary process is that the first and second moments are finite and independent of time. Unless otherwise mentioned, in what follows in this and coming few sections, **we assume that weak stationarity holds**.

A useful **practical aspect** of weak stationarity is that the plausibility of this assumption can be easily investigated by looking at a graph of the observed time series: If the observations vary with constant variance around a fixed level, then weak stationarity appears a plausible assumption.

Empirical example. Let us consider an example related to the U.S. real GDP growth during the period 1985:Q1–2007:Q2, commonly referred to as the *Great Moderation* time period.

- For more on the Great Moderation, see <https://www.federalreservehistory.org/essays/great-moderation>.
- Real Gross Domestic Product (GDP) is the total value of a country's goods and services produced in a specific period, adjusted for inflation.
- This is the dataset GDPC1-qdata.xlsx (source: FRED).

The figure below illustrates that the time series fluctuates around a relatively stable mean and exhibits approximately constant variance over time. As discussed, this behavior is typical for a weakly stationary process.

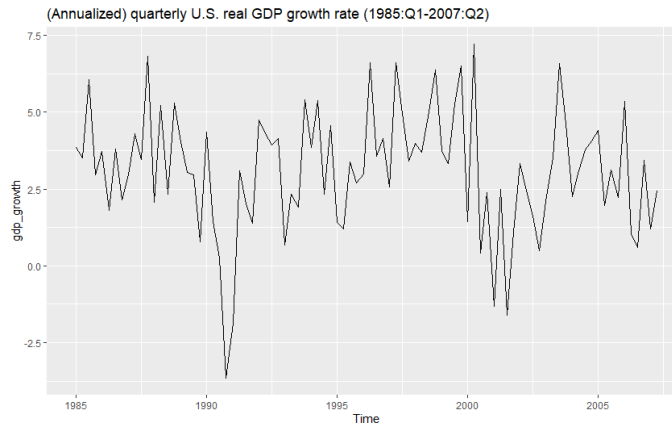


Figure: (Annualized) quarterly U.S. real GDP growth rate (1985:Q1–2007:Q2) (ggplot2 figure).

It is important to note that **weak stationarity does not imply the absence of variability**. Even in a weakly stationary process, individual observations can vary substantially due to random shocks. What matters is that the statistical properties, the mean, variance, and autocovariance structure, remain constant over time. So, **occasional higher volatility**, with high or low values in the series, is entirely consistent with weak stationarity, as long as they are the result of random variation and not due to major structural changes in the process.

- We will soon see more examples of weakly stationary time series through simulations.
- In Section 1, the time series depicting the U.S. inflation rate can also be interpreted as a realization from a weakly stationary process.

More on autocovariance coefficients. Because $\gamma_0 = \text{Var}(y_t)$, it holds that $\gamma_0 \geq 0$. Furthermore, the familiar properties of the covariance function imply that

- $|\gamma_h| \leq \gamma_0$.

- In this univariate (single time series) case, we have (under weak stationarity)

$$\text{Cov}(y_t, y_{t+h}) = \text{Cov}(y_{t-h}, y_t) = \text{Cov}(y_t, y_{t-h}),$$

which also implies that $\gamma_h = \gamma_{-h}$. This expression shows where the term “lag” (“lag length”) h is coming from.

Putting the above properties together, the autocovariance function has the properties

$$\gamma_0 \geq 0, \quad |\gamma_h| \leq \gamma_0 \quad \text{and} \quad \gamma_h = \gamma_{-h}.$$

The case $\gamma_0 = 0$ is obviously not of interest, and in what follows we assume that $\gamma_0 > 0$.

3.4 Autocorrelation function

In practice, a more convenient concept than the autocovariance function is the **autocorrelation function**

$$\rho_h = \text{Cor}(y_t, y_{t+h}) = \text{Cor}(y_t, y_{t-h}) = \gamma_h / \gamma_0.$$

The **autocorrelation coefficients** ρ_h obviously then have the properties

$$\rho_0 = 1, \quad |\rho_h| \leq 1, \quad \text{and} \quad \rho_h = \rho_{-h}.$$

Therefore, it suffices to consider the autocovariance and autocorrelation functions only for the lag lengths $h \geq 0$.

- The latter just for the lags $h > 0$, because $\rho_0 = 1$ always.

Due to the time invariance guaranteed by weak stationarity, it now seems evident that $\mathbf{E}(y_t) = \mu$ and $\text{Cov}(y_t, y_{t+h}) = \gamma_h$ can be estimated. One more problem remains: The autocorrelation function, in general, has an infinite number of quantities to be estimated. In practice, all of these cannot be estimated (without further restrictions). However, typically we can assume that

$$\gamma_h \rightarrow 0, \quad \text{when } h \rightarrow \infty.$$

In this case, the random variables y_t and y_{t+h} become **nearly uncorrelated when the distance h is “large”**.

- That is, when y_t and y_{t+h} are “far” from each other in time, we assume that they are (nearly) uncorrelated.
- Therefore, in practice, it suffices to estimate the autocorrelation function ρ_h , when $h = 1, \dots, H$, and H is so large that $\rho_h \approx 0$ for $h > H$.

3.5 Sample autocorrelation function

Assume that the process y_t is weakly stationary. Then $E(y_1) = \dots = E(y_T) = \mu$. Therefore, a natural estimator of the (population) mean is the **sample mean**

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t.$$

Because (under weak stationarity) $\text{Cov}(y_1, y_{1+h}) = \dots = \text{Cov}(y_{T-h}, y_T) = \gamma_h$ ($h \geq 0$), it also seems natural to estimate the population autocovariance coefficient γ_h using the sample covariance of the observations $(y_1, y_{1+h}), \dots, (y_{T-h}, y_T)$. Because $E(y_t) = E(y_{t+h})$ (under weak stationarity), a commonly used estimator is

$$c_h = \frac{1}{T-h} \sum_{t=1}^{T-h} (y_t - \bar{y})(y_{t+h} - \bar{y}), \quad 0 \leq h < T,$$

which is called the (h th) **sample autocovariance coefficient**.

- Sometimes, the denominator $T - h$ is replaced by T .
- When c_h is interpreted as a function of h (i.e. $c_1, c_2, \dots$), it is called the **sample autocovariance function**.

A natural estimator of the autocorrelation coefficient ρ_h is hence

$$r_h = c_h / c_0, \quad 0 \leq h < T,$$

which is called the **sample autocorrelation coefficient** and $r_1, r_2, \dots$ define the **sample autocorrelation function**.

- Notice that by definition $r_0 = 1$. Some statistical program packages report this uninteresting case.

As an example on sample autocorrelations, consider the quarterly U.S. real GDP growth rate (1985:Q1–2007:Q2) as depicted. Its sample autocorrelation function is plotted below for the lags $h = 1, \dots, 20$. The two horizontal lines represent certain “confidence bands”.

- If $\rho_h = 0$ for all $h > 0$, then any individual estimator r_h has an approximately 95% probability of lying between the confidence bands $(-1.96/\sqrt{T}, 1.96/\sqrt{T})$ where T is the number of observations. These confidence bands are depicted below and typically in corresponding figures.
- If the individual sample autocorrelation coefficient r_h does not fall between these bands, there is statistically significant autocorrelation at the 5% significance level at the lag h .

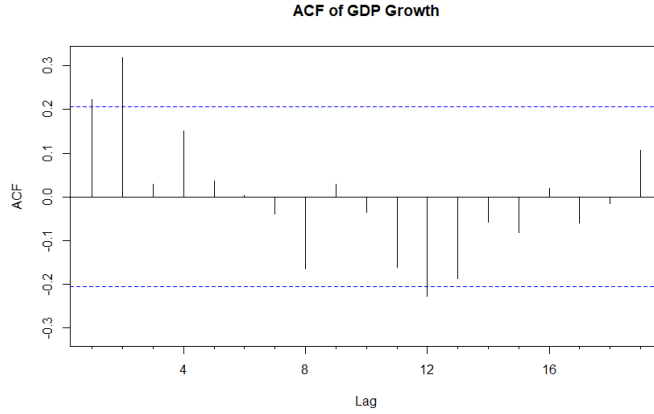


Figure: Sample autocorrelation function (ACF) of the quarterly U.S. real GDP growth rate ($T = 90$, and hence $1.96/\sqrt{T} \approx 0.207$).

This time series exhibits clear autocorrelation, which generally dampens (slowly) as the lag length h increases.

- This dampening effect depends on the time series (process) considered.
- Notice that above the first depicted lag (in this R function) is r_1 . At times in different functions and packages $r_0 = 1$ is depicted, but that is of course not of interest.
- If there is some (remaining) seasonal variation, say in monthly time series after seasonal adjustment (if applicable), it causes rather strong autocorrelations at lags $h = 12, 24$ and 36 . Similar behaviour can also happen if the time series contain some other cyclical pattern.

3.6 White noise process

Weak stationarity is not a strong enough assumption to allow statistical analysis of time series. It alone does not suffice for the construction of a likelihood function, nor is it enough to derive distributions of estimators or test statistics. For this reason, we need to define another concept of stationarity that is stronger than weak stationarity, **strict stationarity**. We will consider that in the next section.

Before that, let us consider first a **white noise process**, which is the simplest possible example of a weakly stationary process. Let u_t be a process for which it holds that

- $E(u_t) = \mu < \infty$, often $E(u_t) = 0$,
- $\text{Var}(u_t) = \sigma^2 < \infty$, and
- $\text{Cov}(u_s, u_t) = 0$, when $s \neq t$.

This process is often called a **(weak) white noise** process and denoted by $u_t \sim \text{wn}(\mu, \sigma^2)$.

- By definition, it is clear that a white noise process is weakly stationary.
- Despite their simplicity, as we will see later on, white noise processes play a central role because they can be used to construct more complicated processes and eventually realistic models for empirical data.

3.7 Strictly stationary process

Let us now define the above-mentioned more stringent form of stationarity. A process $\{y_t; t = 0, \pm 1, \pm 2, \dots\}$ is called **strictly stationary** (sometimes strongly stationary) if the collections of random variables $y_{t_1}, \dots, y_{t_m}$ and $y_{t_1+h}, \dots, y_{t_m+h}$ have the same m -dimensional joint distribution for all integers $t_1, \dots, t_m$, h and m ($m > 0$).

- That is, **the entire probability structure of the process y_t is time-invariant**, not just the first and second-order moments.

Properties. A strictly stationary stochastic process has the following properties (that follow straightforwardly from the definition):

- **SS1:** The random variables y_t have the same distribution for all t .
- **SS2:** The random vectors (y_t, y_{t+h}) and (y_s, y_{s+h}) have the same distribution for all t and s and all (fixed) h .
- **SS3:** y_t is weakly stationary if $E(y_t^2) < \infty$ holds.

It is clear that weak stationarity does not, in general, imply strict stationarity.

- For instance, weak stationarity says nothing about moments of order 3 and above.

i Extra: Gaussian processes

In the case of so-called Gaussian processes, weak stationarity implies strict stationarity. A process y_t (not necessarily stationary in any sense) is called *normal* or *Gaussian*, if the m -dimensional probability distribution of the random variables $y_{t_1}, \dots, y_{t_m}$ is multivariate normal (Gaussian) for

all integers $t_1, \dots, t_m$ and m ($m > 0$) (for simplicity, occasionally we will simply say normal when meaning multivariate normal distribution). Recall that a multivariate normal distribution is fully determined by its first and second moments. Therefore, weak stationarity and normality together do imply strict stationarity.

A very useful property of strict (but not weak) stationarity is that it is conserved in transformations. In other words,

- **SS4:** If x_t is a strictly stationary process, then so is also the process $y_t = g(x_{t+m}, \dots, x_{t-n})$, $m, n \geq 0$, for any “well-behaving” (e.g. continuous or, more generally, measurable) function g . Moreover, m , n , or both can be replaced with ∞ .

“Stationarity” in this material. Unless otherwise stated, stationarity will be interpreted as strict stationarity from now on.

- From a practical time series analysis perspective, as introduced above, visual inspection of the time series is of interest, because it relates directly to the theoretical implications of weak stationarity (i.e. the mean, variance and autocovariances of the series do not depend on time).

3.8 IID and NID processes

Let u_t be a sequence of **independent and identically distributed** (IID) random variables.

- The strict stationarity of this process is easy to check.

i Extra: Some further results

The cumulative distribution function of the random vector $(u_{t_1}, \dots, u_{t_m})$ evaluated at $(x_1, \dots, x_m)$ is $F(x_1) \cdots F(x_m)$. Clearly, this is also the cumulative distribution function of the random vector $(u_{t_1+h}, \dots, u_{t_m+h})$ evaluated at $(x_1, \dots, x_m)$.

If $E(u_t^2) < \infty$, then u_t is also weakly stationary, and in this case it has the same moment properties as the (weak) white noise process above.

This IID process is also called white noise, or sometimes **strong white noise**. We will use the shorthand notation

$$u_t \sim \text{iid}(\mu, \sigma^2) \quad \text{or} \quad u_t \sim \text{IID}(\mu, \sigma^2),$$

where iid refers to “independent and identically distributed”, $\mu = \mathbb{E}(u_t)$ and $\sigma^2 = \text{Var}(u_t)$ (in what follows, most often $\mu = 0$).

If it is additionally assumed that $u_t \sim \text{N}(\mu, \sigma^2)$, one denotes

$$u_t \sim \text{nid}(\mu, \sigma^2) \quad \text{or} \quad u_t \sim \text{NID}(\mu, \sigma^2),$$

where nid refers to **normally and independently distributed**.

i Extra: Some (somewhat peculiar) example cases

The first example demonstrates that the mathematical definition of weak stationarity allows for processes that may feel awkward and surprising. Define the process

$$y_t = A \cos(\lambda t) + B \sin(\lambda t),$$

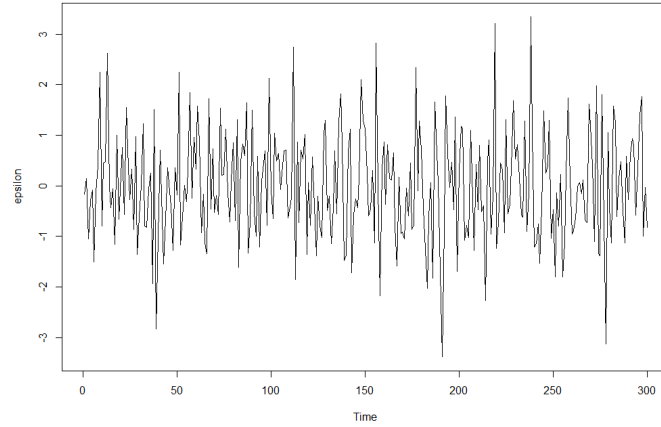
where $\lambda \in [0, \pi)$ is a constant and the random variables A and B satisfy $\mathbb{E}(A) = \mathbb{E}(B) = 0$, $\text{Var}(A) = \text{Var}(B) = \sigma^2$ and $\text{Cov}(A, B) = 0$. Obviously, $\mathbb{E}(y_t) = 0$ holds. Using properties of trigonometric functions, straightforward calculation can be used to establish that

$$\text{Cov}(y_t, y_{t+h}) = \sigma^2 \cos(\lambda h).$$

Therefore, y_t is weakly stationary (details are left as an exercise). This process is peculiar in that its realizations are quite regular functions of time (linear combinations of $\cos(\lambda t)$ and $\sin(\lambda t)$). Note also that for this process, the condition $\gamma_h \rightarrow 0$ as $h \rightarrow \infty$ does not hold. $\square$

The second example: Define the process $y_t = u_t \sqrt{\omega + \alpha u_{t-1}^2}$, where $\omega > 0$, $\alpha \geq 0$ and $u_t \sim \text{iid}(0, \sigma^2)$ with $\mathbb{E}(u_t^2) = \sigma^2 < \infty$ (independence, not just uncorrelatedness, is needed here). It is easy to see that y_t is weak white noise (justification as an exercise). Because $y_{t-1} = u_{t-1} \sqrt{\omega + \alpha u_{t-2}^2}$, it is clear that y_t is not strong white noise (unless $\alpha = 0$). $\square$

As an illustration of an NID process, below we depict a simulated realization from the $\text{nid}(0, 1)$ process. The length of the time series is 300 observations ($T = 300$). In other words, we can state that $u_t \sim \text{nid}(0, 1)$, $t = 1, \dots, 300$. The



simulated time series exhibits completely random variation. As we can see, there are occasionally higher and lower values due to sampling variation.

Figure: A simulated realization of $\text{nid}(0, 1)$ process ($T = 300$).

Chapter 4

Linear process

4.1 MA(1) process

As was mentioned, white noise processes can be used to define more complicated processes. A simple example and special case is the **moving average process of order one**, or **MA(1) process**,

$$y_t = \mu + u_t + \theta_1 u_{t-1}, \quad u_t \sim \text{iid}(0, \sigma^2).$$

The values of the process are hence assumed to be generated as a weighted average of two independent and unobserved random shocks.

- In this process (model equation), we also include the mean of the process $E(y_t) = \mu$. Alternatively, we can write the MA(1) process as

$$y_t - \mu \equiv z_t = u_t + \theta_1 u_{t-1}, \quad u_t \sim \text{iid}(0, \sigma^2).$$

- Notice that possible other deterministic components than a nonzero mean can also be readily included if necessary.

MA(1) processes are strictly stationary (see above and the property SS4) and also weakly stationary. Simple calculations show that

$$E(y_t) = E(\mu) + E(u_t) + \theta_1 E(u_{t-1}) = \mu,$$

$$\begin{aligned} \text{Var}(y_t) &= E\left(y_t - E(y_t)\right)^2 \\ &= \text{Var}(z_t) \\ &\stackrel{\text{iid}}{=} \text{Var}(u_t) + \theta_1^2 \text{Var}(u_{t-1}) \\ &= \sigma^2 (1 + \theta_1^2), \end{aligned}$$

and, for $h > 0$,

$$\begin{aligned}
 \text{Cov}(y_t, y_{t+h}) &= \text{Cov}(z_t, z_{t+h}) \\
 &= \text{E}[(u_t + \theta_1 u_{t-1})(u_{t+h} + \theta_1 u_{t+h-1})] \\
 &= \text{E}(u_t u_{t+h}) + \theta_1 \text{E}(u_t u_{t+h-1}) + \theta_1 \text{E}(u_{t-1} u_{t+h}) + \theta_1^2 \text{E}(u_{t-1} u_{t+h-1}) \\
 &= \begin{cases} \theta_1 \sigma^2, & h = 1, \\ 0, & h > 1. \end{cases}
 \end{aligned}$$

The latter two calculations make use of the independence of the process u_t . These results show the weak stationarity of the MA(1) process.

- The same moment results are also obtained if the assumption $u_t \sim \text{iid}(0, \sigma^2)$ is replaced with the milder assumption $u_t \sim \text{wn}(0, \sigma^2)$, but in this case one cannot deduce the strict stationarity of y_t . The same comment holds also to the more general results to be presented in the next section.

Based on these calculations, the autocorrelation function of an MA(1) process takes the form

$$\rho_h = \begin{cases} 1, & h = 0, \\ \theta_1 / (1 + \theta_1^2), & h = 1, \\ 0, & h > 1. \end{cases}$$

Therefore, a typical feature of an MA(1) process is that **the autocorrelation function drops to zero after lag one**.

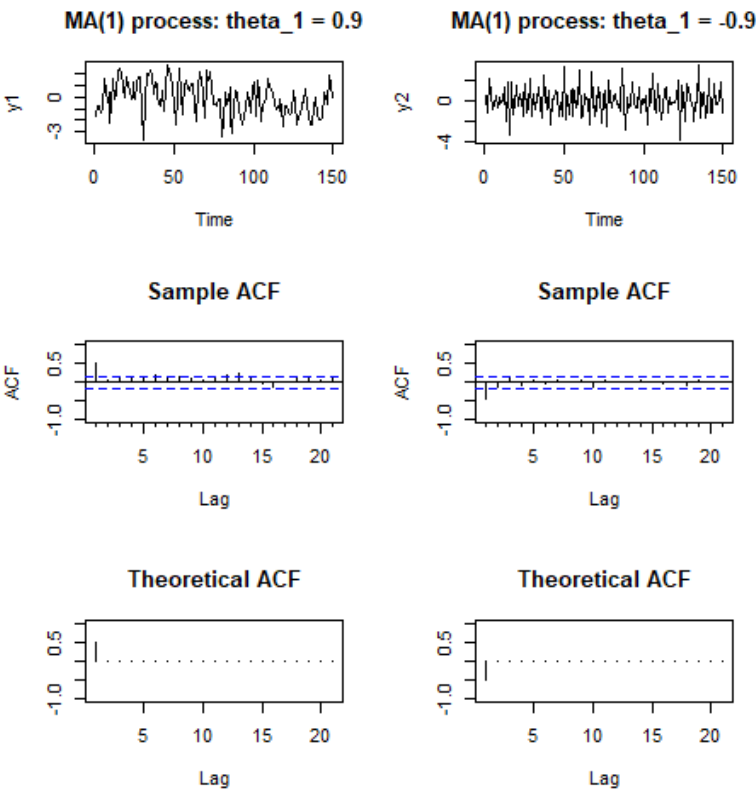
- Thus, observations more than one period apart are uncorrelated and, when assumption $u_t \sim \text{iid}(0, \sigma^2)$ holds, even independent.
- The same conclusion could, of course, be immediately made from the MA(1) model equation above.

The following figure presents two simulated realizations of length 150 ($T = 150$) of an MA(1) process with $u_t \sim \text{nid}(0, 1)$. Sample autocorrelation function based on the observations as well as the theoretical autocorrelation function are also shown.

Figure: Two simulated realizations of the MA(1) process and their (sample) autocorrelations, starting from the first autocorrelations (r_1 and ρ_1).

In the figure, on the left, the MA(1) coefficient θ_1 is 0.9, on the right $\theta_1 = -0.9$. In addition to simulated time series, the figure plots sample autocorrelations (“sample ACF”) and theoretical autocorrelations (“theoretical ACF”) of these two processes. In these figures the values $r_0 = 1$ and $\rho_0 = 1$ are not depicted.

- In both simulated series, the observations vary around their mean (zero) according to their theoretical standard deviation (≈ 1.345).



- In the left panel, the series has positive autocorrelation, which manifests itself as positive observations, typically followed by another positive observation.
- In the right panel, due to negative autocorrelation, positive and negative observations typically alternate.

The estimated sample autocorrelation functions resemble rather closely their theoretical counterparts. In particular, in both cases, the estimated r_1 is well outside the approximate 95% confidence bands implied by the assumption $\rho_h = 0$ ($\forall h > 0$). The remaining estimated sample autocorrelations fall mostly within these bands.

- Due to random variation, some of the remaining 20 estimates may naturally occasionally fall outside these bands.
- Furthermore, as we will discuss later, in the case of an MA(1) process, the confidence bands used above are actually too narrow.

An obvious generalization of the MA(1) is obtained by adding a linear combination of the variables $u_{t-2}, \dots, u_{t-q}$ ($q < \infty$) to the right hand side of the MA(1). This leads to the so-called MA(q) process, to be considered in more detail in Section 5.

4.2 Causal linear process

The MA(1) process introduced in the previous section, and its generalization the MA(q) process ($q < \infty$) to be introduced later on, are special cases of the **linear process**

$$y_t = \mu + \sum_{j=-\infty}^{\infty} \psi_j u_{t-j}, \quad u_t \sim \text{iid}(0, \sigma^2).$$

It is clear that the infinite sum on the right hand side of this equation requires further care and cannot be well defined without suitable further restrictions on the coefficients ψ_j .

- We will assume that

$$\sum_{j=-\infty}^{\infty} \psi_j^2 < \infty.$$

Under this assumption the infinite sum on the right hand side of the linear process is well defined.

The mean and autocovariance function of y_t (general linear process) can be calculated in a similar fashion as for the MA(1) process regardless of the infinite

sum:

$$\mathbb{E}(y_t) = \mu + \sum_{j=-\infty}^{\infty} \mathbb{E}(\psi_j u_{t-j}) = \mu + \sum_{j=-\infty}^{\infty} \psi_j \mathbb{E}(u_{t-j}) = \mu,$$

and denoting (cf. Section 2 on deterministic components) $z_t \equiv y_t - \mu$, we get

$$\text{Var}(y_t) = \text{Var}(z_t) = \sum_{j=-\infty}^{\infty} \text{Var}(\psi_j u_{t-j}) = \sum_{j=-\infty}^{\infty} \psi_j^2 \text{Var}(u_{t-j}) = \sigma^2 \sum_{j=-\infty}^{\infty} \psi_j^2.$$

Moreover, because $\text{Cov}(y_t, y_{t+h}) = \text{Cov}(z_t, z_{t+h})$, for $h > 0$,

$$\begin{aligned} \text{Cov}(y_t, y_{t+h}) &= \mathbb{E} \left(\sum_{j=-\infty}^{\infty} \psi_j u_{t-j} \sum_{i=-\infty}^{\infty} \psi_i u_{t+h-i} \right) \\ &= \sum_{j=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} \mathbb{E}(\psi_j \psi_i u_{t-j} u_{t+h-i}) \\ &= \sum_{j=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} \psi_j \psi_i \mathbb{E}(u_{t-j} u_{t+h-i}) \\ &= \sigma^2 \sum_{j=-\infty}^{\infty} \psi_j \psi_{j+h}, \end{aligned}$$

where the calculations also make use of the properties of the process u_t (compare the results to the MA(1) process). These calculations show that y_t is weakly stationary.

- Notice that the strict stationarity follows from the strict stationarity of u_t and the property SS4.

Causal linear (MA(∞)) process. It typically holds that $\psi_j = 0$ for $j < 0$, in which case the general linear process reduces to

$$y_t = \mu + \sum_{j=0}^{\infty} \psi_j u_{t-j}, \quad u_t \sim \text{iid}(0, \sigma^2).$$

Because y_t no longer depends on future values of the u_t variables, one often speaks of a causal linear process or a causal MA(∞) process.

- The values of the process y_t are assumed to be generated as a weighted sum of (possibly) infinitely many independent and unobserved random shocks.
- In the **noncausal case** the future shocks u_{t+j} ($j > 0$) affect the present value of the process y_t . In this course, we do not consider noncausal models in more detail.

4.3 AR(1) process

A simple special case of a causal linear process containing only one unknown parameter (and constant term) is achieved by assuming that $\psi_j = \phi_1^j$. A process defined like this leads to the autoregressive process of order one, that is an **AR(1) process**

$$y_t = \nu + \phi_1 y_{t-1} + u_t, \quad u_t \sim \text{iid}(0, \sigma^2).$$

Here one interprets the present value of the process to linearly depend on the previous value of the process as well as on an unobserved random shock (or error term) similarly as in the linear process. Furthermore, ν denotes the constant term of the process, whose connection to the mean $E(y_t) = \mu$ will be examined below.

Referring to the causal linear process, for the condition $\sum_{j=0}^{\infty} \psi_j^2 < \infty$ to be satisfied, it needs to be assumed that $|\phi_1| < 1$. The necessity of condition $|\phi_1| < 1$ can be demonstrated by making use of repetitive substitutions.

- First, substitute $y_{t-1} = \nu + \phi_1 y_{t-2} + u_{t-1}$ on the right hand side of the AR(1) equation.
- Then substitute $y_{t-2} = \nu + \phi_1 y_{t-3} + u_{t-2}$ to the resulting expression, and so on.

Continuing in this fashion, we obtain

$$y_t = \phi_1^k y_{t-k} + \nu \sum_{j=0}^{k-1} \phi_1^j + \sum_{j=0}^{k-1} \phi_1^j u_{t-j}.$$

When $|\phi_1| < 1$, this leads us to the limiting solution

$$y_t = \nu \sum_{j=0}^{\infty} \phi_1^j + \sum_{j=0}^{\infty} \phi_1^j u_{t-j}.$$

Because the AR(1) process with $|\phi_1| < 1$ is clearly a special case of the causal linear process, it is both strictly and weakly stationary.

The first and second moments can be deduced from the general formulae derived in the previous section. The expected value, variance and autocovariance functions take the form (given $|\phi_1| < 1$)

$$E(y_t) \equiv \mu = \nu \sum_{j=0}^{\infty} \phi_1^j + E\left(\sum_{j=0}^{\infty} \phi_1^j u_{t-j}\right) = \nu \sum_{j=0}^{\infty} \phi_1^j = \nu / (1 - \phi_1),$$

$$\text{Var}(y_t) = \sigma^2 \sum_{j=0}^{\infty} \phi_1^{2j} = \sigma^2 / (1 - \phi_1^2),$$

and, for $h > 0$,

$$\begin{aligned}
 \gamma_h &= \text{Cov}(y_t, y_{t+h}) \\
 &= \sigma^2 \sum_{j=0}^{\infty} \phi_1^j \phi_1^{j+h} \\
 &= \sigma^2 \phi_1^h / (1 - \phi_1^2) \\
 &= \phi_1 \gamma_{h-1}.
 \end{aligned}$$

The autocorrelation function of the AR(1) process hence becomes

$$\rho_h = \begin{cases} 1, & h = 0, \\ \phi_1^h, & h > 0. \end{cases}$$

Unlike in the case of an MA(1) process, the autocorrelation function differs from zero for all lags (unless $\phi_1 = 0$). Note, however, that the condition $\gamma_h \rightarrow 0$ when $h \rightarrow \infty$, is satisfied.

- Later, we will study a generalization of the AR(1) process, called an AR(p) process, which is obtained by adding a linear combination of the variables $y_{t-2}, \dots, y_{t-p}$, ($p < \infty$) on the right hand side of the AR(1) process.

Persistence. In the AR(1) process, the autoregressive parameter ϕ_1 clearly measures the persistence of a random shock to the time series.

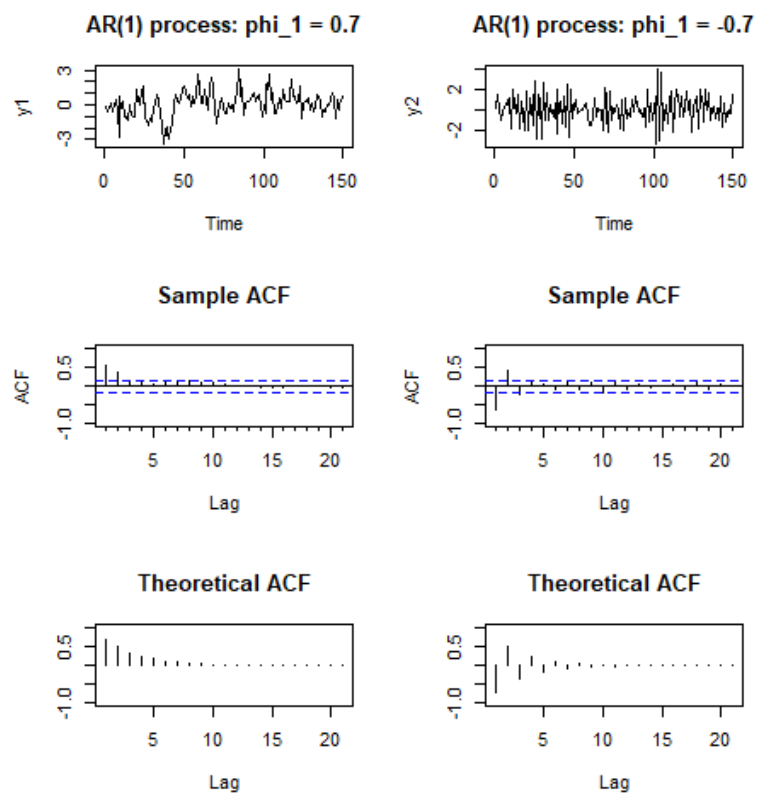
- If ϕ_1 is close to unity, autocorrelation is high and the series (process) is “strongly persistent”.
- If ϕ_1 is close to zero, there is no persistence and the effect of the shock is temporary.

The sign of ϕ_1 determines whether the time series is positively or negatively autocorrelated.

- If ϕ_1 is positive, then the positive values of y_t are tending to follow positive values, and similarly with negative values.
- If ϕ_1 is negative, then positive values tend to follow negative values, and vice versa.

Figure: Two simulated realizations of the AR(1) process and their autocorrelations, starting from the first autocorrelations (r_1 and ρ_1).

The figure presents two simulated realizations of length 150 of an AR(1) process with $u_t \sim \text{nid}(0, 1)$. On the left, the AR coefficient ϕ_1 is 0.7 and on the right -0.7. For simplicity, the constant term (and hence the mean) is set to 0. In addition to simulated time series, the figure plots sample autocorrelations and theoretical autocorrelations of these two processes.



- Being stationary series, the time series vary around their mean (zero) according to their theoretical standard deviation (≈ 1.4).
- When $\phi_1 = 0.7$ (left), the observations have relatively strong positive autocorrelation and, as a consequence, several consecutive observations occur above the mean, as well as below the mean. This gives the time series a “smooth” flavour.
- When $\phi_1 = -0.7$ (right), the sign of the autocorrelation coefficient alternates with the distance between the observations. This alternation between positive and negative autocorrelations gives the observed time series a jagged/zigzag pattern.
- In both cases, the estimated sample autocorrelation functions are rather close to their theoretical counterparts. Moreover, several estimated autocorrelation coefficients are outside of the 95% confidence bands based on the assumption of $\rho_h = 0$ ($\forall h > 0$).

i Extra: Noncausal AR process

The definition of an AR(1) process is usually based on its model equation as described above. In connection with this, the condition $|\phi_1| < 1$ is often called the stationarity condition of an AR(1) process. This terminology is somewhat misleading because AR(1) equation has a stationary solution also in the case $|\phi_1| > 1$, although this solution cannot be represented in the form of linear process. When $|\phi_1| > 1$, and setting $\nu = 0$ for simplicity, the AR(1) equation can be rewritten as

$$y_t = \phi_1^{-1}y_{t+1} - \phi_1^{-1}u_{t+1}$$

by increasing the time index t by one step. Using repetitive substitutions forward in time in a fashion similar as above, this leads to

$$y_t = \phi_1^{-k-1}y_{t+1+k} - \sum_{j=0}^k \phi_1^{-j-1}u_{t+1+j}.$$

This leads to the limiting solution

$$y_t = - \sum_{j=1}^{\infty} \phi_1^{-j}u_{t+j}.$$

Note that we can arrive to the (more or less) same (except for an unimportant minus sign) solution by starting from the linear process and by assuming $\psi_j = \phi_1^j$, when $j < 0$, and $\psi_j = 0$, when $j \geq 0$ ($|\phi_1| > 1$). This solution of the AR(1) equation is called noncausal. This kind of noncausal

AR processes have received some attention in the recent literature. However, in this course we restrict our attention to causal AR processes (as do most textbooks).

4.4 Random walk

For the AR(1) model, and for any initial value y_0 , we obtain a representation (after recursive substitutions)

$$y_t = \phi_1^t y_0 + \nu \sum_{j=0}^{t-1} \phi_1^j + \sum_{j=0}^{t-1} \phi_1^j u_{t-j}, \quad t = 1, 2, \dots$$

If we assume the initial value y_0 to be independent of the variables u_t , $t \geq 1$, one can use the assumption $u_t \sim \text{iid}(0, \sigma^2)$ to deduce

$$\mathbb{E}(y_t) = \phi_1^t \mathbb{E}(y_0) + \nu \sum_{j=0}^{t-1} \phi_1^j$$

and

$$\begin{aligned} \text{Var}(y_t) &= \mathbb{E}\left(y_t - \mathbb{E}(y_t)\right)^2 \\ &= \text{Var}\left(\phi_1^t y_0\right) + \text{Var}\left(\sum_{j=0}^{t-1} \phi_1^j u_{t-j}\right) \\ &= \phi_1^{2t} \text{Var}(y_0) + \sigma^2 \sum_{j=0}^{t-1} \phi_1^{2j}. \end{aligned}$$

When $|\phi_1| = 1$, the expected value of y_t , or at least its variance, clearly depends on t regardless of how y_0 (or its distribution) is chosen.

- In this case, the AR(1) process therefore has no stationary solution.

When $\phi_1 = 1$ and $t \geq 1$, the AR(1) process reduces to

$$y_t = \nu + y_{t-1} + u_t, \quad u_t \sim \text{iid}(0, \sigma^2).$$

This is called a **random walk**. This name is due to the “wandering” nature of the realizations of the process.

- The left panel of the figure below illustrates this. For simplicity, we assume here $\nu = 0$ (that is the random walk without drift) and $u_t \sim \text{iid}(0, 1)$.
- The right panel illustrates the obvious fact that the differences $y_t - y_{t-1} = u_t$ of a random walk without drift, $y_t = y_0 + \sum_{j=0}^{t-1} u_{t-j}$, are stationary.

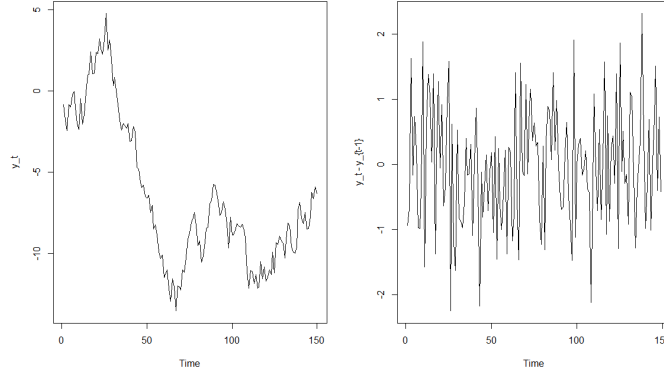


Figure: A simulated realization of the random walk process (assuming $u_t \sim \text{nid}(0, 1)$) (left) and its first-differenced series (right).

The random walk and its generalizations play a central role in the analysis of **nonstationary time series**. We will come back to this in Sections 12–13.

4.5 Lag operator

A concept easing the algebraic manipulations of time series processes is the so-called **backshift operator** or **lag operator**. For any process (or simply a sequence of numbers) x_t , define the operation with the equation $Bx_t = x_{t-1}$. More generally, $B^2x_t = B(Bx_t) = Bx_{t-1} = x_{t-2}$, and inductively define

$$B^k x_t = B(B^{k-1} x_t) = x_{t-k}, \quad B^0 x_t = x_t.$$

- Here k can also be negative, and when $k < 0$, the operator becomes a “forward shift” operator, for example $B^{-1}x_t = x_{t+1}$, $B^{-2}x_t = x_{t+2}$ etc.
- At times backshift or lag operator is denoted by L^k instead of B^k .

Using the lag operator, one can also define polynomials.

- For example, $\theta(B) = 1 + \theta_1 B$ and, e.g., $\psi(B) = \sum_{j=-\infty}^{\infty} \psi_j B^j$.

One can algebraically operate with the lag operator exactly as if B were a real or a complex number.

- For instance, the MA(1) process can be written as

$$y_t = \mu + u_t + \theta_1 u_{t-1} = \mu + \theta(B) u_t.$$

- When differencing (some) process y_t twice, we obtain

$$\begin{aligned}
\Delta^2 y_t &= (1 - B)^2 y_t \\
&= (1 - B) [(1 - B) y_t] \\
&= (1 - B) (y_t - y_{t-1}) \\
&= y_t - y_{t-1} - y_{t-1} + y_{t-2} \\
&= (1 - 2B + B^2) y_t.
\end{aligned}$$

4.6 ARMA(1,1) process

Using the lag operator, the general linear process can be defined by the equation

$$y_t - \mu = \psi(B) u_t, \quad u_t \sim \text{iid}(0, \sigma^2),$$

where $\psi(B) = \sum_{j=-\infty}^{\infty} \psi_j B^j$. The operator $\psi(B)$ is sometimes thought as a **linear filter**, which transforms the white noise sequence $\{u_t\}$ to the process $\{y_t\}$.

In what follows, we will often consider the special case of the (causal) linear process in which the filter $\psi(B)$ is rational, that is,

$$\psi(B) = \sum_{j=0}^{\infty} \psi_j B^j = \theta(B) \phi(B)^{-1},$$

where $\phi(B)$ and $\theta(B)$ are polynomials of finite order. In the simplest case, these polynomials are of order one so that

$$\phi(B) = 1 - \phi_1 B \quad \text{and} \quad \theta(B) = 1 + \theta_1 B.$$

- It is clear that to obtain stationarity, some restrictions have to be placed on the coefficients of the polynomial $\phi(B)$.

Based on our discussion on the AR(1) process above, it is clear that in the first-order case a sufficient stationarity condition is that $|\phi_1| < 1$. Then the condition $\sum_{j=-\infty}^{\infty} \psi_j^2 < \infty$ attached to the linear process is satisfied and the process

$$y_t - \mu = [\theta(B) \phi(B)^{-1}] u_t$$

is well defined. Multiplying both sides of this equation with the polynomial $\phi(B)$, we obtain the representation

$$\phi(B) (y_t - \mu) = \theta(B) u_t$$

or

$$y_t = \nu + \phi_1 y_{t-1} + u_t + \theta_1 u_{t-1}, \quad u_t \sim \text{iid}(0, \sigma^2),$$

where $\nu = \phi(B)\mu \equiv \phi(1)\mu = (1 - \phi_1)\mu$ (see the properties of the backshift operator). A process defined like this is called the **autoregressive moving average process of order one**, or the **ARMA(1,1) process**.

- This combines the AR(1) and MA(1) processes introduced earlier, and these processes can still be obtained as special cases.
- Later we will study a generalization of this process called the ARMA(p, q) process, which can be obtained by generalizing ARMA(1,1) in a similar fashion as discussed around AR(1) and MA(1) processes.
- The coefficients ψ_j of the filter $\psi(B) = \sum_{j=0}^{\infty} \psi_j B^j$ can be solved fairly straightforwardly from equation $\psi(B) = \theta(B)\phi(B)^{-1}$.

The above shows that the autocorrelation function of an ARMA(1,1) process can then be derived by applying the general formulae obtained for the general linear process. Details of these calculations are left as exercises.

4.7 Wold decomposition

The following famous result (named after Herman Wold) shows that every weakly stationary non-deterministic process can be expressed as a sum of a deterministic process and a causal MA(∞) process (the proof of this result is beyond the scope of this course and omitted). In other words, every weakly stationary non-deterministic process y_t ($t = 0, \pm 1, \pm 2, \dots$) has a representation

$$y_t = \sum_{j=0}^{\infty} \psi_j u_{t-j} + d_t,$$

where

- (i) $\psi_0 = 1$, $\sum_{j=0}^{\infty} \psi_j^2 < \infty$,
- (ii) $u_t \sim \text{wn}(0, \sigma^2)$,
- (iii) d_t is deterministic, and
- (iv) $\text{Cov}(u_t, d_s) = 0$ for all t and s . $\square$

Part (iii) means that the process d_t can be predicted linearly using the variables $y_{t-1}, y_{t-2}, \dots$, with no error. This and part (ii) together imply that u_t can be interpreted as a forecast error when forecasting y_t linearly using the lags of y_t . Modelling d_t can thus be thought of as modelling a trend as discussed in Sections 1–2.

- In the case $d_t = 0$ the process y_t is called “purely non-deterministic”.

- A simple example of a process d_t is that it is constant. Such a realization of d_t can in practice be interpreted as the expected value of process y_t , $E(y_t) = \mu$, or at least be included in it.

The task of modelling a weakly stationary process reduces to the task of modelling the linear filter $\psi(B) = \sum_{j=0}^{\infty} \psi_j B^j$.

i Extra: Deterministic and non-deterministic processes/parts

Consider the weakly stationary process

$$y_t = A \cos(\lambda t) + B \sin(\lambda t), \quad t = 0, \pm 1, \pm 2, \dots,$$

where $\lambda \in [0, \pi)$ is a constant and the random variables A and B satisfy the conditions $E(A) = E(B) = 0$, $\text{Var}(A) = \text{Var}(B) = \sigma^2$ and $\text{Cov}(A, B) = 0$. Because

$$y_t + y_{t-2} = A[\cos(\lambda t) + \cos(\lambda(t-2))] + B[\sin(\lambda t) + \sin(\lambda(t-2))],$$

we can use the trigonometric identities

$$\sin(x_1) + \sin(x_2) = 2 \sin\left(\frac{x_1 + x_2}{2}\right) \cos\left(\frac{x_1 - x_2}{2}\right)$$

and

$$\cos(x_1) + \cos(x_2) = 2 \cos\left(\frac{x_1 + x_2}{2}\right) \cos\left(\frac{x_1 - x_2}{2}\right)$$

to derive the result $y_t + y_{t-2} = 2 \cos(\lambda) y_{t-1}$ or, in other words,

$$y_t = 2 \cos(\lambda) y_{t-1} - y_{t-2}, \quad t = 0, \pm 1, \pm 2, \dots$$

The process y_t is somewhat peculiar in that when y_{t-1} and y_{t-2} (and the value of the constant λ) are known, the value for the current period y_t can be predicted using a simple linear formula with perfect precision without any forecast error. A process with such a property is called a *deterministic* process. More generally, the forecast of y_t is allowed to be any linear function of the past values of the process $y_{t-1}, y_{t-2}, \dots$ so that the forecast is a linear combination of the variables $y_{t-1}, \dots, y_{t-n}$ or a mean square limit of such linear combinations as $n \rightarrow \infty$. If a process is not deterministic, then it is called *non-deterministic*.

4.8 Properties of sample mean and sample autocorrelations

Because the sample mean and the sample autocorrelation function are central tools in the analysis of time series, we next briefly discuss some of their statistical properties.

We can show that (see below) that the **sample mean**

$$\bar{y} = \frac{1}{T} (y_1 + \cdots + y_T)$$

is

- **unbiased** and **consistent estimator** of the population mean $\mu = \mathbf{E}(y_t)$, and
- **asymptotically normally distributed**

i Extra: Proof of unbiasedness and consistency of the sample mean

In the following discussion, we assume both weak and strict stationarity of the process.

For the sample mean $\bar{y} = T^{-1} (y_1 + \cdots + y_T)$, it holds that

$$\mathbf{E}(\bar{y}) = \frac{1}{T} (\mathbf{E}(y_1) + \cdots + \mathbf{E}(y_T)) = \mu$$

so that it is an *unbiased estimator* of the population mean $\mu = \mathbf{E}(y_t)$ (recall that in general, an estimator $\hat{\vartheta}$ of a population parameter ϑ is called unbiased if $\mathbf{E}(\hat{\vartheta}) = \vartheta$).

For the variance of $\bar{y}$, we can construct

$$\begin{aligned} \text{Var}(\bar{y}) &= \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \text{Cov}(y_t, y_s) \\ &= \frac{1}{T^2} \sum_{t-s=-(T-1)}^{T-1} (T - |t-s|) \gamma_{t-s} \\ &= \frac{1}{T} \sum_{h=-(T-1)}^{T-1} \left(1 - \frac{|h|}{T}\right) \gamma_h. \end{aligned}$$

The second equality above can be justified by noting that the preceding double sum is the sum of the elements of the matrix $[\gamma_{t-s}]_{t,s=1,\dots,T}$. Now assume that

$$\sum_{h=-\infty}^{\infty} |\gamma_h| < \infty,$$

a requirement more stringent than condition $\gamma_h \rightarrow 0$, when $h \rightarrow \infty$. This condition is satisfied by many processes used in practice (for instance, by the AR(1) and MA(1) processes). Making use of the triangle inequality, we now obtain the result

$$\text{Var}(\bar{y}) = \mathbb{E}(\bar{y} - \mu)^2 \leq \frac{1}{T} \sum_{h=-(T-1)}^{T-1} \left(1 - \frac{|h|}{T}\right) |\gamma_h| \rightarrow 0, \quad \text{as } T \rightarrow \infty.$$

In other words, the sample mean is a consistent estimator for the expected value.

If we strengthen the assumptions made above and additionally assume that the process y_t is Gaussian, the sample mean is also normally distributed with the mean and variance as indicated above. Furthermore, the following asymptotic result can be established (we omit the details)

$$\sqrt{T}(\bar{y} - \mu) \xrightarrow{d} \mathbf{N}\left(0, \sum_{h=-\infty}^{\infty} \gamma_h\right) \quad \text{or} \quad \bar{y} \underset{as}{\sim} \mathbf{N}\left(\mu, \frac{1}{T} \sum_{h=-\infty}^{\infty} \gamma_h\right).$$

This result can be derived without assuming y_t to be Gaussian.

The purpose here is to show the law of large numbers and the central limit theorem to apply to stationary processes with “reasonable assumptions”. To use this result to construct tests and confidence intervals for μ , we also need to estimate the infinite sum $\sum_{h=-\infty}^{\infty} \gamma_h$. A suitable estimator is (compare the expression of $\text{Var}(\bar{y})$ above) $\sum_{h=-K}^K (1 - |h|/T) c_h$, where c_h is the sample autocovariance coefficient defined earlier, and K is a “suitably” chosen number smaller than T (for example, $K \approx \sqrt{T}$). For large values of h , the estimator c_h becomes unprecise, and for this reason K should not be too large compared to T .

The statistical properties of the sample autocorrelation coefficients $r_h = c_h/c_0$ are more complicated to derive than those of the sample mean, so we will not attempt to provide any detailed justifications. Under “reasonably general assumptions”, consistency and asymptotic normality of them can be established.

Important special case: For the important special case of $y_t \sim \text{iid}(\mu, \sigma^2)$, it holds that

$$(r_1, \dots, r_H)' \underset{as}{\sim} \mathbf{N}(\mathbf{0}, T^{-1} \mathbf{I}_H),$$

with a H -dimensional zero mean vector and $\mathbf{I}_H$ ($H \times H$) denotes an identity matrix. This result can be used to test whether it is realistic to consider an observed time series as an uncorrelated time series process. Under the hypothesis

to be tested (uncorrelatedness), the estimators $r_1, \dots, r_H$ are approximately independent with distribution $N(0, T^{-1})$. Based on this, we get

$$P(|r_h| \geq 1.96/\sqrt{T}) \approx 0.05,$$

a result that can be used to evaluate the significance of individual sample autocorrelation coefficients.

To obtain a joint test for several autocorrelation coefficients, that is to test the null hypothesis of no autocorrelation $H_0 : \rho_1 = \dots = \rho_H = 0$, one can use the test statistic

$$Q = T \sum_{h=1}^H r_h^2 \underset{as}{\sim} \chi_H^2,$$

whose large values would lead to rejection. Note that the asymptotic χ_H^2 -distribution follows from the distribution result above for $(r_1, \dots, r_H)$ and the definition of the chi-squared distribution. In practice, an alternative and slightly different test statistic

$$Q_{LB} = T(T+2) \sum_{h=1}^H r_h^2 / (T-h) \underset{as}{\sim} \chi_H^2,$$

called the **Ljung-Box test statistic** is preferred because in small samples its distribution has been found to be closer to the χ_H^2 -distribution than that of the test statistic Q . It should be clear that both tests need H not to be too large compared to T to work well.

Nonlinear dependencies. The autocorrelation function can be used to reveal linear dependencies between the observations, but not nonlinear ones (with the exception of Gaussian processes). To investigate the presence of potential nonlinear dependence over time, one (somewhat limited) approach is to test for autocorrelation in the squared observations.

- Assuming $y_t \sim \text{iid}(\mu, \sigma^2)$ and $E(y_t^4) < \infty$, what was said above about the sample autocorrelations also holds for sample autocorrelations computed from the squared observations y_t^2 . In particular, the asymptotic result(s) remain valid when the autocorrelations are computed from squared observations, and also the Ljung-Box test has its indicated asymptotic distribution. In this context, however, the test is usually called the **McLeod-Li test**.
- Investigating the autocorrelations between squared observations is of great interest, especially in the case of **financial time series**, which themselves often are uncorrelated. We will return to this point later.

Empirical example (continued). As an empirical illustration, let us consider the quarterly U.S. real GDP growth rate. It turns out that we obtain the following results from the Ljung-Box test for different lag lengths H :

Lag (H)	Q_{LB}	p -value
4	16.487	0.002
8	19.494	0.012
12	27.942	0.006
16	32.897	0.008
20	34.679	0.022

These results clearly point out statistically significant autocorrelation (at the 5% significance level) in the real GDP growth rate, which was apparent already based on the estimated autocorrelation coefficients. Furthermore, it turns out that for the McLeod-Li tests, the resulting p -values are high (around 0.5 or higher) for all the lag length selections.

Chapter 5

ARMA processes

5.1 AR(p) process

Earlier, we considered the AR(1) process. This can be generalized to the **AR(p) process**

$$y_t = \nu + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + u_t, \quad u_t \sim \text{iid}(0, \sigma^2),$$

which can be rewritten

$$\phi(B) y_t = \nu + u_t,$$

with the lag-polynomial $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$. The current value of the process hence depends linearly on p past values of the process and the constant term, as well as on an unobserved random shock (or error term or “innovation”, depending on the perspective and context).

- Analogously, we can rewrite (as in some references) AR(p) for the demeaned process as

$$\phi(B) (y_t - \mu) = u_t.$$

- Notice that $\nu \equiv \phi(B) \mu \equiv \phi(1) \mu = (1 - \phi_1 - \cdots - \phi_p) \mu$.

A sufficient condition for the stationarity of an AR(p) process. A sufficient condition for the stationarity of the AR(p) process is that the roots of the polynomial $\phi(z)$ ($z \in \mathbb{C}$) lie outside the unit circle/disk in the complex plane, or equivalently that

$$\phi(z) \neq 0 \text{ for } |z| \leq 1.$$

- As an example, in the AR(1) case the polynomial is $\phi(z) = 1 - \phi_1 z$, whose only root is $z = 1/\phi_1$. Because $|z| = 1/|\phi_1| > 1$ exactly when $|\phi_1| < 1$,

this shows why in the case $p = 1$ the condition $|\phi_1| < 1$ was found sufficient to ensure the existence of a (causal) stationary MA(∞) representation.

- Notice that the resulting root (roots), when $p > 1$, can also be complex numbers. Related to the above stationarity condition, the absolute value or norm of a complex number $z = x + iy$ ($i = \sqrt{-1}$) is defined as $|z| = \sqrt{x^2 + y^2}$. The latter form of the stationarity condition can be further expressed as $\phi(z) = 0 \Rightarrow |z| > 1$.

i Extra: More details on stationarity condition

Still more about the possibility of getting complex roots. The absolute value or norm of a complex number $z = x + iy$ ($i = \sqrt{-1}$) can be identified with the norm of the vector (x, y) .

One way to illustrate this stationarity condition makes use of a well-known result in mathematics called the fundamental theorem of algebra. As a consequence of this result, the polynomial $\phi(z)$ can be written as (assuming that $\phi_p \neq 0$)

$$\phi(z) = (1 - \zeta_1^{-1}z) \cdots (1 - \zeta_p^{-1}z),$$

where for the roots ζ_i it therefore holds that $\phi(\zeta_i) = 0$ and $|\zeta_i| > 1$ ($i = 1, \dots, p$). Therefore, an AR(p) process can be expressed as $(1 - \zeta_1^{-1}B) \cdots (1 - \zeta_p^{-1}B) y_t = u_t$. If all the roots of $\phi(z)$ are real, this equation can be divided by the polynomials $(1 - \zeta_i^{-1}B)$, $i = 1, \dots, p$, one at a time, and in this way it can be seen (similarly as in the case $p = 1$) that the resulting expression is a well-defined linear process. This procedure can also be generalized to the case of complex roots.

Based on the above-mentioned, a stationary AR(p) process has an MA(∞) representation

$$(y_t - \mu) = \phi(B)^{-1} u_t = \psi(B) u_t = \sum_{j=0}^{\infty} \psi_j u_{t-j},$$

where $\psi(B) = \sum_{j=0}^{\infty} \psi_j B^j = \phi(B)^{-1}$. The coefficients ψ_j can be solved as a function of the parameters $\phi_1, \dots, \phi_p$ from equation

$$(1 - \phi_1 B - \cdots - \phi_p B^p) (\psi_0 + \psi_1 B + \psi_2 B^2 + \cdots) = 1$$

by interpreting the right hand side as a power series in B , and setting the coefficients of B^j equal to each other on both sides of the equation.

- Recall that two polynomials are the same if their coefficients are the same.

- For instance, the coefficient of B^0 is 1, so that $\psi_0 = 1$. Next, the coefficient of B^1 is zero, so that $\psi_1 - \phi_1\psi_0 = 0$ from which $\psi_1 = \phi_1$ follows. The general solution is left as an exercise.

Autocorrelation function. The autocorrelation function of an AR(p) process could be derived by making use of its MA(∞) representation. Instead, we present an often used and rather practical alternative approach based on the AR(p) model equation.

Multiplying both sides of the demeaned AR(p) process representation by $y_{t-h} - \mu$ ($h \geq 0$) and taking expectations, we obtain

$$\begin{aligned} \mathbb{E}\left((y_t - \mu)(y_{t-h} - \mu)\right) &= \phi_1 \mathbb{E}\left((y_{t-1} - \mu)(y_{t-h} - \mu)\right) + \cdots \\ &+ \phi_p \mathbb{E}\left((y_{t-p} - \mu)(y_{t-h} - \mu)\right) + \mathbb{E}\left(u_t(y_{t-h} - \mu)\right). \end{aligned}$$

Because y_{t-h} (and likewise $y_{t-h} - \mu$) is a linear function of the innovation terms $u_{t-h}, u_{t-h-1}, \dots$, the variables y_{t-h} and u_t are independent when $h > 0$. Therefore, we get

- $\mathbb{E}\left(u_t(y_{t-h} - \mu)\right) = 0, \quad h > 0,$
- For $h = 0$, we get $\mathbb{E}\left(u_t(y_t - \mu)\right) = \mathbb{E}\left(u_t^2\right) = \sigma^2$.

As $\gamma_h = \gamma_{-h}$, we get

$$\gamma_h = \begin{cases} \phi_1\gamma_1 + \cdots + \phi_p\gamma_p + \sigma^2, & h = 0 \\ \phi_1\gamma_{h-1} + \cdots + \phi_p\gamma_{h-p}, & h > 0. \end{cases}$$

Dividing this (in the case $h > 0$) by the variance γ_0 , we get the autocorrelation function of an AR(p) process

$$\rho_h = \phi_1\rho_{h-1} + \cdots + \phi_p\rho_{h-p}, \quad h > 0,$$

so that it satisfies a difference equation similar to the one the AR(p) process itself satisfies.

- When the roots of $\phi(z)$ lie outside the unit disk in the complex plane, the solution ρ_h to this difference equation decays exponentially to zero as the lag length h increases.
- The solution ϕ_1^h for the AR(1) is an example of this (this solution can also be easily obtained by solving the difference equation above in the case $p = 1$ using the initial value $\rho_0 = 1$).

Partial autocorrelation function. We next define the partial autocorrelation function, which is, among other things, a useful tool for model selection. Assume

that α_h equals the conventional partial correlation coefficient that measures the correlation between the random variables y_t and y_{t-h} when the linear effect of the random variables $y_{t-1}, \dots, y_{t-h+1}$ has been first eliminated.

- Therefore, a particular consequence is that $|\alpha_h| \leq 1$.

In the case of an $\text{AR}(p)$ process, the partial autocorrelation function α_h has a special useful feature: For $m > p$, an $\text{AR}(p)$ process can be interpreted as an $\text{AR}(m)$ process with $\phi_{p+1} = \dots = \phi_m = 0$, which makes it clear that the partial autocorrelation function of an $\text{AR}(p)$ process satisfies (also $\alpha_0 = 1$):

$$y_t \sim \text{AR}(p) \Rightarrow \alpha_p = \phi_p \quad \text{and} \quad \alpha_h = 0 \text{ for } h > p.$$

In other words, **the partial autocorrelation function of an $\text{AR}(p)$ process drops to zero after lag p** (assuming $\phi_p \neq 0$).

- The sample counterpart of the (population) partial autocorrelation function is obtained by using the sample autocovariances $\mathbf{c}_h$.
- The (sample) partial autocorrelation function is typically estimated using recursive algorithms or regression techniques, which are closely related to the Yule-Walker framework (see details in the Extra section below).

Model selection implications. An important detail for model selection is that if an observed time series has been generated by an $\text{AR}(p)$ process, its estimated autocorrelation function should decay to zero as the lag length increases with no apparent breaks, whereas the estimated partial autocorrelation function should have a visible break after lag p^{**} .

To identify the location of the break point in the estimated partial autocorrelation function, one can make use of the following result:

- In the case of an $\text{AR}(p)$ process, the estimators $\hat{\alpha}_h$, $h > p$, are approximately independent with $\text{N}(0, T^{-1})$ -distribution. Therefore,

$$\text{P}(|\hat{\alpha}_h| \geq 1.96/\sqrt{T}) \approx 0.05$$

when $h > p$.

- On the other hand, because the estimators $\mathbf{c}_1, \dots, \mathbf{c}_p$ are consistent, the estimator $\hat{\alpha}_p$ converges in probability to the value of the theoretical partial autocorrelation coefficient α_p which for an $\text{AR}(p)$ process is nonzero. Therefore, for $h = p$, $\text{P}(|\hat{\alpha}_h| \geq 1.96/\sqrt{T})$ approaches one as the sample size increases.
- These remarks justify depicting the sample partial autocorrelation coefficients similarly to the usual autocorrelation coefficients, and adding horizontal critical/confidence bands to make their interpretation easier.

i Extra: Yule-Walker equations and autocorrelation and partial autocorrelation coefficients

Choosing $h = 1, \dots, p$ in the AR(p) autocorrelation function leads to the equations ($\rho_0 = 1$ and $\rho_h = \rho_{-h}$)

$$\begin{aligned}\rho_1 &= \phi_1 + \phi_2\rho_1 + \dots + \phi_p\rho_{p-1} \\ \rho_2 &= \phi_1\rho_1 + \phi_2 + \dots + \phi_p\rho_{p-2} \\ &\vdots \\ \rho_p &= \phi_1\rho_{p-1} + \phi_2\rho_{p-2} + \dots + \phi_p,\end{aligned}$$

which collectively are called the **Yule-Walker equations**. In matrix notation, these can be expressed as

$$\boldsymbol{\rho} = \mathbf{P}\boldsymbol{\phi},$$

where $\boldsymbol{\rho} = [\rho_1 \dots \rho_p]'$, $\boldsymbol{\phi} = [\phi_1 \dots \phi_p]'$ and $\mathbf{P} = [\rho_{i-j}]_{i,j=1,\dots,p}$ is a $p \times p$ matrix, whose row i and column j element equals ρ_{i-j} . This makes it clear that the parameter vector $\boldsymbol{\phi}$ can be expressed as a function of the autocovariance or autocorrelation coefficients. The result is the Yule-Walker equation

$$\boldsymbol{\phi} = \mathbf{P}^{-1}\boldsymbol{\rho} = \boldsymbol{\Gamma}^{-1}\boldsymbol{\gamma},$$

where $\boldsymbol{\gamma} = [\gamma_1 \dots \gamma_p]'$, $\boldsymbol{\gamma} = \gamma_0\boldsymbol{\rho}$ and $\boldsymbol{\Gamma} = [\gamma_{i-j}]_{i,j=1,\dots,p} = \gamma_0\mathbf{P}$. From the equations derived above it follows that also the parameter σ^2 can be expressed as a function of the autocovariance coefficients and the parameters $\phi_1, \dots, \phi_p$ as

$$\sigma^2 = \gamma_0 - \phi_1\gamma_1 - \dots - \phi_p\gamma_p.$$

A note on $\mathbf{P}^{-1}$: It is rather clear that this inverse matrix exists here. If not, there would need to be an exact linear relationship between the variables $y_{t-1}, \dots, y_{t-p}$, which (except for the case $\sigma^2 = 0$) is not possible. The sufficient condition for stationarity of an AR(p) process guarantees that $\mathbf{P}$ is invertible, although seeing this is somewhat technical.

To be more precise, let us denote $\boldsymbol{\gamma} = \boldsymbol{\gamma}_p$ and $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}_p$ in the Yule-Walker equation. Now, the partial autocorrelation function with lag h is defined as

$$\alpha_h = \begin{cases} 1, & \text{for } h = 0 \\ \text{the last component of the vector } \boldsymbol{\Gamma}_h^{-1}\boldsymbol{\gamma}_h, & \text{when } h \geq 1. \end{cases}$$

Because the second and third expressions of the Yule-Walker equation can be defined for any weakly stationary (non-deterministic) process, so can also the partial autocorrelation function.

The sample counterpart of the (population) partial autocorrelation function is straightforward to define, simply estimating the vector γ_h and matrix Γ_h using the obvious estimators $\mathbf{c}_h = [\mathbf{c}_1 \cdots \mathbf{c}_h]'$ and $\mathbf{C}_h = [\mathbf{c}_{i-j}]_{i,j=1,\dots,h}$. Therefore, the sample partial autocorrelation coefficient $\hat{\alpha}_h$ equals 1 for $h = 0$, and the last component of the Yule-Walker estimate $\hat{\phi}_{YW} = \mathbf{C}_h^{-1} \mathbf{c}_h$ of the parameter ϕ for $h \geq 1$. As a remark, we note that there exist recursive formulas that can be used to compute both theoretical and sample partial autocorrelation coefficients that avoid computing the inverse of the (potentially large) matrix $\mathbf{C}_h$, but we omit the details of this.

Empirical example. As an example, below we present the estimated sample partial autocorrelation functions (with lag lengths $h = 1, \dots, 19$) of the quarterly U.S. real GDP growth series, together with the above-mentioned critical/confidence bands.

- The corresponding sample autocorrelation coefficients are shown already above.
- The first two sample partial autocorrelation coefficients $\hat{\alpha}_1$ and $\hat{\alpha}_2$ are outside the confidence bands, whereas the rest lie within these bands (except the 11th lag), pointing towards an AR(2) process. The sample autocorrelation function partly supports this conclusion, while an MA(2) or some other ARMA model might also be a potential candidate for the GDP growth.
- We will come back to empirical model selection later in this material.

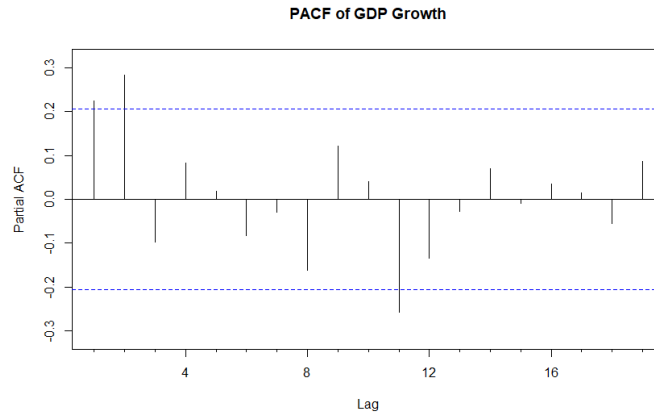


Figure: Sample partial autocorrelation function (PACF) of the quarterly U.S. real GDP growth rate.

5.2 MA(q) process and invertibility

The MA(q) process is defined as

$$y_t = \mu + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}, \quad u_t \sim \text{iid}(0, \sigma^2),$$

or alternatively using the lag operator as

$$(y_t - \mu) = \theta(B) u_t,$$

with the lag-polynomial $\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$. Therefore, the current value of the process is assumed to depend linearly on the present and last q unobservable random shocks. Because an MA(q) process is a special case of the (causal) linear process, it is always both weakly and strictly stationary. Moreover, we get

- $E(y_t) = \mu$,
- $\text{Var}(y_t) = \gamma_0 = (1 + \theta_1^2 + \cdots + \theta_q^2) \sigma^2$.

Autocorrelation function. By making use of the general results of the linear process, the MA(q) process has the autocovariance function given by

$$\gamma_h = \begin{cases} \sigma^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h}, & \text{for } 0 \leq h \leq q, \\ 0, & \text{for } h > q, \end{cases}$$

where $\theta_0 = 1$. The autocorrelation function is then obtained via the formula $\rho_h = \gamma_h / \gamma_0$.

- Therefore, the autocorrelation function of an MA(q) process drops to zero after lag q (assuming that $\theta_q \neq 0$).
- If one observes a similar feature in a sample autocorrelation function, one can consider an MA(q) process as a good candidate to model the time series.

When considering the suitability of an MA(q) process, one can make use of the following result: In the case of an MA(q) process, the estimators r_h , $h > q$, are approximately normally distributed with mean zero and variance $(1 + 2\rho_1^2 + \cdots + 2\rho_q^2) / T$. Therefore, if one wants to test whether an individual autocorrelation coefficient is statistically different from zero at the 5% significance level, the estimates r_h , $h > q$, should be compared with the critical bounds

$$\pm 1.96 \sqrt{\widehat{w}_q / T}, \quad \widehat{w}_q = 1 + 2r_1^2 + \cdots + 2r_q^2.$$

In this case, $P(|r_h| \geq 1.96 \sqrt{\widehat{w}_q / T}) \approx 0.05$. Note also that these bounds are wider than the ones obtained in the case $y_t \sim \text{iid}(\mu, \sigma^2)$, where $\widehat{w}_q$ is replaced by 1.

Invertibility. It can be shown that in a (causal) $AR(p)$ process there exists a one-to-one correspondence between the parameters $\phi = (\phi_1, \dots, \phi_p)'$ and σ^2 , and the autocovariance function. For an $MA(q)$ process, a similar result does not hold.

- To see this, consider as an example the $MA(1)$ process for which it holds that (assume that $\theta_1 \neq 0$)

$$\gamma_0 = \sigma^2 (1 + \theta_1^2) = \sigma^2 \theta_1^2 (1 + 1/\theta_1^2) \text{ and } \gamma_1 = \sigma^2 \theta_1 = \theta_1^2 \sigma^2 (1/\theta_1).$$

Define $\theta_1^* = 1/\theta_1$ and $\sigma_*^2 = \theta_1^2 \sigma^2$, so that $u_t^* = \theta_1 u_t \sim \text{iid}(0, \sigma_*^2)$. Now, it can be seen that the two $MA(1)$ processes $y_t = u_t + \theta_1 u_{t-1}$ and $y_t^* = u_t^* + \theta_1^* u_{t-1}^*$ have exactly the same autocovariance function, so that they cannot be distinguished from each other based on autocovariances.

- If in addition $u_t \sim \text{iid}(0, \sigma^2)$, the entire probability structures of these two processes are indistinguishable.
- A consequence is that the parameters cannot be estimated from an observed time series unless one “tells” the method which of the parameter combinations (θ_1, σ^2) and (θ_1^*, σ_*^2) (that fit the data equally well) should be chosen.

The typical solution to the above problem is to set the condition $|\theta_1| < 1$. When one assumes $|\theta_1| < 1$ in an $MA(1)$ process, one can use the equation $u_t = y_t - \mu - \theta_1 u_{t-1}$ and repetitive substitutions to first obtain $u_t = (y_t - \mu) - \theta_1(y_{t-1} - \mu) + \theta_1^2 u_{t-2}$, and eventually

$$(y_t - \mu) = - \sum_{j=1}^k (-\theta_1)^j (y_{t-j} - \mu) - (-\theta_1)^{k+1} u_{t-k-1} + u_t.$$

Similarly as when considering the stationarity of an $AR(1)$ process, it can be concluded, given the condition $|\theta_1| < 1$,

$$y_t - \mu = - \sum_{j=1}^{\infty} (-\theta_1)^j (y_{t-j} - \mu) + u_t.$$

That is, when $|\theta_1| < 1$ holds, an $MA(1)$ process can be “inverted” to an $AR(\infty)$ process, which explains why in the case $|\theta_1| < 1$ an $MA(1)$ process is called **invertible**.

Invertibility condition of an $MA(q)$ process. An $MA(q)$ process is invertible, if the roots of the polynomial $\theta(z)$ ($z \in \mathbb{C}$) lie outside the unit circle/disk in the complex plane or, equivalently, if

$$\theta(z) \neq 0 \text{ for } |z| \leq 1.$$

Similarly as in the case of the stationarity condition of an AR(p) process, invertibility ensures that there exists an AR(∞) representation also for MA(q) processes. In other words, when the invertibility condition holds, one can formally solve u_t from the equation $(y_t - \mu) = \theta(B) u_t$ by multiplying both sides by $\theta(B)^{-1}$. The solution is

$$\pi(B)(y_t - \mu) = u_t \quad \text{or} \quad \sum_{j=0}^{\infty} \pi_j (y_{t-j} - \mu) = u_t,$$

where $\pi(B) = \sum_{j=0}^{\infty} \pi_j B^j = \theta(B)^{-1}$ and the coefficients π_j can be solved as a function of the parameters $\theta_1, \dots, \theta_q$ from the equation

$$(1 + \theta_1 B + \dots + \theta_q B^q)(\pi_0 + \pi_1 B + \pi_2 B^2 + \dots) = 1$$

by interpreting the right hand side as a power series in B , setting the coefficients of B^j equal on both sides of the equation (and $\pi_0 = 1$).

As in the MA(1) case, invertibility ensures that there exists a one-to-one correspondence between the autocovariance function of an MA(q) process and the parameters $\boldsymbol{\theta} = (\theta_1, \dots, \theta_q)'$ and σ^2 .

- Except for the first-order case, this correspondence is rather complicated, and we omit the details.
- In what follows (unless otherwise mentioned), MA processes are always assumed to be invertible.

Partial autocorrelation function of the MA(q) process. The general definition of the partial autocorrelation function presented above in connection with the AR(p) process can also be applied in the case of an MA(q) process, but the calculations involved would become rather complicated. However, because an MA(q) process has (due to invertibility and assuming here $\theta_q \neq 0$) an AR(∞) representation, and based on what was said above about the partial autocorrelation function of an AR(p) process, it is intuitively clear that the partial autocorrelation function of an MA(q) process does not drop to zero at any point, but rather smoothly decays towards zero.

- It can be shown that the partial autocorrelation function of an MA(1) process is

$$\alpha_h = -(-\theta_1)^h / (1 + \theta_1^2 + \dots + \theta_1^{2h}).$$

Because $|\theta_1| < 1$, the partial autocorrelation function of an MA(1) process decays to zero at an exponential rate as the lag length h increases.

- For a general MA(q) process, it can be shown that the partial autocorrelation function decays to zero at an exponential rate, potentially following the shape of a dampening sine wave.

5.3 ARMA(p, q) process

An ARMA(p, q) process can be characterized as a combination of AR(p) and MA(q) processes. It is defined by the equation

$$y_t = \nu + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q},$$

where $u_t \sim \text{iid}(0, \sigma^2)$. Similarly as in the case of an AR(p) process, the current value y_t is assumed to depend linearly on p past values of the process. Unlike the AR(p) process, the error term is now (generally) not an independent random shock, but instead an autocorrelated MA(q) process.

Using the lag operator and the lag-polynomials $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$ and $\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$, the ARMA(p, q) process can be expressed as

$$\phi(B)(y_t - \mu) = \theta(B)u_t.$$

- If $\theta(B) = 1$ (that is, $q = 0$), one obtains the AR(p) process as a special case.
- If $\phi(B) = 1$ (that is, $p = 0$), one obtains the MA(q) process.

Stationarity and invertibility. Based on what was said for linear processes, it is clear that an ARMA(p, q) process has a well-defined (causal) MA(∞) representation if the polynomial $\phi(z)$ satisfies the sufficient stationarity condition of an AR(p) process (what was said in the case of an AR(p) process holds, but now with an MA(q) error term).

- A sufficient condition for the stationarity of the ARMA(p, q) process is that the roots of the polynomial $\phi(z)$ ($z \in \mathbb{C}$) lie outside the unit circle/disk in the complex plane, or equivalently that

$$\phi(z) \neq 0 \text{ for } |z| \leq 1.$$

- An ARMA(p, q) process is invertible, if the roots of the polynomial $\theta(z)$ ($z \in \mathbb{C}$) lie outside the unit circle/disk in the complex plane, or equivalently if

$$\theta(z) \neq 0 \text{ for } |z| \leq 1.$$

The general discussion concerning invertibility in the previous subsection in connection with the MA(1) process also holds here and generalizes to the case of an ARMA(p, q) process.

As can be deduced from what was discussed above, when the stationarity condition holds, an ARMA(p, q) process has an MA(∞) representation (cf. the AR(p) case)

$$(y_t - \mu) = \frac{\theta(B)}{\phi(B)} u_t = \psi(B) u_t,$$

where $\psi(B) = \sum_{j=0}^{\infty} \psi_j B^j = \theta(B) \phi(B)^{-1}$ and the coefficients ψ_j can be solved as a function of the parameters $\phi_1, \dots, \phi_p$ and $\theta_1, \dots, \theta_q$ by equating the coefficients of B^j on both sides of the equation

$$(1 - \phi_1 B - \dots - \phi_p B^p)(\psi_0 + \psi_1 B + \psi_2 B^2 + \dots) = 1 + \theta_1 B + \dots + \theta_q B^q.$$

- From this equation, it can be solved that $\psi_0 = 1$, $\psi_1 = \phi_1 \psi_0 + \theta_1$, and in general,

$$\psi_j = \sum_{i=1}^p \phi_i \psi_{j-i} + \theta_j, \quad j = 0, 1, 2, \dots,$$

where $\theta_0 = 1$, $\theta_j = 0$, $j > q$, and $\psi_j = 0$, $j < 0$.

On the other hand, when the invertibility condition holds, then an ARMA(p, q) process has an AR(∞) representation

$$\frac{\phi(B)}{\theta(B)}(y_t - \mu) = \pi(B)(y_t - \mu) = u_t,$$

where $\pi(B) = \sum_{j=0}^{\infty} \pi_j B^j = \phi(B) \theta(B)^{-1}$. The coefficients π_j can be solved similarly as the ψ_j 's above (just exchange the roles of the polynomials $\phi(B)$ and $\theta(B)$). The end result can be expressed as

$$\pi_j = -\sum_{i=1}^q \theta_i \pi_{j-i} - \phi_j, \quad j = 0, 1, 2, \dots,$$

where $\phi_0 = -1$, $\phi_j = 0$, $j > p$, and $\pi_j = 0$, $j < 0$. In particular, it holds that $\pi_0 = 1$ and, as in the case of the coefficients ψ_j , $\pi_j \rightarrow 0$ at an exponential rate as $j \rightarrow \infty$.

Identification condition. Earlier in connection to MA processes, the invertibility condition was motivated by the fact that it guarantees the existence of a one-to-one correspondence between the autocovariance function of a process and its model parameters. For general ARMA(p, q) processes, invertibility alone is not sufficient to guarantee the existence of such a correspondence, which is also required for maximum likelihood estimation of the parameters.

- Consider as an example the simple case of an ARMA(1,1) process with the linear representation

$$(y_t - \mu) = \frac{1 + \theta_1 B}{1 - \phi_1 B} u_t.$$

In the special case $\phi_1 = -\theta_1$, it is clear that the polynomials on the right hand side can be cancelled out, resulting in the equation $y_t - \mu = u_t$ so that the process is not a “real” ARMA(1,1) process, but instead just white noise. This implies that the parameters ϕ_1 and θ_1 are not identified in

the sense that maximum likelihood estimation will not work because this method cannot distinguish different pairs of parameter values that satisfy the restriction $\phi_1 = -\theta_1$ (and $|\phi_1| < 1$, $|\theta_1| < 1$).

Identification (or uniqueness) condition of an ARMA(p, q) process. The polynomials $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \dots + \theta_q z^q$ of a stationary and invertible ARMA(p, q) process are assumed to have no common roots and $\phi_p \neq 0$ or $\theta_q \neq 0$.

- Quite often, the requirement that $\phi_p \neq 0$ or $\theta_q \neq 0$ is not explicitly mentioned but is understood to be contained in the condition ruling out common roots.
- Unless otherwise mentioned, in what follows, we assume that this identification condition holds.

Autocorrelation function of an ARMA(p, q) process can be derived in a manner similar to the AR(p) case.

- In what follows, we only present the general principle of the solution.

Multiplying both sides of the demeaned ARMA(p, q) model equation by $y_{t-h} - \mu$ ($h \geq 0$) and taking expectations yields

$$\begin{aligned} \mathbb{E}\left((y_t - \mu)(y_{t-h} - \mu)\right) &= \phi_1 \mathbb{E}\left((y_{t-1} - \mu)(y_{t-h} - \mu)\right) + \dots \\ &+ \phi_p \mathbb{E}\left((y_{t-p} - \mu)(y_{t-h} - \mu)\right) + \mathbb{E}\left(u_t(y_{t-h} - \mu)\right) \\ &+ \theta_1 \mathbb{E}\left(u_{t-1}(y_{t-h} - \mu)\right) + \dots + \theta_q \mathbb{E}\left(u_{t-q}(y_{t-h} - \mu)\right). \end{aligned}$$

Making use of the MA(∞) representation and the definition of autocovariance this further leads to

$$\gamma_h = \begin{cases} \phi_1 \gamma_{h-1} + \dots + \phi_p \gamma_{h-p} + \sigma^2 \sum_{j=0}^{\infty} \theta_{h+j} \psi_j, & 0 \leq h < \max\{p, q+1\}, \\ \phi_1 \gamma_{h-1} + \dots + \phi_p \gamma_{h-p}, & h \geq \max\{p, q+1\}, \end{cases}$$

where $\theta_0 = 1$ and $\theta_j = 0$ for $j \notin \{0, \dots, q\}$. The autocorrelation function is then obtained using the formula $\rho_h = \gamma_h / \gamma_0$. Moreover, the coefficients ψ_j can be expressed as a function of the parameters $\phi_1, \dots, \phi_p$ and $\theta_1, \dots, \theta_q$.

- Because this recursive solution is obtained from a difference equation similar to the one the autocovariances of an AR(p) process satisfy, it is rather clear that as the lag length h increases, the autocorrelation function of an ARMA(p, q) process decays exponentially to zero, potentially following the shape of a dampening sine wave (in the case $p = 1$ this is rather easy to verify).
- Because an ARMA(p, q) process has an AR(∞) representation when the invertibility condition holds, one can use the general definition of the

partial autocorrelation function to conclude that the general shape of the partial autocorrelation function of an ARMA(p, q) process is comparable to that of the autocorrelation function.

To conclude and importantly, for a genuine ARMA(p, q) process (that is, with $p \geq 1$ and $q \geq 1$), neither the autocorrelation nor the partial autocorrelation function ever drops to zero, but instead both steadily decay towards zero.

5.4 Briefly on ARIMA(p, d, q) process

In many applications, time series often exhibit trends, implying that their properties cannot be modeled by stationary processes. (Note that the term “trend” is used quite loosely here.) However, as already discussed in connection to transformations in Section 1, quite often the differences of such time series $y_t - y_{t-1} \equiv \Delta y_t$ (that is, $\Delta = 1 - B$) look stationary and hence could be modeled as stationary processes.

- If the differenced series is trending, then one may consider modeling the change of the difference $\Delta^2 y_t$.
- One might apply the same idea to justify using higher order differences, but this is very rare in practice (even second differences are already rare).

Generally, a process y_t is called an ARIMA(p, d, q) process if it becomes a stationary and invertible ARMA(p, q) process after differencing d times, i.e., $\Delta^d y_t \sim \text{ARMA}(p, q)$.

- In practice, the case $d = 1$ is by far the most common.
- When $d = 0$, the special case is an ARMA(p, q) process, and thus also an AR(p) process when $q = 0$.

We will consider nonstationary processes more detail in Section 12.

Chapter 6

Parameter estimation

In this section, we will consider how to estimate (unknown) values of the parameters of an ARMA(p, q) model (or ARIMA(p, d, q) model). In a nutshell, we observe the estimation to be possible with different methods depending on the model at hand:

- AR(p): estimation can be done by (conditional) ordinary least squares (OLS).
- ARMA(p, q): due to the MA(q) part, estimation is more complicated. However, estimation can be carried out by the method of maximum likelihood, which is naturally also available for the AR(p) model as well.

6.1 Estimation of AR models with OLS

Estimation of the AR(p) model can be carried out with ordinary least squares (OLS), conditioning on the first p observations. The AR(p) model (including constant term) can be rewritten as

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + u_t,$$

where $\mathbf{x}_t = [1 \ y_{t-1} \cdots y_{t-p}]'$ and $\boldsymbol{\beta} = [\nu \ \phi_1 \cdots \phi_p]'$. The OLS estimators of $\boldsymbol{\beta}$ (and σ^2) can be constructed in the usual way familiar from past studies related to linear regression models. That is, the conditional OLS estimator of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}} = \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right)^{-1} \sum_{t=1}^T \mathbf{x}_t y_t.$$

In this case (notation), it is assumed that the required p initial values $y_{-p+1}, \dots, y_0$ are available. Furthermore, the OLS estimator for σ^2 is

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T \left(y_t - \mathbf{x}'_t \hat{\boldsymbol{\beta}} \right)^2.$$

- This is the (conditional) maximum likelihood form of the variance estimator; in practice the degrees-of-freedom corrected version, obtained by dividing by $T - k$ with $k = p + 1$ the number of estimated regression coefficients, is often reported instead. The two are of course asymptotically ($T \rightarrow \infty$) equivalent.
- Notice that the summation in the OLS estimator starts from $t = 1, 2, \dots$ due to these available initial values. Depending on, mainly notational choices, the OLS estimator is alternatively at times written as

$$\hat{\boldsymbol{\beta}} = \left(\sum_{t=p+1}^T \mathbf{x}_t \mathbf{x}'_t \right)^{-1} \sum_{t=p+1}^T \mathbf{x}_t y_t,$$

if treating observations $y_1, \dots, y_p$ as initial values.

The usual (asymptotic) statistical properties of the resulting OLS estimator hold. In particular, it can be shown that:

- As the error term u_t is uncorrelated with the lags of y_t , we have $\mathbf{E}(\mathbf{x}_t u_t) = \mathbf{0}$. That is, $\mathbf{E}(u_t y_{t-j}) = 0$ for $j = 1, 2, \dots, p$, and
- the OLS estimator $\hat{\boldsymbol{\beta}}$ is a consistent estimator of $\boldsymbol{\beta}$.
- Note that consistency relies on the error term being uncorrelated with the regressors. If autocorrelation remains in the residuals, then u_t is correlated with the lags of y_t and the OLS estimator becomes *inconsistent*. Remaining residual autocorrelation is therefore a sign of misspecification that has to be fixed by respecifying the model (typically by increasing the lag length p), and it cannot be cured by adjusting the standard errors.
- Conditional heteroskedasticity, on the other hand, does not threaten consistency but does invalidate the conventional standard errors. Heteroskedasticity consistent (White) or heteroskedasticity-autocorrelation consistent (HAC, e.g. Newey–West) standard errors can then be used to take it into account for detailed statistical inference (if this is the main objective in the empirical analysis). Modelling the conditional variance explicitly (see the AR-GARCH models later on) is the alternative route.

6.2 Maximum likelihood estimation

Estimation of ARMA (and ARIMA) models with a moving average component is somewhat more complicated than the OLS based estimation of autoregressive models. The essential problem in estimating these models stems from the fact that u_t is not observable.

The estimation of an ARMA(p, q) model, including also the AR(p), can be based on (conditional) method of maximum likelihood by making an assumption of the distribution of the error u_t . A typical case would be to assume the error term is Gaussian $u_t \sim \text{nid}(0, \sigma^2)$.

- It is worth noting that regardless of the errors being exactly Gaussian or not, the maximum likelihood estimator (MLE) can be interpreted as a **quasi-MLE (QMLE)** where the possible error in the model specification can be taken into consideration by using a robust parameter covariance matrix in formulation of the standard errors.
- We will come back to this QMLE interpretation within the context of AR-GARCH model. Similar argumentation as there can be used also in this context (without the GARCH errors).

We will next consider details on so called conditional maximum likelihood technique. The exact method (exact method of maximum likelihood) will be introduced at the end of next section in Extra material.

Conditional maximum likelihood method. The (exact) ML estimation of ARMA models requires the necessary selections for the initial values of y_t and u_t . By conditioning to the initial values, we get the conditional maximum likelihood technique. If the sample size (length of the time series) is large, the conditional and exact MLEs are close to each other (both lead to the same asymptotic distributions). Moreover, in practice, numerical methods can be used to maximize the log-likelihood function in both cases.

Let us consider a concrete example case of an ARMA(1,1) model

$$y_t = \nu + \phi_1 y_{t-1} + u_t + \theta_1 u_{t-1},$$

where $u_t \sim \text{nid}(0, \sigma^2)$. That is, the normality assumption of the error term is assumed. The goal is to estimate the vector of parameters $\boldsymbol{\vartheta} = (\nu, \phi_1, \theta_1, \sigma^2)'$.

- We use the notation $\boldsymbol{\vartheta}$ (“vartheta”) to denote a general parameter vector containing all the parameters of the model.
- Even though we concentrate on the ARMA(1,1), the following argumentation generalizes readily to more general ARMA processes, which are of

the form $y_t = \mathbf{E}_{t-1}(y_t) + u_t$ where $\mathbf{E}_{t-1}(y_t)$ corresponds to the employed model structure (systematic part of the model). Here $\mathbf{E}_{t-1}(\cdot)$ denotes the conditional expectation and it will be introduced more detail in Forecasting section. In this ARMA(1,1) case, we get $\mathbf{E}_{t-1}(y_t) = \nu + \phi_1 y_{t-1} + \theta_1 u_{t-1}$, whereas, e.g., in the AR(p) case we get $\mathbf{E}_{t-1}(y_t) = \nu + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p}$.

In the ARMA(1,1) model, taking the initial values for y_0 and u_0 as given, the sequence of $u_1, \dots, u_T$ can be constructed from $y_1, \dots, y_T$ by iterating

$$u_t = y_t - \nu - \phi_1 y_{t-1} - \theta_1 u_{t-1}, \quad t = 1, \dots, T.$$

When conditioning on y_{t-1} and u_{t-1} , which are included in the information available at time $t-1$, the normality assumption of u_t implies that the conditional distribution of y_t is also normal:

$$y_t | (y_{t-1}, u_{t-1}) \sim \mathbf{N}(\nu + \phi_1 y_{t-1} + \theta_1 u_{t-1}, \sigma^2).$$

The specific form of the conditional density function of y_t , conditional on y_{t-1} and u_{t-1} , is hence

$$f_{y_t | y_{t-1}, u_{t-1}}(y_t | y_{t-1}, u_{t-1}; \boldsymbol{\vartheta}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_t - \nu - \phi_1 y_{t-1} - \theta_1 u_{t-1})^2}{2\sigma^2}\right).$$

Construction of the log-likelihood function. The (conditional) log-likelihood function can be based on the conditional density functions of single observations. Here we are not restricting ourselves to the ARMA(1,1) and/or Gaussian case: the following steps generalize also to other modelling choices, such as other error distributions or models with a time-varying conditional variance (GARCH).

Denote $\mathbf{Y}_t = (y_1 \dots y_t)$ and, therefore, the sample of observations is denoted by $\mathbf{Y}_T = (y_1, \dots, y_T)$. The joint conditional density function (without arguments) $f_{\mathbf{Y}_T}(\mathbf{Y}_T)$ is obtained as the product of conditional density functions

$$f_{\mathbf{Y}_T} = f_{y_T | \mathbf{Y}_{T-1}} \cdot f_{\mathbf{Y}_{T-1}} = f_{y_T | \mathbf{Y}_{T-1}} f_{y_{T-1} | \mathbf{Y}_{T-2}} \cdot f_{\mathbf{Y}_{T-2}} = \dots = \prod_{t=1}^T f_{y_t | \mathbf{Y}_{t-1}} \cdot f_{\mathbf{Y}_0}.$$

By conditioning on $\mathbf{Y}_0$, which contains the necessary initial values (depending on, e.g., the lag lengths p and q of the ARMA(p, q) model), the conditional density function gets the form

$$\prod_{t=1}^T f_{y_t | \mathbf{Y}_{t-1}},$$

which then leads to the (conditional) log-likelihood function

$$l(\boldsymbol{\vartheta}) = \sum_{t=1}^T \log f_{y_t | \mathbf{Y}_{t-1}}(y_t | \mathbf{Y}_{t-1}; \boldsymbol{\vartheta}).$$

- Going back to the specific example of Gaussian ARMA(1,1) model, the (conditional) log-likelihood function is hence

$$l(\boldsymbol{\vartheta}) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T u_t^2,$$

where $u_t = y_t - \nu - \phi_1 y_{t-1} - \theta_1 u_{t-1}$ in this ARMA(1,1) case.

The maximum likelihood estimate (MLE) of $\boldsymbol{\vartheta}$, that is $\hat{\boldsymbol{\vartheta}}$, is obtained by maximizing the (conditional) log-likelihood function.

- This requires numerical optimization methods. These numerical (iterative) algorithms require initial values and/or possibly also preliminary estimation of the parameters, but econometric and statistical program packages are carrying out these steps straightforwardly.

AR(p) case. We will next show that, in the case of the AR(p) model, the conditional maximum likelihood estimation actually leads to the (conditional) least squares estimates when the error term is Gaussian ($u_t \sim \text{nid}(0, \sigma^2)$).

Since y_t is a linear combination of $u_t, u_{t-1}, \dots$ (given stationarity and corresponding MA(∞) representation), it follows that u_s and $\mathbf{x}_t$ are independent $\forall s \geq t$, where, as above in the AR(p) case, $\mathbf{x}_t = [1 \ y_{t-1} \ \dots \ y_{t-p}]'$. By assuming the normality of the error term, we get

$$y_t | \mathbf{Y}_{t-1} \sim \text{N}(\mathbf{x}_t' \boldsymbol{\beta}, \sigma^2),$$

and the conditional density function of y_t is

$$f_{y_t | \mathbf{Y}_{t-1}}(y_t | \mathbf{x}_t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_t - \mathbf{x}_t' \boldsymbol{\beta})^2\right).$$

From this we get the (sample) log-likelihood function

$$l(\boldsymbol{\vartheta}) \equiv l(\boldsymbol{\beta}, \sigma^2) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T (y_t - \mathbf{x}_t' \boldsymbol{\beta})^2.$$

Here the first term can also be removed because it does not depend on the parameters in $\boldsymbol{\vartheta}$. This corresponds to the (conditional) log-likelihood function of a Gaussian linear regression model, but now for the time series observations $(y_1, \dots, y_T)$. Maximizing this gives us the same conditional least squares estimator $\hat{\boldsymbol{\beta}}$ (and $\hat{\sigma}^2$) as obtained above.

Exact method of maximum likelihood. The exact log-likelihood function contains also the impact of initial values when constructing the log-likelihood function. Some details on this technique are provided below.

i Extra: Exact maximum likelihood estimation of ARMA models

Let us take a closer look at exact maximum likelihood (ML) based estimation.

The exact likelihood function. For the purpose of estimation based on the exact method of maximum likelihood, consider the *Gaussian* zero-mean ARMA(p, q) process

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}, \quad u_t \sim \text{nid}(0, \sigma^2),$$

and assume the stationarity, invertibility and the identification condition of the ARMA process hold. Suppose $\mathbf{y} = (y_1, \dots, y_T)'$ is a vector of observations generated by this process. As before, denote $\boldsymbol{\phi} = (\phi_1, \dots, \phi_p)'$, $\boldsymbol{\theta} = (\theta_1, \dots, \theta_q)'$ and, for notational simplicity in this zero-mean case, $\boldsymbol{\beta} = (\boldsymbol{\phi}', \boldsymbol{\theta}')$ (the constant term is left out here and can be added as discussed below). Making use of the general formulas for the mean and autocovariance function of a linear process, it can be seen that $\mathbf{E}(\mathbf{y}) = \mathbf{0}$ and $\text{Cov}(\mathbf{y}) = \sigma^2 \boldsymbol{\Sigma}$, where the (positive definite) matrix $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}(\boldsymbol{\beta})$ ($T \times T$) is a function of the parameter $\boldsymbol{\beta}$ (but not of the parameter σ^2).

Since linear transformations of Gaussian distributions are also Gaussian (this holds also for infinite dimensional transformations), $\mathbf{y} \sim \mathbf{N}(\mathbf{0}, \sigma^2 \boldsymbol{\Sigma})$.

- The zero-mean assumption is only a notational convenience: a constant term, and even a trend component, can be included without affecting the asymptotic properties of the ML estimators of $\boldsymbol{\beta}$ and σ^2 . In the stationary case, working with y_t demeaned by the (estimated) mean $\mu = \nu/(1 - \phi_1 - \cdots - \phi_p)$ leads to the same asymptotic results.

The density function of the random vector $\mathbf{y}$ is

$$f_{\mathbf{y}}(\mathbf{y}) = (2\pi)^{-T/2} \det(\sigma^2 \boldsymbol{\Sigma})^{-1/2} \exp \left\{ -\frac{1}{2\sigma^2} \mathbf{y}' \boldsymbol{\Sigma}^{-1} \mathbf{y} \right\}.$$

Making use of properties of the determinant, $\det(\sigma^2 \boldsymbol{\Sigma}) = \sigma^{2T} \det(\boldsymbol{\Sigma})$, so that the log likelihood function becomes

$$l(\boldsymbol{\beta}, \sigma^2) = -\frac{T}{2} \log \sigma^2 - \frac{1}{2} \log \det(\boldsymbol{\Sigma}(\boldsymbol{\beta})) - \frac{1}{2\sigma^2} \mathbf{y}' \boldsymbol{\Sigma}(\boldsymbol{\beta})^{-1} \mathbf{y}.$$

Because the matrix $\boldsymbol{\Sigma}(\boldsymbol{\beta})$ is a rather complicated function of the parameter $\boldsymbol{\beta}$, the log-likelihood function cannot be expressed in a simple form and, consequently, the above expression is difficult to use directly for maximizing the likelihood function. In practice, one has to resort to numerical methods to maximize the likelihood function. This involves computing the inverse matrix $\boldsymbol{\Sigma}(\boldsymbol{\beta})^{-1}$ and the determinant $\det(\boldsymbol{\Sigma}(\boldsymbol{\beta}))$ for a range of given values of $\boldsymbol{\beta}$. For a given value of $\boldsymbol{\beta}$, the matrix $\boldsymbol{\Sigma}(\boldsymbol{\beta})$ can be computed for ARMA processes.

Many different algorithms have been proposed to solve the maximization problem described above.

Statistical inference. Collect the parameters of the conditional mean into the $p + q + 1$ -dimensional vector

$$\boldsymbol{\beta} = (\nu, \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q)',$$

so that the full parameter vector of the model is $\boldsymbol{\vartheta} = (\boldsymbol{\beta}', \sigma^2)'$.

- The AR(p) model is obtained as the special case $q = 0$, in which case $\boldsymbol{\beta} = (\nu, \phi_1, \dots, \phi_p)'$ coincides with the regression parameter vector used above in the OLS discussion. Likewise, the pure MA(q) model is obtained with $p = 0$.
- If the model is parameterized through the mean μ instead of the constant term ν (recall $\nu = \mu(1 - \phi_1 - \dots - \phi_p)$ in the stationary case), the results below apply as such with ν replaced by μ in $\boldsymbol{\beta}$.

It can be shown that when the stationarity, invertibility, and identification conditions hold (and $\sigma^2 > 0$), the “usual” asymptotic properties of a maximum likelihood estimator (exact or conditional) hold. In particular, the estimators $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ are consistent, asymptotically normally distributed, and asymptotically independent of each other. Formally,

$$\sqrt{T} \left(\hat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta} \right) \xrightarrow{d} \mathbf{N} \left(\mathbf{0}, \mathcal{I}(\boldsymbol{\vartheta})^{-1} \right), \quad \text{where} \quad \mathcal{I}(\boldsymbol{\vartheta}) = \lim_{T \rightarrow \infty} \frac{1}{T} \mathbf{E} \left[-\frac{\partial^2 l(\boldsymbol{\vartheta})}{\partial \boldsymbol{\vartheta} \partial \boldsymbol{\vartheta}'} \right]$$

is the (average) Fisher information matrix. In practice it is convenient to state the same result in the “unscaled” form

$$\hat{\boldsymbol{\beta}}_{as} \sim \mathbf{N} \left(\boldsymbol{\beta}, \mathbf{V}(\boldsymbol{\vartheta})^{-1} \right), \quad \mathbf{V}(\boldsymbol{\vartheta}) = \mathbf{E} \left[-\frac{\partial^2 l(\boldsymbol{\beta}, \sigma^2)}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \right],$$

where $\mathbf{V}(\boldsymbol{\vartheta})$ is the (sample) Fisher information matrix of the parameter $\boldsymbol{\beta}$

i Extra: Some additional details on asymptotic distribution

First, note that $\mathbf{V}(\boldsymbol{\vartheta}) \approx T \mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}}(\boldsymbol{\vartheta})$, so that its inverse is of order $1/T$ and shrinks towards zero as the sample size grows, as it should.

Second, because the information matrix is block diagonal with respect to $\boldsymbol{\beta}$ and σ^2 , the $\boldsymbol{\beta}$ -block of $\mathcal{I}(\boldsymbol{\vartheta})^{-1}$ equals the inverse of the $\boldsymbol{\beta}$ -block of $\mathcal{I}(\boldsymbol{\vartheta})$. This is why inference on the conditional mean parameters $\boldsymbol{\beta}$ can be based on $\mathbf{V}(\boldsymbol{\vartheta})^{-1}$ alone, and the estimation of σ^2 can be ignored (asymptotically).

Furthermore, even though $\mathbf{V}$ concerns $\boldsymbol{\beta}$ only, it does depend on σ^2 as well. Indeed, differentiating the (conditional) log-likelihood function derived

above gives

$$\mathbf{V}(\boldsymbol{\vartheta}) = \frac{1}{\sigma^2} \sum_{t=1}^T \mathbf{E} \left[\frac{\partial u_t}{\partial \boldsymbol{\beta}} \frac{\partial u_t}{\partial \boldsymbol{\beta}'} \right],$$

which shows concretely how the AR and MA parts enter the result. In the AR(p) case $\partial u_t / \partial \boldsymbol{\beta} = -\mathbf{x}_t$, and hence $\mathbf{V}(\boldsymbol{\vartheta})^{-1} = \sigma^2 (\sum_{t=1}^T \mathbf{E}(\mathbf{x}_t \mathbf{x}_t'))^{-1}$, whose empirical counterpart $\hat{\sigma}^2 (\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t')^{-1}$ is exactly the familiar OLS covariance matrix estimator. In the ARMA case the derivatives $\partial u_t / \partial \boldsymbol{\beta}$ are obtained recursively (they themselves depend on their own lags through the MA part), which is one reason why no closed-form expression is available.

The asymptotic distribution result holds also without the normality assumption. That is, for the QMLE, provided the errors are iid with $\mathbf{E}(u_t) = 0$ and $\mathbf{E}(u_t^2) = \sigma^2 < \infty$ (and $\mathbf{E}(u_t^4) < \infty$ for inference on σ^2). In that case the estimators are no longer (asymptotically) efficient, but the standard errors of $\hat{\boldsymbol{\beta}}$ obtained from $\mathbf{V}(\boldsymbol{\vartheta})^{-1}$ remain asymptotically valid.

- If, however, the errors are conditionally heteroskedastic (as is typical with financial time series), the information matrix equality fails and the above covariance matrix is no longer valid. A robust (“sandwich”) covariance matrix estimator should then be used. This is exactly the situation we will return to in the context of AR-GARCH models.
- The asymptotic independence of $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ is automatic under normality. For the QMLE it additionally requires the error distribution to be symmetric in the sense $\mathbf{E}(u_t^3) = 0$; with skewed errors the two estimators are asymptotically correlated (which, however, does not affect inference on $\boldsymbol{\beta}$ alone).
- It should be noted that the stationarity, invertibility, and identification conditions are required for the above asymptotic distribution result, which can give a poor approximation also when these conditions are “nearly violated”. Three cases are worth distinguishing:
 - *Near-nonstationarity*: the autoregressive polynomial has a root close to the unit circle (e.g. $\phi_1 \approx 1$ in the AR(1) case). The normal approximation is then poor and, in the limiting unit root case, the asymptotic distribution is nonstandard altogether (unit root tests, considered later, are needed).
 - *Near-noninvertibility*: the moving average polynomial has a root close to the unit circle (e.g. $\theta_1 \approx \pm 1$ in the MA(1) case). Since the parameter is then close to the boundary of the admissible parameter space, the estimator has a nonstandard finite-sample distribution and the estimate tends to “pile up” at the boundary.

- *Near-violation of the identification condition:* the AR and MA polynomials have (approximately) a common factor, e.g. $\phi_1 \approx -\theta_1$ in the ARMA(1,1) case. The information matrix is then close to singular, so the standard errors become very large and the estimates numerically unstable, a symptom of an overparameterized model rather than of genuinely uncertain parameters.
- The exact and the conditional ML estimators have the same asymptotic distribution, but they can differ noticeably in small samples, and the more so the closer the roots are to the unit circle. In such cases the exact method is preferable.

Hypothesis testing. Using the empirical counterpart of $\mathbf{V}(\boldsymbol{\vartheta})$, namely

$$\hat{\mathbf{V}} \equiv \mathbf{V}(\hat{\boldsymbol{\vartheta}}) = -\frac{\partial^2 l(\hat{\boldsymbol{\beta}}, \hat{\sigma}^2)}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'}$$

(the so-called empirical or observed information matrix of the parameter $\boldsymbol{\beta}$, obtained by evaluating the Hessian of the log-likelihood function at the ML estimates and by dropping the expectation), the asymptotic distribution result above can be used to construct statistical Wald tests and confidence intervals concerning the parameter $\boldsymbol{\beta}$.

- The partial derivatives appearing in this expression can in practice be approximated with their numerical counterparts.

In particular, the square roots of the diagonal elements of the matrix $\hat{\mathbf{V}}^{-1}$ can be used as approximate standard errors of the components of the estimator $\hat{\boldsymbol{\beta}}$.

- For instance, an individual (i th) estimated component of $\hat{\boldsymbol{\beta}}$, that is $\hat{\beta}_i$, would be interpreted as “significantly” different from zero if its absolute value is at least 2 (specifically 1.96) times the size of its approximate standard error.

Formally, let us denote $\mathbf{V}(\boldsymbol{\vartheta})^{-1} = [v^{ij}]$ and its empirical counterpart $\hat{\mathbf{V}}^{-1} = [\hat{v}^{ij}]$. Then, componentwise,

$$\hat{\beta}_i \underset{as.}{\sim} \mathbf{N}(\beta_i, v^{ii}),$$

and $\text{s.e.}(\hat{\beta}_i) = \sqrt{\hat{v}^{ii}}$. Thus the null hypothesis $\beta_i = 0$ gets rejected at the 5% significance level if the t -ratio (cf. t -test)

$$\left| \frac{\hat{\beta}_i}{\sqrt{\hat{v}^{ii}}} \right| \geq 1.96 \iff |\hat{\beta}_i| \geq 1.96 \sqrt{\hat{v}^{ii}},$$

and the 95% confidence interval for β_i is obtained as

$$\hat{\beta}_i \pm 1.96 \cdot \text{s.e.}(\hat{\beta}_i) = \hat{\beta}_i \pm 1.96 \sqrt{\hat{v}^{ii}}.$$

- The critical value 1.96 is the 97.5th percentile of the standard normal distribution, as the result is asymptotic. The name “ t -ratio” is used by analogy with the linear regression model; many software packages report p -values based on the $t(T - k)$ distribution instead, which makes little difference unless T is small.

Wald test statistic. More generally than above, linear restrictions of the form $H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, where the $N \times (p + q + 1)$ matrix $\mathbf{R}$ has full row rank N , can be tested with the Wald test statistic

$$W = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})' (\mathbf{R}\hat{\mathbf{V}}^{-1}\mathbf{R}')^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \stackrel{H_0}{\underset{as}{\sim}} \chi_N^2$$

under the null hypothesis, so that H_0 is rejected for large values of W . The squared t -ratio above is the special case $N = 1$ with $\mathbf{R} = (0, \dots, 1, \dots, 0)$ and $\mathbf{r} = 0$ (and, accordingly, $1.96^2 \approx 3.84$ is the 5% critical value of the $\chi^2(1)$ distribution).

LR test statistic. Tests constructed using likelihood ratio principles can also be used in a conventional manner. Denoting by $\hat{\boldsymbol{\theta}}$ the ML estimator obtained under the N restrictions of H_0 and by $\tilde{\boldsymbol{\theta}}$ the unrestricted one, the likelihood ratio test statistic (under the null hypothesis) is

$$LR = 2 \left(l(\hat{\boldsymbol{\theta}}) - l(\tilde{\boldsymbol{\theta}}) \right) \stackrel{H_0}{\underset{as}{\sim}} \chi_N^2.$$

The Wald, likelihood ratio, and Lagrange multiplier (score) tests are asymptotically equivalent, but their finite-sample properties differ.

- When applying the LR test in practice, the restricted and unrestricted log-likelihood functions must be computed from *exactly the same* observations (and with the same treatment of the initial values).
- With conditional ML this is easy to violate: an ARMA(3,0) and an ARMA(1,0) model estimated by conditional ML “as such” use effective samples of different length, in which case the log-likelihood values are not comparable.

Empirical example. In the previous section, we concluded an AR(2) model to be one possible candidate for the quarterly U.S real GDP growth (1985:Q1–2007:Q2). The conditional maximum likelihood estimation of this model gives us

$$y_t = \underset{(0.197)}{1.693} + \underset{(0.101)}{0.160}y_{t-1} + \underset{(0.101)}{0.287}y_{t-2} + \hat{u}_t, \quad \hat{\sigma}^2 = 3.536,$$

where in parentheses under the estimates are the approximate standard errors. The corresponding t -ratios are $1.693/0.197 \approx 8.6$, $0.160/0.101 \approx 1.6$, and $0.287/0.101 \approx 2.8$.

- The coefficient of the second lag is statistically significant at the 5% significance level ($2.8 > 1.96$), and hence the model cannot be reduced to an AR(1) model. Its 95% confidence interval is $0.287 \pm 1.96 \cdot 0.101$, that is, roughly $[0.089, 0.485]$.
- The coefficient of the first lag is, in turn, *not* significant at the 5% level ($1.6 < 1.96$). Note, however, that dropping the first lag while keeping the second one would be unconventional: the model would no longer be a “complete” AR(2) specification, and such lag restrictions should be tested jointly (and supported by the residual diagnostics) rather than by looking at a single t -ratio.

Alternatively, if reporting the estimation result for the demeaned time series, we then obtain

$$\bar{y}_t = \underset{(0.101)}{0.160} \bar{y}_{t-1} + \underset{(0.102)}{0.287} \bar{y}_{t-2} + \hat{u}_t,$$

where $\bar{y}_t = y_t - 3.061$ is the demeaned GDP growth series. The value 3.061 is the estimated mean of the process, obtained from the estimates as follows

$$\hat{\mu} = \hat{\nu} / (1 - \hat{\phi}_1 - \hat{\phi}_2) = 1.693 / (1 - 0.160 - 0.287) \approx 3.061.$$

Chapter 7

ARMA model selection

7.1 Starting point

Consider an observed time series $y_1, \dots, y_T$, which is assumed, based on its graph or other diagnostics, to be generated by a stationary process, making an ARMA(p, q) specification reasonable.

- As mentioned earlier, achieving stationarity may require preliminary transformations: differencing (possibly combined with a logarithmic transformation), which amounts to selecting the differencing order d of an ARIMA(p, d, q) model and then modelling the differenced data as an ARMA process, or removing deterministic trends by an appropriate detrending method. Nonstationary processes with unit roots are considered later.
- For example, if the process follows a deterministic linear trend, the model can be written as

$$y_t = \mu_0 + \mu_1 t + z_t, \quad t = 1, \dots, T,$$

where z_t is an ARMA(p, q) process (cf. the decomposition $y_t = \mu_t + z_t$ in Section 2). This is an example of a **trend-stationary process**. The usual approach is to estimate μ_0 and μ_1 by least squares and then analyse the residual series, that is, to estimate the deterministic part and the ARMA(p, q) parameters of z_t separately. An obvious alternative is to estimate all the parameters jointly.

Finding a suitable ARMA(p, q) process, or, using terminology commonly employed in statistical modelling, an **ARMA(p, q) model**, traditionally involves the following interrelated stages:

- 1) Specifying (selecting) the model orders p and q
- 2) Estimating the model parameters (which may include preliminary estimation steps)
- 3) Evaluating the adequacy of the estimated model through diagnostic checks and, in some cases, assessing forecasting performance

If the model is found to be inadequate in Step 3, one must return to Step 1 to select new orders p and q , re-estimate the parameters, and re-evaluate the model. As noted, Steps 1–3 are interdependent and are not always performed in a strictly sequential manner.

In practice, one can never know with certainty whether the true (correct) orders of the ARMA(p, q) model have been identified.

- If the orders are chosen too small, part of the autocorrelation structure of the time series remains unmodelled. This is clearly suboptimal from a modelling standpoint, and potentially harmful for forecasting as well. One might then be tempted, but erroneously so, to select overly large orders to avoid underfitting.
- However, choosing orders that are too large leads to inefficiencies in parameter estimation, which can degrade forecasting accuracy. This is especially problematic when both p and q are chosen too large, as it may result in a model where $\phi_p = 0 = \theta_q$, violating the identification condition of the ARMA process. In such cases, the model parameters are not identifiable, making meaningful estimation impossible.

For these reasons, it is generally advisable to follow the **principle of parsimony**: select a model that is as simple as possible while still being sufficiently rich to capture the essential dynamics of the data.

In what follows in this section, we assume that the error term satisfies $u_t \sim \text{iid}(0, \sigma^2)$.

- In the context of maximum likelihood estimation, we adopt the stronger assumption $u_t \sim \text{nid}(0, \sigma^2)$.
- Unless otherwise stated, we also assume that the stationarity and invertibility conditions hold.

Constant term or demeaning? As previously discussed, the mean $E(y_t) = \mu$ is typically nonzero, which means that we need to include a constant term in the model, or alternatively to work with a demeaned time series. A few clarifying points from an empirical modelling perspective:

- The mean is typically estimated using the sample mean $\bar{y}$, after which one may consider the **centered time series** $y_t - \bar{y}$, $t = 1, \dots, T$, and

proceed as if $\bar{y}$ exactly equals the unknown population mean μ . While this introduces a small approximation error, it can be shown that the error becomes negligible in large samples.

- **If the time series is not centered, you should always include the constant term** in the model equation.

7.2 Sample autocorrelations and partial autocorrelations

A good second step after visualizing the series is to investigate the autocorrelation structure of the time series using the estimated (sample) autocorrelation function (ACF) and partial autocorrelation function (PACF). In many cases, these functions may contain clues regarding the suitable lag lengths p and q concerning an adequate ARMA(p, q) model.

In particular, recall the theoretical properties of AR, MA and ARMA processes: **one should look for potential cutoffs (“sudden drops to zero”) in the sample ACF and sample PACF.**

- A sharp cutoff in the ACF with a gradually decaying PACF suggests an MA(q) process.
- A sharp cutoff in the PACF with a gradually decaying ACF suggests an AR(p) process.
- Gradual decay in both ACF and PACF may indicate an ARMA(p, q) process.

This approach is part of the **Box-Jenkins model selection procedure**. While this method does not always point to a single, definitive model, it is useful for **ruling out implausible alternatives** and narrowing down the set of candidate models. It is often used in conjunction with other model selection criteria, which will be introduced in the following sections.

7.3 Information criteria and sequential tests in model selection

As with the use of the sample ACF and PACF, the goal here is to select the orders p and q of the ARMA(p, q) model after one has first set “sufficiently large” maximum lag lengths, denoted p^* and q^* . This preliminary selection can be guided, for instance, by examining the sample autocorrelation and partial autocorrelation functions.

- Let $\tilde{\sigma}_{p,q}^2$ ($0 \leq p \leq p^*$, $0 \leq q \leq q^*$) be the estimator of the innovation variance σ^2 of an ARMA(p, q) process.
- Suppose also that the quantity $\max(p^*, q^*)$ used in this estimation method is held constant for all attempted values of p and q .

As discussed, it is preferable to avoid choosing p and q too large.

- One possible way to choose p and q would be to minimize $\tilde{\sigma}_{p,q}^2$ over the possible values $0 \leq p \leq p^*$, $0 \leq q \leq q^*$. This does not work, however: by the nature of the least squares and maximum likelihood methods, $\tilde{\sigma}_{p,q}^2$ decreases as the orders increase, so one would always end up choosing orders that are too large.

To fix this obvious problem, one approach to select the orders p and q is to minimize the function

$$C(p, q) = \log \tilde{\sigma}_{p,q}^2 + \frac{(p + q + 1) g(T)}{T}, \quad 0 \leq p \leq p^*, \quad 0 \leq q \leq q^*,$$

where the so-called **penalty function** $g(\cdot)$ is positive valued and satisfies $g(T)/T \rightarrow 0$ as $T \rightarrow \infty$ (“+1” due to the inclusion of the constant term ν). The idea behind the penalty function is to penalize for using an unnecessarily large model. If increasing the order p or q does not make $\tilde{\sigma}_{p,q}^2$ sufficiently much smaller, then one does not choose the larger model.

Typically used penalty functions (which have been derived based on different principles) are:

- AIC: $g(T) = 2$ (Akaike information criterion)
- HQ: $g(T) = 2 \log(\log T)$ (Hannan and Quinn information criterion)
- BIC: $g(T) = \log T$ (Schwarz information criterion/Bayesian information criterion)

The first of these (AIC) penalizes the least (favors larger models) and the last (BIC) the most (favors smaller models).

- Note that for the HQ penalty function there exist also other versions, in which the constant 2 is replaced by some other constant.

Some notes on empirical model selection. In practice, it is often advisable to use the criteria described above only as one tool in model selection, rather than relying solely on the mechanical minimization of the criterion $C(p, q)$. Ideally, the final model selection should be made after successful parameter estimation (i.e. when there is clearly no suspicious behaviour in the estimates) and after conducting diagnostic checks to assess the adequacy of the model (see the next subsection).

However, from a modern and partly **machine learning-oriented perspective**, information criteria and similar tools are often applied in a more automated fashion to select the lag lengths of ARMA models. While not always ideal, this reflects a practical compromise: with large and complex datasets, detailed diagnostic checks at every step of the selection process may not be feasible, or even reasonable.

Neighbouring models. For time series that appear difficult to model, it may be useful to consider as alternatives also models in which the lag lengths p and q are one larger than in the model minimizing the criterion function.

- However, it is important to keep in mind that selecting both orders too large can lead to identification problems (cf. the identification condition of the ARMA processes).

Another practical approach is to select a small set of models corresponding to the lowest values of the information criterion function and subject them to more detailed investigation. It is also common to use multiple information criteria simultaneously and examine whether they point to the same model.

Sequential tests. One way to choose the $AR(p)$ or $ARMA(p, q)$ model orders is based on sequential tests. In what follows, we consider an $AR(p)$ model for simplicity, but the procedure generalizes to ARMA models as well.

- Start by choosing a relatively large model order p^* .
- Estimate an $AR(p^*)$ model

$$y_t = \nu + \phi_1 y_{t-1} + \cdots + \phi_{p^*-1} y_{t-p^*+1} + \phi_{p^*} y_{t-p^*} + u_t.$$

- Test the restriction $H_0 : \phi_{p^*} = 0$.
- If the null hypothesis cannot be rejected, estimate an $AR(p^* - 1)$ model and test $H_0 : \phi_{p^*-1} = 0$.
- Continue until the null hypothesis is rejected for the first time.

The procedure above can be applied mechanically using t -tests or Wald test statistics, or using information criteria. Note the obvious multiple testing problem and its effect on the p -values, although this is often ignored in this context.

7.4 Evaluating the adequacy of the estimated model

In the previous sections, it is assumed that the lag lengths (orders) p and q of the $ARMA(p, q)$ model have been chosen, in one way or another, so that the model

is adequate. This means that the error term u_t satisfies at least the assumption $u_t \sim \text{iid}(0, \sigma^2)$, while p and q are not, at least simultaneously, unnecessarily large.

- If the normality of the error term is also assumed, the error term should preferably satisfy $u_t \sim \text{nid}(0, \sigma^2)$.

As discussed above, one often follows **the principle of parsimony** when choosing the model, which can sometimes lead to a model in which at least one of the orders is chosen too small. In such a case, the error term of the selected model remains autocorrelated, and moving to a larger model to capture this remaining correlation may improve, for instance, the forecasting performance of the model.

Residuals. A natural way to investigate the adequacy of an estimated model is to use residuals, whose properties should resemble those of the theoretical error terms u_t . As with linear regression models, the residuals are obtained as the difference between the observed values y_t and the fitted values $\hat{y}_t$ of the selected and estimated ARMA model

$$\hat{u}_t = y_t - \hat{y}_t.$$

In practice, one can also scale further and divide the residuals $\hat{u}_t$ by their estimated standard deviation $\hat{\sigma}$. The **standardized residuals** obtained in this way, $\hat{u}_t/\hat{\sigma}$, should in the case of a correctly specified model be approximately independent with mean zero and variance one.

Graphical and formal residual analysis. The first step in the investigation of the properties of the residuals is to plot their time series graph. If the estimated model is correct (or, empirically more realistically, “adequate”), the residual series should not exhibit trends, cyclical components, systematic variation in its level or variance over time (that is, heteroskedasticity), too many outlying observations, and so on.

- This is because the residuals $\hat{u}_t$ should reflect the assumptions made on the error term u_t .

The next step is to examine whether the residuals are uncorrelated. This is done similarly to how autocorrelation is assessed in the original time series—by analyzing the sample ACF and PACF of the residuals. If significant autocorrelation remains, it suggests that the model has not fully captured the dynamics of the data, and further refinement may be necessary.

- It should be noted, though, that for the autocorrelation coefficients computed from residuals, the critical bounds $\pm 1.96/\sqrt{T}$ are not valid, not even asymptotically.

- Asymptotically correct bounds do exist but are complicated, although some computer programs plot them automatically. If they are not readily available, the somewhat incorrect bounds $\pm 1.96/\sqrt{T}$ are sometimes used as a rough guide to the significance of the residual autocorrelations.

For testing remaining autocorrelation in the residuals, the **Ljung-Box test statistic** (introduced earlier) can also be used.

- However, when considering an $\text{ARMA}(p, q)$ model with a constant term, instead of the (asymptotic) χ^2_H -distribution used earlier, one should now use the $\chi^2_{H-p-q-1}$ -distribution where one has to choose H “large enough” for this (asymptotic) distribution to be valid.
- In practice, H is chosen somewhere around 10–20 depending on the sample size T . Moreover, typically a couple of different selections of H are considered to see whether the testing results are similar for different selections.

Possible (remaining) nonlinearities. Although the errors of the chosen model could be considered to have no autocorrelation, they are not necessarily independent. In line with the earlier discussion on possible nonlinearities, one (admittedly restricted) way of investigating potential nonlinear dependence in the errors is to use the autocorrelations of the squared residuals.

It can be shown that when errors satisfy the assumption $u_t \sim \text{nid}(0, \sigma^2)$, the sample autocorrelation coefficients of the squared residuals $\hat{u}_t^2$ ($t = 1, \dots, T$) are approximately independent and $N(0, 1/T)$ -distributed. Therefore, to these coefficients one can apply the critical bounds $\pm 1.96/\sqrt{T}$ as well as the **McLeod-Li test** presented earlier (that is, the Ljung-Box test applied to the squared residuals).

- If one notices that the errors are not autocorrelated, but that their squares are autocorrelated, one can also conclude that the errors cannot be Gaussian (for Gaussian processes, being uncorrelated is equivalent to being independent). Although the asymptotic results of the estimators presented in the previous section do hold even without the normality of the errors, it is still useful to investigate how realistic the normality assumption is by checking the histogram of the residuals or using a quantile-quantile plot.
- If the errors are only serially uncorrelated, but not independent, the covariance matrix of the maximum likelihood estimator $\hat{\beta}$ is not necessarily the one given in the parameter estimation section. The consistency and asymptotic normality of $\hat{\beta}$ can still hold under more general assumptions, but the estimated standard errors or test statistics are not necessarily valid.

Empirical example. Let us take a look at the sufficiency of the $\text{AR}(2)$ model estimated for the U.S. real GDP growth series. Below we depict the residual

time series from the fitted model, as well as their sample autocorrelations and histogram.

- It turns out that there is no (substantial) statistically significant residual autocorrelation left in the residuals. In addition to the sample autocorrelations, the Ljung-Box test statistics for different lag lengths H generally also confirm this conclusion (at least at the 5% significance level).
- As an example, the Ljung-Box test statistic based on the first $H = 16$ lags takes the value 17.546, and the corresponding p -value from the χ^2_{13} -distribution is 0.176 (degrees of freedom $H - p - q - 1 = 13$).

The histogram of the residuals suggests that the normality assumption is reasonable (see bottom right). Residual histograms rarely match the normal density function (also shown in the figure below) exactly, but in this case the normality assumption seems relatively adequate.

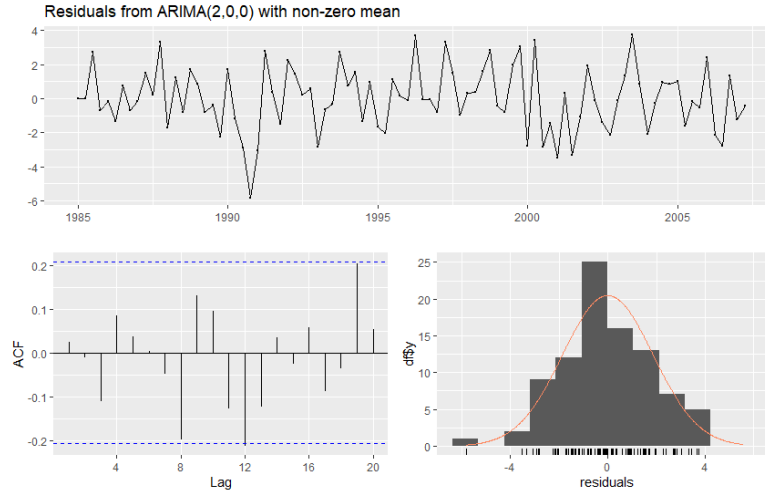


Figure: The residual time series of the AR(2) model estimated for the U.S. real GDP growth series (upper figure), their estimated autocorrelation function ($h = 0, \dots, 20$, bottom left) and their histogram (bottom right).

The main conclusions from residual diagnostics do not substantially change if using the AIC to select the lag length of the AR model. In the resulting AR(3) model, the third lag (y_{t-3}) is not a statistically significant predictor in terms of its estimated t -value. Moreover, when ARMA(p, q) models are considered, the AIC selects an MA(2) model, with essentially the same conclusions from residual diagnostics as in the AR(2) model.

When considering squared residuals and the McLeod-Li test statistics for different lag lengths, it turns out that with essentially any lag length the resulting p -

values are very high (typically higher than 0.50). Therefore, for the Great Moderation time period in the U.S. economy, there is no initial evidence for strong nonlinearities in the real GDP growth.

Chapter 8

Forecasting ARMA processes

8.1 Properties of the conditional expectation

Before considering forecasting with an ARMA(p, q) process, we examine a general situation where the objective is to forecast the (scalar) random variable Y using the realized value $\mathbf{X} = \mathbf{x}$ of a random vector $\mathbf{X}$. A result from probability theory tells us that the optimal forecast, in the sense of minimum mean square forecast error, is the conditional expected value of Y given $\mathbf{X} = \mathbf{x}$. In other words,

$$\mathbb{E} \left[(Y - \mathbb{E}(Y | \mathbf{X} = \mathbf{x}))^2 \right] \leq \mathbb{E} \left[(Y - g(\mathbf{x}))^2 \right]$$

for any function $g(\mathbf{x})$ of $\mathbf{x}$ (assuming the expectations above are finite).

i Extra: Some details on the conditional expectations

In the case of continuous distributions, the conditional expectation of Y given $\mathbf{X} = \mathbf{x}$ is defined by the equation

$$\mathbb{E}(Y | \mathbf{X} = \mathbf{x}) = \int_{-\infty}^{\infty} y f_{Y|\mathbf{X}}(y; \mathbf{x}) dy = \int_{-\infty}^{\infty} y \frac{f_{Y, \mathbf{X}}(y, \mathbf{x})}{f_{\mathbf{X}}(\mathbf{x})} dy,$$

where

- $f_{Y, \mathbf{X}}(y, \mathbf{x})$ is the joint density function of the random vector $(Y, \mathbf{X})$,
- $f_{\mathbf{X}}(\mathbf{x}) = \int_{-\infty}^{\infty} f_{Y, \mathbf{X}}(y, \mathbf{x}) dy$ is the marginal density function of $\mathbf{X}$, and
- $f_{Y|\mathbf{X}}(y; \mathbf{x}) = f_{Y, \mathbf{X}}(y, \mathbf{x}) / f_{\mathbf{X}}(\mathbf{x})$ defines the conditional density function of Y given $\mathbf{X} = \mathbf{x}$.

When $\mathbf{x}$ varies over the possible values of the random vector $\mathbf{X}$, $E(Y|\mathbf{X} = \mathbf{x})$ as a function of $\mathbf{x}$ defines a random variable for which it is natural to use the notation $E(Y|\mathbf{X})$.

The above definition can be generalized to the case where the random vector $\mathbf{X}$ might be infinite-dimensional. This case will be encountered in the ARMA(p, q) context assuming that all the values of the process $\{y_t, t = 0, \pm 1, \dots\}$ that precede the **forecast origin** (the time when the forecast is constructed) are known.

For our purposes, it suffices to know some simple properties of the conditional expected value (which also hold in the case of an infinite-dimensional conditioning variable). In this course, we will use the following properties of the conditional expected value:

- **CEV1:** $E(aY_1 + bY_2|\mathbf{X}) = aE(Y_1|\mathbf{X}) + bE(Y_2|\mathbf{X})$, when a and b are constants.
- **CEV2:** $E(Y|\mathbf{X}) = E(Y)$, when Y and $\mathbf{X}$ are independent random variables.
- **CEV3:** $E(Y) = E[E(Y|\mathbf{X})]$ (the so-called law of iterated expectations).
- **CEV4:** $E[h(\mathbf{X})Y|\mathbf{X}] = h(\mathbf{X})E(Y|\mathbf{X})$ for any function h (assuming the expected value of $h(\mathbf{X})Y$ exists and is finite).

8.2 Forming forecasts

Forecasting an ARMA(p, q) process. As the discussion above suggests, we consider forecasting an ARMA(p, q) process assuming that the entire infinitely long history of the process is known.

In what follows, we again assume that the constant term is included in the model equation to control a nonzero mean of y_t (i.e. $E(y_t) = \mu \neq 0$).

- When we are interested in forecasting the levels of y_t , which is typically the case, it is important to consider how forecasts for the original levels are recovered if the series has been demeaned and/or detrended. That is, all such transformations must be carefully taken into account in forecast computations.
- We will also assume that the ARMA(p, q) process under consideration is stationary and invertible and that the innovation term satisfies the condition $u_t \sim \text{iid}(0, \sigma^2)$.

In the calculations that follow, we repeatedly make use of the above-mentioned properties of the conditional expected value (CEV1–CEV4).

AR(1) case. Consider forecasting an $\text{AR}(p)$ process; for simplicity, we first assume that $p = 1$.

- In one-step forecasting, the aim is to forecast the value y_{t+1} , when the preceding history $\{y_t, y_{t-1}, \dots\}$ of the process is known.
- More generally, in h -step forecasting, the aim is to forecast the value of y_{t+h} , when $\{y_t, y_{t-1}, \dots\}$ is known.

For brevity, denote the conditional expectation as

$$\mathbf{E}(y_{t+h} | y_s, s \leq t) \equiv \mathbf{E}_t(y_{t+h}), \quad (h \geq 1).$$

Taking conditional expectations of both sides of the $\text{AR}(1)$ equation (forwarded first by one period)

$$y_{t+1} = \nu + \phi_1 y_t + u_{t+1}$$

leads to (see CEV1)

$$\mathbf{E}_t(y_{t+1}) = \nu + \phi_1 \mathbf{E}_t(y_t) + \mathbf{E}_t(u_{t+1}).$$

- Under the stationarity condition $|\phi_1| < 1$, the variables $y_t, y_{t-1}, \dots$ depend only on the variables $u_t, u_{t-1}, \dots$. Recall the $\text{MA}(\infty)$ representation of y_t to see this.
- Therefore, in the conditional expectation $\mathbf{E}_t(u_{t+1})$, the conditioning random variables $\{y_s, s \leq t\}$ are independent of u_{t+1} . Following the property CEV2, we then obtain

$$\mathbf{E}_t(u_{t+1}) = \mathbf{E}(u_{t+1}) = 0.$$

More generally, it holds that

$$\mathbf{E}_t(u_{t+k}) = \mathbf{E}(u_{t+k}) = 0, \quad k \geq 1.$$

- Moreover, $\mathbf{E}_t(y_t) = y_t$ (see CEV4: y_t is included in the information set at time t).

Putting the above results together, we obtain

$$\mathbf{E}_t(y_{t+1}) = \nu + \phi_1 y_t.$$

When forecasting y_{t+2} , we obtain

$$\begin{aligned} \mathbf{E}_t(y_{t+2}) &= \nu + \phi_1 \mathbf{E}_t(y_{t+1}) + \mathbf{E}_t(u_{t+2}) \\ &= \nu + \phi_1 \mathbf{E}_t(y_{t+1}) \\ &= (1 + \phi_1)\nu + \phi_1^2 y_t, \end{aligned}$$

and inductively (given $|\phi_1| < 1$), h -period forecast

$$\mathbf{E}_t(y_{t+h}) = (1 + \phi_1 + \cdots + \phi_1^{h-1})\nu + \phi_1^h y_t = \mu + \phi_1^h(y_t - \mu),$$

where the last equality uses $\mu = \nu/(1 - \phi_1)$.

AR(p) case. In the case of a general AR(p) process, we use similar steps to obtain forecasts. Taking conditional expectations of both sides of

$$y_{t+1} = \nu + \phi_1 y_t + \cdots + \phi_p y_{t+1-p} + u_{t+1},$$

and using similar arguments as in the case $p = 1$, including the results $\mathbf{E}_t(u_{t+1}) = 0$ and $\mathbf{E}_t(y_{t-j}) = y_{t-j}$ ($j \geq 0$), we obtain

$$\mathbf{E}_t(y_{t+1}) = \nu + \phi_1 y_t + \cdots + \phi_p y_{t+1-p}.$$

When forecasting y_{t+2} , we similarly obtain

$$\mathbf{E}_t(y_{t+2}) = \nu + \phi_1 \mathbf{E}_t(y_{t+1}) + \phi_2 y_t + \cdots + \phi_p y_{t+2-p},$$

where the conditional expected value on the right hand side could be replaced with the expression obtained for it above.

- In other words, the variable y_{t+1} , unknown at time t , has been replaced with its forecast $\mathbf{E}_t(y_{t+1})$.
- The variables $y_t, \dots, y_{t+2-p}$ are known at time t . Therefore, they remain on the right hand side as they are included in the information set at the forecast origin.

Inductively, it is straightforward to see that the above generalizes to forecasting h periods ahead

$$\mathbf{E}_t(y_{t+h}) = \nu + \phi_1 \mathbf{E}_t(y_{t+h-1}) + \phi_2 \mathbf{E}_t(y_{t+h-2}) + \cdots + \phi_p \mathbf{E}_t(y_{t+h-p}), \quad h \geq 1,$$

where $\mathbf{E}_t(y_{t+h-j}) = y_{t+h-j}$ for $h \leq j$. This means that forecasts can be computed recursively, starting with the one-step-ahead case $h = 1$, and proceeding one step at a time to the forecast horizons $h = 2, h = 3, \dots$

ARMA(p, q) case. Obtaining forecasting formulae for an ARMA(p, q) process is quite similar to forecasting with an AR(p) process.

- When the stationarity condition holds, $y_t - \mu = \sum_{j=0}^{\infty} \psi_j u_{t-j}$, from which it follows that u_{t+i} , $i \geq 1$, is independent of the variables $\{y_t, y_{t-1}, \dots\}$. Therefore, $\mathbf{E}_t(u_{t+i}) = \mathbf{E}(u_{t+i}) = 0$ for all $i \geq 1$ (see CEV2).
- On the other hand, when the invertibility condition holds, $u_t = \sum_{j=0}^{\infty} \pi_j (y_{t-j} - \mu)$, and therefore $\mathbf{E}_t(u_{t-i}) = u_{t-i}$ for all $i \geq 0$ (see CEV4).

Taking conditional expectations of both sides of the equation defining an ARMA(p, q) and using the above-mentioned properties, we obtain h -period-ahead forecast

$$\begin{aligned} E_t(y_{t+h}) &= \nu + \phi_1 E_t(y_{t+h-1}) + \cdots + \phi_p E_t(y_{t+h-p}) \\ &\quad + \theta_1 E_t(u_{t+h-1}) + \cdots + \theta_q E_t(u_{t+h-q}), \quad h \geq 1, \end{aligned}$$

where $E_t(y_{t+h-j}) = y_{t+h-j}$ for $h \leq j$, and

$$E_t(u_{t+h-j}) = \begin{cases} u_{t+h-j}, & \text{when } h \leq j, \\ 0, & \text{when } h > j. \end{cases}$$

Using this forecast formula, forecasts can again be computed recursively, starting with the one-step-ahead case $h = 1$, and proceeding one step at a time to the horizons $h = 2, h = 3, \dots$

In practice. The discussion above makes the unrealistic assumption that the parameters of the examined process are known. In practice, unknown parameters are replaced by their estimates, and in this case the above-mentioned optimality of forecasts (forecast construction) only holds approximately.

Moreover, for ARMA models, the innovation terms appearing in the forecasting formulae above (not in the AR(p) case) have to be computed using the observed time series, so in the formula $u_t = \sum_{j=0}^{\infty} \pi_j (y_{t-j} - \mu)$ we have to truncate the sum at some point (e.g., $\sum_{j=0}^{t-1} \pi_j (y_{t-j} - \mu)$).

- A popular alternative is to calculate u_t using the difference equation

$$u_t = y_t - \nu - \phi_1 y_{t-1} - \cdots - \phi_p y_{t-p} - \theta_1 u_{t-1} - \cdots - \theta_q u_{t-q}, \quad t = 1, 2, \dots,$$

where the initial values $y_0, \dots, y_{1-p}$ can be assumed known (observed), and the initial values $u_0, \dots, u_{1-q}$ can be set to $u_0 = \cdots = u_{1-q} = 0$ (that is, to the unconditional mean of u_t). When t is large, the effect of initial values is negligible.

8.3 Prediction intervals

An alternative way to approach forecasting an ARMA(p, q) process is via the MA(∞) representation. Taking conditional expectations of both sides of the equation

$$y_{t+h} = \mu + \sum_{j=0}^{\infty} \psi_j u_{t+h-j},$$

and using results of conditional expectations used above, we obtain

$$\mathbf{E}_t(y_{t+h}) = \mu + \sum_{j=h}^{\infty} \psi_j u_{t+h-j}.$$

This formula is not useful in practice, but it is convenient for investigating the properties of the forecast error $y_{t+h} - \mathbf{E}_t(y_{t+h})$. From the two equations above, we obtain

$$y_{t+h} - \mathbf{E}_t(y_{t+h}) = \sum_{j=0}^{h-1} \psi_j u_{t+h-j}.$$

Note, in particular, that the forecast error of the one-step forecast is u_{t+1} .

- Taking (unconditional) expectations from both sides of the last equation shows that the forecast $\mathbf{E}_t(y_{t+h})$ is unbiased in the sense that the forecast error has an expected value of zero:

$$\mathbf{E}(y_{t+h} - \mathbf{E}_t(y_{t+h})) = 0.$$

- Furthermore, we can compute the variance of the forecast error. With straightforward computation, we obtain

$$\text{Var}(y_{t+h} - \mathbf{E}_t(y_{t+h})) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \equiv \sigma_h^2.$$

- Importantly, note that as h increases, the forecast converges (in mean square) to the mean of the process, whereas the variance of the forecast error σ_h^2 converges to the variance of the process being predicted (y_t), namely $\sigma^2 \sum_{j=0}^{\infty} \psi_j^2$ (cf. the properties of the ARMA process).

In addition to the results obtained above, if one assumes that $u_t \sim \text{nid}(0, \sigma^2)$, then

$$y_{t+h} - \mathbf{E}_t(y_{t+h}) \sim \mathbf{N}(0, \sigma_h^2),$$

and, therefore, y_{t+h} is contained in the interval

$$\mathbf{E}_t(y_{t+h}) \pm 1.96\sigma_h$$

with 95% probability (in repeated sampling). In practice, the unknown parameters in the forecast $\mathbf{E}_t(y_{t+h})$ and in the standard deviation σ_h are replaced by their estimates, and hence the normality of the forecast errors holds only approximately.

Empirical example (continued). In the previous two sections, we concluded that an AR(2) model is one possible candidate for the U.S. real GDP growth (1985:Q1–2007:Q2). The estimated AR(2) model

$$y_t = \underset{(0.197)}{1.693} + \underset{(0.101)}{0.160}y_{t-1} + \underset{(0.101)}{0.287}y_{t-2} + \hat{u}_t, \quad \hat{\sigma}^2 = 3.536,$$

is now used to compute 8-quarter-ahead predictions. Starting from the last observation (2007:Q2), forecasts are obtained with the forecasting formulae above applied in the case of $p = 2$ (and $q = 0$) and $h = 8$. The resulting (point) forecasts of the estimated AR(2) model and their 80% and 95% prediction intervals are presented below.

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2007 Q3	2.427727	0.01787587	4.837578	-1.2578223	6.113277
2007 Q4	2.781453	0.34098863	5.221917	-0.9509150	6.513820
2008 Q1	2.834604	0.28036903	5.388840	-1.0717615	6.740971
2008 Q2	2.944709	0.38002888	5.509388	-0.9776305	6.867048
2008 Q3	2.977582	0.40040373	5.554759	-0.9638718	6.919035
2008 Q4	3.014464	0.43506842	5.593860	-0.9303814	6.959310
2009 Q1	3.029805	0.44884169	5.610767	-0.9174375	6.977047
2009 Q2	3.042852	0.46149492	5.624209	-0.9049929	6.990696

The figure below depicts the 8-step forecasts and the prediction intervals. As we can see, the forecast converges towards the mean $\hat{\nu}/(1 - \hat{\phi}_1 - \hat{\phi}_2) \approx 3.06$ as the forecast horizon lengthens. The prediction intervals are quite broad.

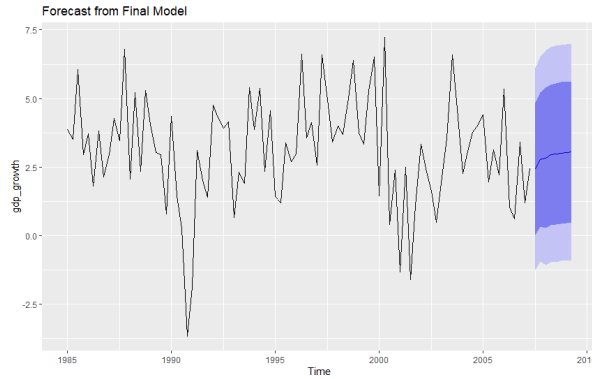


Figure: 8-step-ahead forecasts for the U.S. real GDP growth and their 80% and 95% prediction intervals from the AR(2) process.

Obviously, these forecasts, constructed at 2007:Q2, turned out to be far too optimistic for the years 2008 and 2009 because of the Great Financial Crisis (2008–2009) and the possible end of the Great Moderation. It can generally be concluded that no econometric model was able to predict such a downturn.

These forecasts should therefore be seen more as a scenario of how the U.S. economy might have evolved without the financial crisis.

8.4 Introduction to out-of-sample forecasting

In various fields, particularly in economics and finance, it is crucial to assess the reliability and accuracy of forecasts generated by estimated time series models. A popular method for this evaluation is (pseudo) **out-of-sample forecasting**. This approach aims to mimic a real-time forecasting situation as closely as possible, using only information that would have been available at the time the forecast was made.

The fundamental idea is to evaluate the forecast performance by systematically reserving a portion of the data, unknown to the model during its estimation, to serve as a “future” against which forecasts are compared. This method is at times termed “pseudo” out-of-sample forecasting because, while we simulate real-time forecasting, we do have full knowledge of the entire dataset beforehand. In true real-time forecasting, future observations are, by definition, unknown.

- The test sample period (forecast evaluation sample) is typically around 30–50% of the (last) observations. This amount, however, varies in different applications.
- Notice that **backtesting** in the financial industry is essentially a form of out-of-sample forecasting applied specifically to financial data and trading strategies. The underlying principle is the same, but the terminology and focus differ slightly: a time series analyst might use out-of-sample forecasting to evaluate how accurately a stock’s price can be predicted, whereas a financial analyst would use backtesting to see whether a strategy based on those predictions would actually have made money.

Out-of-sample forecasting generally involves the following steps:

1. **Define the initial estimation sample:** An initial segment of the time series, say the first T observations $(y_1, \dots, y_T)$, is used to estimate the model’s parameters. This period is often referred to as the **training sample** or **estimation sample**.
2. **Define the forecast horizon h :** Like above where we introduced how to construct forecasts with ARMA models, we specify how many periods ahead we want to forecast (e.g., $h = 1$ for one-period-ahead, $h = 2$ for two-periods-ahead, etc.).
3. **Define the forecasting evaluation period:** A subsequent segment of the data, known as the **test sample** or **forecasting sample**, is set aside. This

period spans from y_{T+1} up to $y_{T+m+h-1}$, where m is the number of distinct **forecast origins** (i.e., the number of times we re-estimate the model and generate a new h -step forecast). In other words, with this notation, we assume that we have observations $y_1, \dots, y_T, y_{T+1}, \dots, y_{T+m+h^{\max}-1}$ where $h^{\max}$ is the maximum forecast horizon considered.

4. **Iterative forecasting and parameter updating:** The core of (pseudo) out-of-sample forecasting is its iterative nature. We start by estimating the model using the initial T observations, then we construct forecasts h periods ahead. We then “advance time” by one period, potentially re-estimate the model with updated information, and generate a new h -period-ahead forecast. This process is repeated m times.

Let’s clarify the notation with an example. Suppose we want to evaluate two-period-ahead forecasts ($h = 2$).

- Let T denote the end of our *initial* estimation sample.
- Let j be an index for the forecast origin, running from $j = 0, 1, \dots, m - 1$. For simplicity, let $m = 4$, **which is far too short in practice** but is used here for pedagogical reasons.

The sequence of forecasts at different forecast origins and their corresponding actual values would be:

- Forecast origin T (i.e., $j = 0$):
 - Estimate the model using data $y_1, \dots, y_T$.
 - Generate the $h = 2$ step forecast, denoted $\hat{y}_{T+2|T}$.
 - This forecast is compared against the actual observation y_{T+2} .
 - The forecast error is $\hat{e}_{T+2} = y_{T+2} - \hat{y}_{T+2|T}$.
- Forecast origin $T + 1$ (i.e., $j = 1$):
 - Update the estimation sample. This is where rolling vs. expanding parameter updating window (see below) becomes relevant.
 - Generate the $h = 2$ step forecast, denoted $\hat{y}_{T+3|T+1}$.
 - This forecast is compared against the actual observation y_{T+3} .
 - The forecast error is $\hat{e}_{T+3} = y_{T+3} - \hat{y}_{T+3|T+1}$.
- Forecast origins $T + 2$ and $T + 3$ (i.e., $j = 2$ and $j = 3$): proceed in the same way, which gives the forecasts $\hat{y}_{T+4|T+2}$ and $\hat{y}_{T+5|T+3}$, and the corresponding forecast errors $\hat{e}_{T+4}$ and $\hat{e}_{T+5}$.

In general, for $j = 0, 1, \dots, m - 1$, we compute the h -period forecast from the information available at $T + j$, denoted $\hat{y}_{T+h+j|T+j}$. The corresponding forecast error is:

$$\hat{e}_{T+h+j} = y_{T+h+j} - \hat{y}_{T+h+j|T+j}$$

Parameter updating strategies. In a real forecasting situation, the information set used for model estimation typically includes the most recent observations. The way this information set is updated over time defines different parameter estimation strategies:

- **Expanding window:** In this approach, the estimation sample grows by one observation in each forecast computation iteration. If the initial estimation sample is $y_1, \dots, y_T$, the next estimation sample will be $y_1, \dots, y_{T+1}$, then $y_1, \dots, y_{T+2}$, and so on. This means older observations are always retained. This is suitable when one believes that all past data is equally relevant for estimating the current parameters.
- **Rolling window:** Here, the estimation sample has a fixed size (e.g., T observations) and “rolls” forward by one observation in each iteration. If the initial sample is $y_1, \dots, y_T$, the next sample will be $y_2, \dots, y_{T+1}$, then $y_3, \dots, y_{T+2}$, and so on, with the forecast origin moving forward as in the expanding window approach. This approach gives more weight to recent observations and is useful when model parameters are suspected to change over time (i.e., the presence of structural breaks or regime changes).

The accompanying R code (R Lab below) allows you to select either a **rolling** or **expanding** window strategy using the `window_type` parameter.

Evaluating forecast accuracy. Once the forecast errors ($\hat{e}_{T+h+j}$) for the entire test sample are computed, a typical measure for evaluating forecast accuracy is the **Mean Squared Forecast Error (MSFE)**:

$$\text{MSFE} = \frac{1}{m} \sum_{j=0}^{m-1} \hat{e}_{T+h+j}^2.$$

The MSFE penalizes larger errors more heavily because the forecast errors are squared. Its square root, the **Root Mean Squared Forecast Error (RMSFE)**, is also commonly used as it is in the same units as the original series, making it easier to interpret.

Another popular alternative is the **Mean Absolute Forecast Error (MAFE)**:

$$\text{MAFE} = \frac{1}{m} \sum_{j=0}^{m-1} |\hat{e}_{T+h+j}|.$$

The MAFE is less sensitive to outliers than the MSFE.

The objective of this exercise is to identify the model(s) that produce the smallest MSFE and/or MAFE values, indicating superior forecasting performance.

Empirical example (continued). Let us continue with an illustration of out-of-sample forecasting of the quarterly U.S. real GDP growth. Assume that

we start out-of-sample forecasting in 2010:Q1 and construct one-quarter ($h = 1$) and four-quarter ($h = 4$) ahead out-of-sample forecasts for the forecast evaluation (test) sample period 2010:Q1–2018:Q4. We compare the performance of AR(1), AR(2), AR(3) and AR(4) models.

- The first estimation sample is between 1985:Q1–2009:Q4.
- Note that the previous results (for the sample period excluding the Great Financial Crisis) suggested the AR(2) model.
- Parameter updating is done via expanding and rolling window approaches separately.
- The MSFEs suggest that the AR(3) model yields the smallest MSFE in all the comparisons, except for the $h = 4$ horizon with the rolling window, where the AR(4) model does.

```
--- MSFE Results for AR(1), AR(2), AR(3), AR(4) (h=1, expanding window) ---
```

```
$`AR(1)`  
[1] 3.010018  
$`AR(2)`  
[1] 2.810865  
$`AR(3)`  
[1] 2.760154  
$`AR(4)`  
[1] 3.451758
```

```
--- MSFE Results for AR(1), AR(2), AR(3), AR(4) (h=1, rolling window) ---
```

```
$`AR(1)`  
[1] 3.032567  
$`AR(2)`  
[1] 2.793521  
$`AR(3)`  
[1] 2.731007  
$`AR(4)`  
[1] 3.481733
```

```
--- MSFE Results for AR(1), AR(2), AR(3), AR(4) (h=4, expanding window) ---
```

```
$`AR(1)`  
[1] 2.628715  
$`AR(2)`  
[1] 2.662022  
$`AR(3)`  
[1] 2.457973  
$`AR(4)`  
[1] 2.467844
```

```
--- MSFE Results for AR(1), AR(2), AR(3), AR(4) (h=4, rolling window) ---  
$`AR(1)`  
[1] 2.602836  
$`AR(2)`  
[1] 2.633853  
$`AR(3)`  
[1] 2.496958  
$`AR(4)`  
[1] 2.478106
```

Chapter 9

Volatility modelling: AR-GARCH model

9.1 Modelling conditional variance

ARMA models can be characterized as models for the conditional mean of a stationary process.

- When the invertibility condition holds, ARMA processes have the $AR(\infty)$ representation from which it can be seen that the conditional expected value of y_t , conditional on the past of the process $\{y_{t-1}, y_{t-2}, \dots\}$, is $E_{t-1}(y_t) = \mu - \sum_{j=1}^{\infty} \pi_j (y_{t-j} - \mu)$.

The conditional variance of an ARMA process (model) y_t is

$$\begin{aligned}\text{Var}_{t-1}(y_t) &= E_{t-1}(y_t - E_{t-1}(y_t))^2 \\ &= E_{t-1}(u_t^2) \\ &= E(u_t^2) \quad (\text{CEV2}) \\ &= \sigma^2.\end{aligned}$$

This shows that evident systematic variation in conditional variance, that several time series contain, cannot be taken into consideration with an ARMA model. Typical examples of such time series are financial time series and especially different asset return series.

Empirical example. The NASDAQ 100 (ticker symbol ^NDX) is a major stock market index that tracks the performance of 100 of the largest non-financial companies listed on the NASDAQ stock market (U.S. stock market).

- It includes 100 of the largest domestic and international companies listed on NASDAQ, weighted by a modified market capitalization method to limit the influence of the very large firms. The index excludes financial companies and is heavily concentrated in the technology sector and dominated by a few extremely large companies like (as in 2025) NVIDIA Corporation (NVDA), Microsoft Corp. (MSFT), Apple Inc. (AAPL) and Amazon.com Inc. (AMZN).
- The NASDAQ 100 is widely viewed as the benchmark for large-cap growth stocks.
- The data source: quantmod package in R, which accesses publicly available financial data, seemingly sourced from Yahoo Finance.

Let us consider excess stock returns of the NASDAQ 100 over the 3-Month Treasury Bill rate. As a simple approximation, we can think that we are interested in the percentage changes (log-differences) of the NASDAQ 100 index adjusted for risk-free rate return (see Transformations in Section 1). Below we depict the daily excess stock returns between 1.1.2003–30.9.2025.

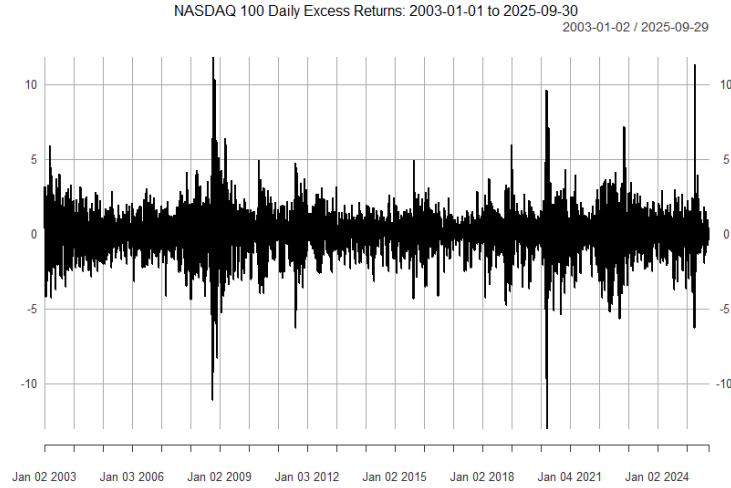


Figure: Excess stock returns of the NASDAQ 100 stock market index. (source: R package quantmod)

The return series seems quite stationary in terms of its level. Below we depict the sample autocorrelation and partial autocorrelation functions for the first 40 lags and their approximate 95% critical bounds. These suggest that there is some autocorrelation but its degree is not very high.

- Typically asset returns are almost non-autocorrelated
- For this sample period and NASDAQ returns, the resulting Ljung-Box

test statistics reject the null hypothesis of no autocorrelation with all the relevant statistical significance levels.

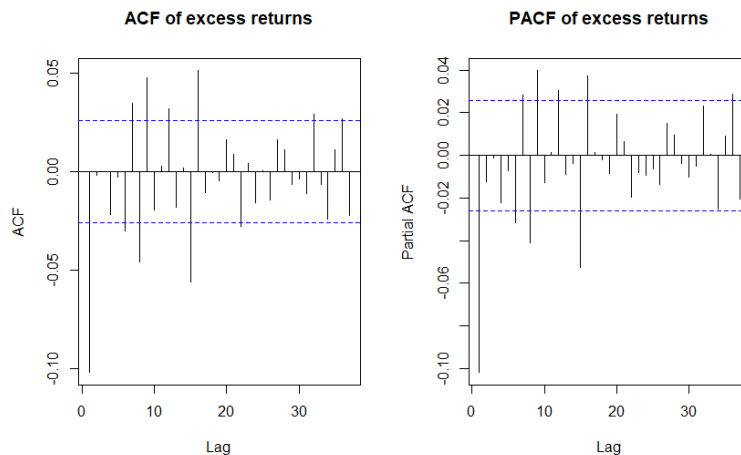


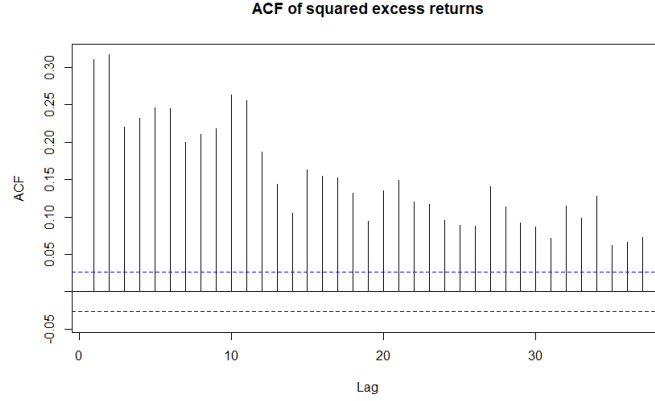
Figure: Sample autocorrelations and partial autocorrelations of the NASDAQ 100 excess returns.

The asset return series exhibits, however, some even more clear and typical variation, which is reflected by the autocorrelation function of the squared observations: All the reported autocorrelation coefficients of squared observations exhibit clear positive autocorrelation exceeding the 95%-critical bound. The approximate p -value of the McLeod-Li test with $H = 10$, and also with other lag length selections, is zero with four decimal precision.

- There are periods (see above) during which the variation of the series is either larger or smaller than on average. This “**volatility clustering**” is a certain type of heteroskedasticity.
- When the definition of variance is taken into account, it is natural to investigate this using the autocorrelation function of the squared observations. Notable is also the large difference between large and small absolute values, which suggests a more fat-tailed distribution than the normal distribution.

Figure: Sample autocorrelations for squared observations (squared stock returns).

In what follows, heteroskedasticity similar to that of the stock return series above is thought to be not related with changes in unconditional variance, but rather with changes in the **conditional variance** when conditioning on the past values of the series. Next, we will first consider some general aspects of models used in



modelling conditional variance, and then focus on some particular models that are most common in practice.

9.2 Model formulation

In this section, we consider an AR-GARCH model determined by the following two equations

$$y_t = \nu + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + u_t,$$

$$u_t = h_t^{1/2} \varepsilon_t, \quad \varepsilon_t \sim \text{iid}(0, 1),$$

where h_t is a function of the variables u_{t-j} , $j > 0$, and the error term ε_t is assumed to be independent of the variables y_{t-j} , $j > 0$, and hence also of the variables u_{t-j} , $j > 0$. In other words, we consider an **AR(p) model whose error term is conditionally heteroskedastic**.

- The coefficients $\phi_1, \dots, \phi_p$ are assumed to satisfy the (sufficient) stationarity condition of the AR(p) process: $1 - \phi_1 z - \cdots - \phi_p z^p = \phi(z) \neq 0$, when $|z| \leq 1$.

Regarding the conditional variance, by its definition, the variance of a random variable is associated with the squares of the random variable. Therefore, it seems natural that the conditional variance h_t would depend on the past squared values of the process. For concreteness sake, consider a general **GARCH(r,s) model**

$$h_t = \omega + \beta_1 h_{t-1} + \cdots + \beta_r h_{t-r} + \alpha_1 u_{t-1}^2 + \cdots + \alpha_s u_{t-s}^2,$$

whose parameters are assumed to satisfy the required conditions for non-negativeness, identification, and strict stationarity (to be considered below).

As before, we denote the conditional expectation as $E_{t-1}(\cdot) = E(\cdot | y_s, s \leq t-1)$. Because u_t is a function of the variables $y_t, \dots, y_{t-p}$, **the conditional variance h_t is a function of the variables $y_{t-j}, j > 0$** . Therefore, we can use the same arguments as considered, e.g., in forecast construction (see CEV4) to conclude that in the AR-GARCH model

$$E_{t-1}(u_t) = h_t^{1/2} E_{t-1}(\varepsilon_t) = h_t^{1/2} E(\varepsilon_t) = 0.$$

Together with this result, the model equations of the AR-GARCH model can be used to justify the following two results

$$E_{t-1}(y_t) = \nu + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} \quad \text{and} \quad \text{Var}_{t-1}(y_t) = h_t.$$

- The latter result can be justified by noticing that $y_t = E_{t-1}(y_t) + u_t$ and that

$$\begin{aligned} \text{Var}_{t-1}(y_t) &= E_{t-1}(y_t - E_{t-1}(y_t))^2 \\ &= E_{t-1}(u_t^2) \\ &= h_t E(\varepsilon_t^2) \\ &= h_t, \end{aligned}$$

where the result CEV2 and the identity $u_t^2 = h_t \varepsilon_t^2$ are used (see the general model definition above).

These two results demonstrate that the conditional mean and conditional variance of the process y_t depend on the past values of the process. Based on the above points, **building a volatility model**, that is selecting the model equation for h_t , roughly consists of the following steps:

- Finding an adequate specification for the conditional mean $E_{t-1}(y_t)$ (e.g., to select a suitable AR or ARMA model, and including also possible deterministic terms) is necessary to obtain a suitable specification for the conditional variance.
- Checking for the (conditional) heteroskedasticity of the error term. This can be based on residuals, as they are empirical counterparts of the error terms. That is, testing for “ARCH effects” using, e.g., the McLeod-Li test or the ARCH-LM test described below.
- Finding a sufficient specification for the conditional variance h_t . There are a lot of different alternative model specifications to $\text{GARCH}(r, s)$ suggested in the econometric literature. Replacing it with one of the alternatives yields a straightforward extension of the AR-GARCH model introduced above.
- Estimation of the full model can be carried out by the method of maximum likelihood (see details below).

i Extra: Relation to risk management and Value-at-Risk (VaR)

The core motivation for modeling conditional heteroskedasticity (time-varying volatility h_t) in asset returns y_t is its direct and profound impact on risk management, particularly in calculating measures like Value-at-Risk (VaR).

The question that often arises in risk management is: What is the maximum expected loss over a specific time horizon with a given confidence level? This loss is the Value-at-Risk (VaR) threshold c . The VaR threshold c is the loss level such that the probability of the actual loss y_{t+1} being less than or equal to c is a fixed, small probability π (e.g., 1% or 5%):

$$P_t(y_{t+1} \leq c) = G\left(\frac{c - E_t(y_{t+1})}{h_{t+1}^{1/2}}\right) = \pi,$$

where $P_t(\cdot)$ denotes the conditional probability given information up to time t , $E_t(y_{t+1})$ is the conditional mean (return forecast), h_{t+1} is the conditional variance forecast (volatility), and $G(\cdot)$ is the cumulative distribution function (CDF) of the standardized innovation $\frac{y_{t+1} - E_t(y_{t+1})}{h_{t+1}^{1/2}}$.

The Value-at-Risk is the most commonly used risk measure. For the VaR model to be a useful and reliable tool in practice, the specification of the conditional variance h_t is critical. Ignoring or otherwise misspecifying the time-varying nature of volatility (for example, treating h_t as a constant) leads to severely flawed risk estimates.

9.3 GARCH(1,1) and ARCH(s) models

Instead of the general GARCH(r, s) model, the **GARCH(1,1) model**

$$h_t = \omega + \beta_1 h_{t-1} + \alpha_1 u_{t-1}^2$$

has been found adequate for most (financial) time series data.

- In this GARCH(1,1) case, the non-negativeness of h_t requires $\omega > 0$ and $\alpha_1, \beta_1 \geq 0$. Moreover, for β_1 to be identified, α_1 must be strictly positive $\alpha_1 > 0$.
- In the GARCH(r, s) model, these non-negativeness conditions are clearly more complicated.

Another special case of GARCH models is obtained when $r = 0$. That is the “GARCH-part” is missing and the GARCH(r, s) model reduces to an **ARCH(s)**

model

$$h_t = \omega + \sum_{i=1}^s \alpha_i u_{t-i}^2.$$

In line with the idea of capturing volatility clustering with ARCH and GARCH models, the periods of high (and low) conditional variance tend to persist. This is easy to see with the **ARCH(1) model** ($s = 1$)

$$h_t = \omega + \alpha_1 u_{t-1}^2,$$

where $\omega, \alpha_1 \geq 0$. A large shock (u_{t-1}^2) increases h_t , which then subsequently increases u_t^2 and h_{t+1} . This same logic holds also for more general ARCH and GARCH models. Moreover, in the ARCH(1) model, assuming normality of ε_t , the (unconditional) kurtosis of u_t is

$$\frac{E(u_t^4)}{E(u_t^2)^2} = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2}.$$

The kurtosis is finite (i.e. the 4th moment exists) if $3\alpha_1^2 < 1$, and it then always exceeds the value 3 implied by the normal distribution. The ARCH(1) model is hence capable of capturing excess kurtosis often present in financial data.

- Large outliers (in absolute value) appear in (observed) asset returns more often than implied by iid innovations.
- Similar result on excess kurtosis holds also for more general ARCH and GARCH models, but the formulae become more complicated.

As a summary of large past research, the GARCH(1,1) model has generally been found a successful and parsimonious alternative to the ARCH(s) model with a typically large s for asset returns. Why? By recursive substitutions, we get

$$\begin{aligned} h_t &= \omega + \beta_1 h_{t-1} + \alpha_1 u_{t-1}^2 \\ &= \omega + \beta_1 \omega + \beta_1^2 h_{t-2} + \alpha_1 \beta_1 u_{t-2}^2 + \alpha_1 u_{t-1}^2 \\ &\vdots \\ &= \omega \sum_{j=0}^k \beta_1^j + \alpha_1 \sum_{j=0}^k \beta_1^j u_{t-1-j}^2 + \beta_1^{k+1} h_{t-k-1}. \end{aligned}$$

This suggests that in the case $\beta_1 < 1$,

$$h_t = \omega \sum_{j=0}^{\infty} \beta_1^j + \alpha_1 \sum_{j=0}^{\infty} \beta_1^j u_{t-1-j}^2,$$

which is a particular kind of ARCH(∞) form. Therefore, we can conclude that GARCH(1,1) is able to capture volatility clustering in a parsimonious way via only three parameters.

Another perspective on the GARCH models, and specifically on the GARCH(1,1) model, is obtained by substituting $h_t = u_t^2 - \xi_t$ (and $h_{t-1} = u_{t-1}^2 - \xi_{t-1}$) into the model equation. Therefore, the GARCH(1,1) can be rewritten as

$$u_t^2 = \omega + (\alpha_1 + \beta_1)u_{t-1}^2 + \xi_t - \beta_1\xi_{t-1},$$

where $\xi_t = u_t^2 - h_t = h_t(\varepsilon_t^2 - 1)$ and $E(\xi_t) = 0$. This is an ARMA(1,1) type of presentation for u_t^2 . Note, however, that ξ_t is only a martingale difference sequence (uncorrelated with its own past, but heteroskedastic), not an iid error term.

- As for the ARMA(1,1) process, the condition $\alpha_1 + \beta_1 < 1$ (together with $\omega > 0$ and $\alpha_1, \beta_1 \geq 0$, which guarantee the non-negativity of h_t) is necessary and sufficient for the existence of a weakly stationary solution, i.e., for a finite unconditional variance

$$E(u_t^2) = \frac{\omega}{1 - \alpha_1 - \beta_1}.$$

- Strict stationarity holds under the weaker condition $E[\log(\alpha_1\varepsilon_t^2 + \beta_1)] < 0$, which also allows $\alpha_1 + \beta_1 = 1$. The latter borderline case is known as the integrated GARCH (IGARCH) model, in which the conditional variance process is strictly stationary but has no finite unconditional variance.

i Extra: Special case: AR-GARCH model with zero conditional mean

The AR-GARCH model presentation above contains also the special case of $E_{t-1}(y_t) = 0$ often considered in various references (books etc.). This leads to the notation

$$u_t = y_t = h_t^{1/2}\varepsilon_t, \quad \varepsilon_t \sim \text{iid}(0, 1).$$

Therefore, in this case the GARCH(1,1) model reduces to

$$h_t = \omega + \beta_1 h_{t-1} + \alpha_1 y_{t-1}^2$$

and in the ARCH(s) case, we get

$$h_t = \omega + \sum_{i=1}^s \alpha_i y_{t-i}^2.$$

9.4 Parameter estimation

If it is assumed that the error term is Gaussian

$$\varepsilon_t \sim \text{nid}(0, 1),$$

the (conditional) likelihood function can be derived following analogous principles as in ARMA models. In other words, assuming $\varepsilon_t \sim \text{nid}(0, 1)$, we get

$$u_t = h_t^{1/2} \varepsilon_t, \quad \varepsilon_t \sim \text{nid}(0, 1),$$

where $h_t = h_t(y_{t-1}, y_{t-2}, \dots)$, ε_t and vector $(y_{t-1}, y_{t-2}, \dots)$ are independent, and $u_t = y_t - \mathbf{E}_{t-1}(y_t)$. Assume also that y_t is stationary. Given the above assumptions and the conditional moments (conditional mean and variance) of the AR-GARCH model, we get the conditional density function of the observation y_t as

$$y_t | \mathbf{Y}_{t-1} \sim \text{N}(\nu + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p}, h_t).$$

Therefore, the conditional distribution of y_t , conditional on $\mathbf{Y}_{t-1} = (y_1, \dots, y_{t-1})$, $t = 1, \dots, T$, is Gaussian with the conditional mean (see above) $\nu + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p}$ and conditional variance h_t where the model equation for h_t needs to be specified.

Using the observed time series and required initial values (depending on the specific model) $\dots, y_{-1}, y_0, y_1, \dots, y_T$, as in the parameter estimation of ARMA models, this leads to the conditional joint density function

$$\prod_{t=1}^T f_{y_t | \mathbf{Y}_{t-1}} = (2\pi)^{-T/2} \cdot \prod_{t=1}^T h_t^{-1/2} \cdot \exp\left(-\frac{1}{2} \sum_{t=1}^T \frac{u_t^2}{h_t}\right).$$

Denote the parameter vector of the AR-GARCH model by $\boldsymbol{\vartheta} = (\boldsymbol{\phi}', \boldsymbol{\lambda}')'$ where

- $\boldsymbol{\phi} = (\nu, \phi_1, \dots, \phi_p)'$ contains the parameters related to the conditional mean, and
- $\boldsymbol{\lambda} = (\omega, \alpha_1, \dots, \alpha_s, \beta_1, \dots, \beta_r)'$ contains the parameters of the model for the conditional variance. As an example, in the GARCH(1,1) case ($r = s = 1$), $h_t = \omega + \beta_1 h_{t-1} + \alpha_1 u_{t-1}^2$, and hence $\boldsymbol{\lambda} = (\omega, \alpha_1, \beta_1)'$.
- A remark on the notation: in the estimation of ARMA models (see Section 6) the parameters of the conditional mean were collected in $\boldsymbol{\beta}$, so that the full parameter vector was $\boldsymbol{\vartheta} = (\boldsymbol{\beta}', \sigma^2)'$. Here the symbol β is reserved for the GARCH coefficients $\beta_1, \dots, \beta_r$, and hence the conditional mean parameters are denoted by $\boldsymbol{\phi}$ (which now contains the constant term ν as well). Otherwise the two setups are completely parallel: the single variance parameter σ^2 of the ARMA model is replaced by the vector $\boldsymbol{\lambda}$ that determines the time-varying conditional variance h_t .

The log-likelihood then becomes (omitting the constant term $-(T/2)\log(2\pi)$ that does not depend on the model parameters)

$$l(\boldsymbol{\vartheta}) = \sum_{t=1}^T l_t(\boldsymbol{\vartheta}) = -\frac{1}{2} \sum_{t=1}^T \log h_t(\boldsymbol{\phi}, \boldsymbol{\lambda}) - \frac{1}{2} \sum_{t=1}^T \frac{u_t(\boldsymbol{\phi})^2}{h_t(\boldsymbol{\phi}, \boldsymbol{\lambda})},$$

where the notation $u_t(\boldsymbol{\phi}) = y_t - \nu - \phi_1 y_{t-1} - \dots - \phi_p y_{t-p}$ and $h_t(\boldsymbol{\phi}, \boldsymbol{\lambda})$ emphasizes that both quantities are functions of the parameters.

- Note that the conditional variance h_t depends on the conditional mean parameters $\boldsymbol{\phi}$ as well, because it is a function of the lagged errors $u_{t-j}(\boldsymbol{\phi})$, $j > 0$. This is why the estimation of the two equations cannot, in general, be separated.
- The connection to the ARMA case is seen by setting $h_t \equiv \sigma^2$: the log-likelihood above then reduces to $-(T/2)\log(\sigma^2) - (1/2\sigma^2) \sum_{t=1}^T u_t(\boldsymbol{\phi})^2$, which is exactly the conditional Gaussian log-likelihood function of Section 6 (there written in terms of $\boldsymbol{\beta}$).
- Just as in the ARMA case, the recursions require initial values: in the GARCH(1,1) case one needs u_0 and h_0 . A common choice is to set h_0 equal to the sample variance of the series (or of the residuals of the conditional mean model). In the stationary case the effect of the initial values vanishes as T increases, so that the conditional and exact ML estimators have the same asymptotic distribution.

Maximization of the likelihood function is performed using numerical methods, similarly as with ARMA models. In addition, the non-negativity and stationarity restrictions of the conditional variance parameters have to be imposed during the optimization.

Statistical inference. If the normality assumption of the error term $\varepsilon_t \sim \text{nid}(0, 1)$ made above holds, the usual asymptotic properties of the maximum likelihood estimator (consistency and asymptotic normality) hold, as described for ARMA models in Section 6. If the Gaussianity assumption is found inappropriate, one could either (or both):

- Use a non-Gaussian distribution, like the t -distribution for the error term. This leads to the different form for the log-likelihood function, but the general lines to obtain it are the same as above.
- Modify the asymptotic distribution and interpret the maximum likelihood estimator (MLE) as **quasi-MLE (QMLE)**.

i Extra: Details on the QMLE

Concerning the QMLE approach, if the likelihood function is maximized at $\hat{\boldsymbol{\vartheta}} = (\hat{\boldsymbol{\phi}}', \hat{\boldsymbol{\lambda}}')'$, it can be shown, even without the normality assumption and under general assumptions and regularity conditions, that

$$\hat{\boldsymbol{\vartheta}} \underset{as}{\sim} \mathbf{N} \left(\boldsymbol{\vartheta}, \mathbf{V}(\boldsymbol{\vartheta})^{-1} \mathbf{B}(\boldsymbol{\vartheta}) \mathbf{V}(\boldsymbol{\vartheta})^{-1} \right),$$

where

$$\mathbf{V}(\boldsymbol{\vartheta}) = \mathbf{E} \left[-\frac{\partial^2 l(\boldsymbol{\vartheta})}{\partial \boldsymbol{\vartheta} \partial \boldsymbol{\vartheta}'} \right]$$

and

$$\mathbf{B}(\boldsymbol{\vartheta}) = \mathbf{E} \left[\sum_{t=1}^T \left(\frac{\partial}{\partial \boldsymbol{\vartheta}} l_t(\boldsymbol{\vartheta}) \right) \left(\frac{\partial}{\partial \boldsymbol{\vartheta}} l_t(\boldsymbol{\vartheta}) \right)' \right],$$

where $l_t(\boldsymbol{\vartheta})$ is as described above and its specific form naturally depends on the selected AR-GARCH model.

- The matrix $\mathbf{V}(\boldsymbol{\vartheta})$ is the same (Fisher information) matrix as in Section 6. The matrix $\mathbf{B}(\boldsymbol{\vartheta})$ is the expected outer product of the scores, and the covariance matrix above is the corresponding “sandwich” matrix.

Using the empirical counterparts of the matrices $\mathbf{V}(\boldsymbol{\vartheta})$ and $\mathbf{B}(\boldsymbol{\vartheta})$, namely

$$\hat{\mathbf{V}} = -\frac{\partial^2 l(\hat{\boldsymbol{\vartheta}})}{\partial \boldsymbol{\vartheta} \partial \boldsymbol{\vartheta}'} \quad \text{and} \quad \hat{\mathbf{B}} = \sum_{t=1}^T \left(\frac{\partial}{\partial \boldsymbol{\vartheta}} l_t(\hat{\boldsymbol{\vartheta}}) \right) \left(\frac{\partial}{\partial \boldsymbol{\vartheta}} l_t(\hat{\boldsymbol{\vartheta}}) \right)'$$

(both evaluated at the estimates and, as in Section 6, computed in practice with numerical derivatives), the asymptotic distribution of the estimator $\hat{\boldsymbol{\vartheta}}$ presented above can be used to form approximate standard errors and Wald tests concerning the parameter $\boldsymbol{\vartheta}$. Specifically, the approximate standard errors are the square roots of the diagonal elements of the estimated sandwich matrix

$$\hat{\mathbf{V}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{V}}^{-1} = [\hat{v}^{ij}], \quad \text{s.e.}(\hat{\vartheta}_i) = \sqrt{\hat{v}^{ii}},$$

and individual t -ratios, confidence intervals, and (joint) Wald tests are formed exactly as described for ARMA models in Section 6: the null hypothesis $\vartheta_i = 0$ is rejected at the 5% level if $|\hat{\vartheta}_i / \sqrt{\hat{v}^{ii}}| \geq 1.96$, and the 95% confidence interval is $\hat{\vartheta}_i \pm 1.96 \sqrt{\hat{v}^{ii}}$.

- Cf. the maximum likelihood estimation of AR (and ARMA) models.
- If the normality assumption (i.e. $\varepsilon_t \sim \text{nid}(0, 1)$) holds, then $\mathbf{V}(\boldsymbol{\vartheta}) = \mathbf{B}(\boldsymbol{\vartheta})$ (the information matrix equality), and the expressions above for the (QMLE) asymptotic distribution and the standard errors and Wald tests simplify to the usual MLE (maximum likelihood estimator) case $\mathbf{V}(\boldsymbol{\vartheta})^{-1}$, that is, to the result of Section 6.

- Unlike the Wald test, the likelihood ratio test is *not* robust to the misspecification of the error distribution: under the QMLE interpretation the LR statistic no longer has the χ^2 limiting distribution. When the normality assumption is questionable, hypotheses concerning $\boldsymbol{\vartheta}$ should therefore be tested with Wald tests based on the robust covariance matrix above.
- The asymptotic results require the parameters to be interior points of the admissible parameter space. In particular, testing $\alpha_1 = 0$ (“no ARCH effects”) with a t -ratio is *not* a standard testing problem, because $\alpha_1 \geq 0$ is a boundary value and β_1 is not identified when $\alpha_1 = 0$. This is one reason why the presence of conditional heteroskedasticity is tested with the ARCH-LM (or McLeod-Li) test described below rather than with a t -ratio. Similarly, the results require strict stationarity, and the normal approximation deteriorates when $\alpha_1 + \beta_1$ is very close to unity (cf. the “nearly violated” conditions in Section 6).
- If the distribution of ε_t is symmetric, the information matrix is block diagonal with respect to $\boldsymbol{\phi}$ and $\boldsymbol{\lambda}$, so that the estimators of the conditional mean and conditional variance parameters are asymptotically independent. With asymmetric (skewed) innovations this is no longer the case, and the full covariance matrix has to be used.

The practical message from above is that it is often reasonable to rely on the QMLE-based asymptotic distribution result and resulting robust standard errors in the first place, such as so called Bollerslev-Wooldridge standard errors, for different parameter estimates of the AR-GARCH model.

- The rationale is that the model specification, and especially the distributional assumption $\varepsilon_t \sim \text{nid}(0, 1)$, is rarely exactly correct. Even though the model is not entirely correctly specified, the QMLE-based standard errors allow for additional robustness against such misspecification, which makes their use advisable.
- Using the QMLE specifically means that we obtain the same estimates $\hat{\boldsymbol{\vartheta}}$ but we rely on the robust (sandwich) asymptotic covariance matrix and hence on the resulting robust standard errors.

Empirical example (continue). Consider the estimation result of the AR-GARCH model with Gaussian error term for the excess stock returns of the NASDAQ 100 index. For simplicity, fit a relatively simple AR(1)-GARCH(1,1) model (see above with selections $p = r = s = 1$). That is, we specify

- AR(1) model for the conditional mean.

- GARCH(1,1) model for the conditional variance

That is, we estimate an AR(1) model with GARCH(1,1) errors. The estimation result of the maximum likelihood estimation, based on the normality assumption of ε_t , yields

$$y_t = \underset{(0.013)}{0.087} - \underset{(0.013)}{0.044}y_{t-1} + \hat{u}_t,$$

$$\hat{h}_t = \underset{(0.006)}{0.035} + \underset{(0.013)}{0.876}\hat{h}_{t-1} + \underset{(0.012)}{0.105}\hat{u}_{t-1}^2,$$

where we report the robust standard errors under the parameter estimates (see the QMLE asymptotic distribution result). The full estimation result provided by **rugarch package** in R yields the following results:

```
*-----*
*           GARCH Model Fit           *
*-----*

Conditional Variance Dynamics
-----
GARCH Model : sGARCH(1,1)
Mean Model  : ARFIMA(1,0,0)
Distribution : norm

Optimal Parameters
-----
      Estimate Std. Error  t value Pr(>|t|)
mu      0.087229   0.012889   6.7679 0.000000
ar1     -0.043797   0.014149  -3.0954 0.001965
omega    0.035129   0.004868   7.2160 0.000000
alpha1   0.104804   0.008170  12.8272 0.000000
beta1    0.876000   0.008975  97.6078 0.000000

Robust Standard Errors:
      Estimate Std. Error  t value Pr(>|t|)
mu      0.087229   0.012567   6.9412 0.000000
ar1     -0.043797   0.012630  -3.4677 0.000525
omega    0.035129   0.006497   5.4066 0.000000
alpha1   0.104804   0.012428   8.4331 0.000000
beta1    0.876000   0.012869  68.0716 0.000000

LogLikelihood : -9030.906

Information Criteria
-----
```

```

Akaike          3.1583
Bayes           3.1641
Shibata         3.1583
Hannan-Quinn    3.1603

```

Weighted Ljung-Box Test on Standardized Residuals

```

-----
                        statistic p-value
Lag[1]                  0.1471  0.7014
Lag[2*(p+q)+(p+q)-1] [2] 0.6119  0.9317
Lag[4*(p+q)+(p+q)-1] [5] 2.0557  0.6952
d.o.f=1
H0 : No serial correlation

```

Weighted Ljung-Box Test on Standardized Squared Residuals

```

-----
                        statistic p-value
Lag[1]                  1.998  0.1575
Lag[2*(p+q)+(p+q)-1] [5] 4.486  0.1993
Lag[4*(p+q)+(p+q)-1] [9] 6.661  0.2292
d.o.f=2

```

Weighted ARCH LM Tests

```

-----
Statistic Shape Scale P-Value
ARCH Lag[3]    0.09937 0.500 2.000 0.7526
ARCH Lag[5]    4.68135 1.440 1.667 0.1217
ARCH Lag[7]    5.49080 2.315 1.543 0.1794

```

The estimation result shows well the point using the QMLE:

- The estimated coefficients (“AR”, “ARCH” and “GARCH” estimates) are the same for the “optimal parameters” (that is, when we assume that the normality assumption ε_t holds) and for QMLE (resulting robust estimation results), but the estimated standard errors are different. Especially for the GARCH part this means, as often in these circumstances, that the robust standard errors are somewhat higher, reflecting the additional uncertainty coming from possible model misspecification.
- All in all, all the estimated coefficients are statistically significant at the conventional significance levels. Moreover, the typical pattern of the GARCH(1,1) model is also present where the GARCH parameter (here β_1) is larger than the ARCH coefficient α_1 , and their sum is relatively close to 1. Estimated volatility, which is here the conditional standard deviation $\hat{h}_t^{1/2}$ of the estimated model and is at times interpreted as a measure of

risk in financial markets, is depicted below.

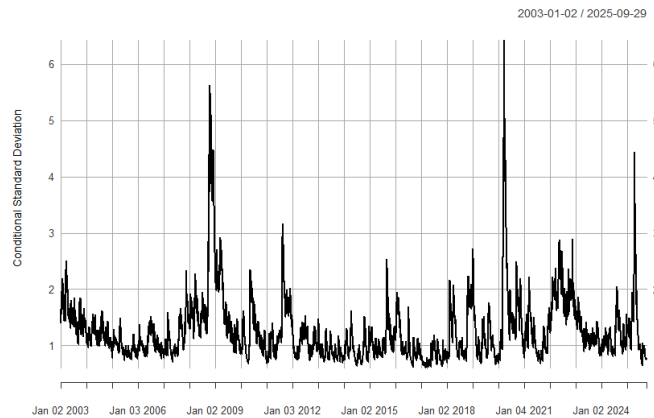


Figure: Estimated conditional standard deviation based on the AR(1)-GARCH(1,1) model for the NASDAQ 100 excess returns.

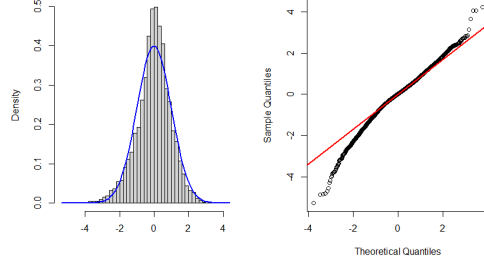
Residual diagnostics show that:

- There is no remaining autocorrelation in the residuals and squared residuals (based on the Ljung-Box and the McLeod-Li tests and residual autocorrelation coefficients).
- The ARCH LM tests do not reject the null hypothesis of no remaining conditional heteroskedasticity in the standardized residuals $\hat{\varepsilon}_t = \hat{u}_t / \hat{h}_t^{1/2}$ either. Together with the previous point, the p -values reported above show that the parsimonious AR(1)-GARCH(1,1) is an adequate model for the NASDAQ 100 excess returns.
- There is some deviation from the normality assumption in the (standardized) residuals. This suggests that the estimates should be interpreted as QMLEs, as above.

Figure: Histogram and Q-Q plot of (standardized) residuals.

i Extra: Details on the ARCH-LM test and “weighted” test statistics reported by rugarch

Testing for ARCH effects: the ARCH-LM test. Recall the second step of building a volatility model above: before (and after) specifying a



model for h_t , one has to check whether the error term really is conditionally heteroskedastic. The most common test for this purpose, alongside the McLeod-Li test, is the **ARCH-LM test** of Engle (1982).

Let $\hat{u}_t$, $t = 1, \dots, T$, denote the residuals of the estimated conditional mean model (here the $AR(p)$ model, estimated by OLS or ML). Under the null hypothesis there are no ARCH effects, that is, the conditional variance is constant, $h_t = \sigma^2$, whereas under the alternative h_t follows an $ARCH(m)$ model. The test is based on the auxiliary regression

$$\hat{u}_t^2 = \gamma_0 + \gamma_1 \hat{u}_{t-1}^2 + \dots + \gamma_m \hat{u}_{t-m}^2 + e_t, \quad t = m+1, \dots, T,$$

in which the null hypothesis of no ARCH effects is

$$H_0 : \gamma_1 = \dots = \gamma_m = 0.$$

The Lagrange multiplier (LM) test statistic is obtained from the coefficient of determination R^2 of this auxiliary regression as

$$ARCH-LM(m) = TR^2 \xrightarrow{d} \chi^2(m)$$

under the null hypothesis, so that H_0 is rejected for large values of the statistic (small p -values), indicating the presence of conditional heteroskedasticity. Equivalently, the conventional F -test of the joint significance of the lagged squared residuals in the auxiliary regression can be used; the two versions are asymptotically equivalent.

- The intuition is transparent: if the squared errors are predictable from their own past, the conditional variance is time varying. Note the analogy with the ACF of the squared observations examined in the empirical example above.
- The test is of the Lagrange multiplier type, which means that only the *restricted* model has to be estimated: the residuals $\hat{u}_t$ are obtained from the conditional mean model alone, and no GARCH model needs to be estimated. This makes the test convenient at the specification stage. As mentioned above, a Wald (t -ratio) test of $\alpha_1 = 0$ would

not be valid here, because the null hypothesis lies on the boundary of the parameter space.

- Although the alternative hypothesis is an ARCH(m) model, the test also has good power against GARCH alternatives, because a GARCH(1,1) model can be written as an ARCH(∞) model (see above). The lag length m is chosen by the user, and it is advisable to report the results for a few different values of m .
- **Post-estimation use.** The same test can be applied to the **standardized residuals** $\hat{\varepsilon}_t = \hat{u}_t / \hat{h}_t^{1/2}$ of the estimated AR-GARCH model, in which case it tests whether there is any conditional heteroskedasticity *left* in the residuals, that is, whether the specified model for h_t is adequate. Rejection suggests increasing the orders r and s or changing the specification of the conditional variance.
- A caveat concerning this post-estimation use: because $\hat{h}_t$ depends on the estimated parameters, the estimation of the model affects the asymptotic distribution of the statistic and the $\chi^2(m)$ approximation is no longer exact. The correct limiting distribution for the squared standardized residuals of a fitted GARCH model was derived by Li and Mak (1994). The tests reported by the rugarch package are based on this line of work (see the next remark).

Weighted test statistics. The tests in the output above are *weighted* versions of the Ljung-Box and ARCH-LM tests. The reason for the modification is twofold.

- *The effect of estimation.* The Ljung-Box and ARCH-LM statistics introduced above have their χ^2 limiting distributions when they are computed from the true errors. When they are computed from the residuals of an *estimated* model, the estimation of the parameters changes the asymptotic distribution of the (squared) residual autocorrelations. The conventional remedy is a degrees-of-freedom adjustment, $\chi^2(H - k)$, where k is the number of estimated parameters. This is what the rows “d.o.f=1” and “d.o.f=2” in the output refer to: $k = 1$ for the standardized residuals (the AR(1) coefficient ϕ_1) and $k = 2$ for the squared standardized residuals (α_1 and β_1). For squared standardized residuals of a fitted GARCH model, the appropriate statistic was derived by Li and Mak (1994).
- *Power.* The conventional Ljung-Box statistic weights all the autocorrelations equally,

$$Q(H) = T(T+2) \sum_{j=1}^H \frac{\hat{\rho}_j^2}{T-j},$$

so that its power deteriorates as H grows: the informative low-order autocorrelations get diluted by a large number of uninformative

higher-order ones. Fisher and Gallagher (2012) therefore propose weighting the squared autocorrelations by weights that decrease in the lag length,

$$Q_W(H) = T(T+2) \sum_{j=1}^H \frac{H-j+1}{H} \cdot \frac{\hat{\rho}_j^2}{T-j},$$

and an analogous weighting is applied to the ARCH-LM statistic. The weighted statistic behaves asymptotically as a linear combination of χ^2 distributed random variables, whose distribution is approximated by a **gamma distribution**. This is why the weighted ARCH LM output reports the columns “Shape” and “Scale” (the parameters of the gamma distribution) instead of degrees of freedom.

What this means in practice:

- The numerical values of the weighted statistics are **not** comparable with the values of the plain Ljung-Box, McLeod-Li, or ARCH-LM statistics computed as described above; only the p -values and the conclusions drawn from them are comparable. In the example above, both lead to the same conclusion that nothing systematic is left in the residuals.
- The labels of the type “Lag[2($p+q$) + ($p+q$) - 1]” indicate that rugarch selects the lag lengths as a function of the fitted model orders instead of letting the user pick them; the figure in the second brackets is the lag length actually used.
- For teaching purposes, the unweighted statistics remain the reference case: they are transparent and have the familiar χ^2 limiting distribution. In applied work, when the tests are computed from the residuals of an estimated GARCH model, the weighted versions have better size and power properties. They are also directly available in the R package **WeightedPortTest** (functions `Weighted.Box.test` and `Weighted.LM.test`).

9.5 Extra: Forecasting

Assume observations are available up to time t and the object of interest is to forecast the future values of y_{t+k} ($k \geq 1$). In addition to stationarity, assume that $E(y_t^2) < \infty$.

Forecasts for y_t . From the first equation of the AR(p)-GARCH model, that is, the model for the conditional mean, it can be seen that the optimal (in the mean square sense) one-step-ahead forecast is

$$E_t(y_{t+1}) = \nu + \phi_1 y_t + \cdots + \phi_p y_{t+1-p}.$$

When $k \geq 2$, it can be seen that (cf. forecasting formulae for AR(MA) models)

$$\mathbb{E}_t(u_{t+k}) = \mathbb{E}_t[\mathbb{E}_{t+k-1}(u_{t+k})] = 0,$$

so that

$$\mathbb{E}_t(y_{t+k}) = \nu + \phi_1 \mathbb{E}_t(y_{t+k-1}) + \cdots + \phi_p \mathbb{E}_t(y_{t+k-p}), \quad k = 1, 2, \dots,$$

where $\mathbb{E}_t(y_{t+k-j}) = y_{t+k-j}$ for $j \geq k$.

- In conclusion, from these and the forecasting formulae for an AR(p) process with a homoskedastic error term, it can be concluded that the optimal forecasts can be formed recursively exactly in the same manner as in the (conditionally) homoskedastic case.
- In practice, when computing forecasts, unknown parameters naturally need to be replaced with their estimates, which are based on the finite sample sizes, and hence numerically here forecasts can be slightly different than obtained with an AR(p) model with conditionally homoskedastic errors.

As the above shows, conditional heteroskedasticity does not affect the forecasting formulae, which are the same as in the AR case discussed earlier. However, the presence of conditional heteroskedasticity changes how the prediction intervals are computed (see Extra below).

i Extra: Prediction intervals under conditional heteroskedasticity

To see this, note that because the forecasts can be derived exactly as in the homoskedastic AR(p) model, the same also holds for the k -step forecast errors. Based on the calculations above,

$$\mathbb{E}_t(u_{t+k-j}) = \begin{cases} u_{t+k-j}, & \text{when } k \leq j \\ 0, & \text{when } k > j, \end{cases}$$

so that

$$y_{t+k} - \mathbb{E}_t(y_{t+k}) = \sum_{j=0}^{k-1} \psi_j u_{t+k-j} = \sum_{j=0}^{k-1} \psi_j h_{t+k-j}^{1/2} \varepsilon_{t+k-j},$$

where ψ_j are again the coefficients of the power series $\psi(z) = \phi(z)^{-1} = \sum_{j=0}^{\infty} \psi_j z^j$ ($\psi_0 = 1$).

Let us focus our attention on the one-step-ahead forecast error, whose conditional expected value becomes

$$\mathbb{E}_t[y_{t+1} - \mathbb{E}_t(y_{t+1})] = \mathbb{E}_t(u_{t+1}) = 0$$

with conditional variance

$$\mathbb{E}_t[(y_{t+1} - \mathbb{E}_t(y_{t+1}))^2] = \mathbb{E}_t(u_{t+1}^2) = h_{t+1}.$$

This shows that the one-step-ahead prediction error is conditionally heteroskedastic, that is, the forecast accuracy depends on what kind of values the process attained at the forecast origin (at time of forecast computation) and just before it.

- As can be seen from the expression of the forecast error, a similar result holds also for forecast horizons longer than one. Therefore, any analysis of forecast accuracy, such as prediction intervals is sensible to be based on the conditional distribution with conditioning on the history up until the date of forecasting.

If it is assumed that $\varepsilon_t \sim \text{nid}(0, 1)$, then the conditional distribution of the one-step-ahead forecast error u_t (conditioned on $\{y_{t-1}, y_{t-2}, \dots\}$) is Gaussian with mean zero and variance h_t .

- Using this result, one can form prediction intervals for the one-step-ahead forecast exactly as for ARMA process.
- For multiple step ahead forecasts, the situation is not that straightforward, because the conditional distributions of multiple step forecast errors are not Gaussian and have no simple expressions. Therefore, forming prediction intervals for multiple step ahead forecasts is more complicated.

Volatility forecasts. Concerning forecasting the conditional variance, we assume that observations up to and including time t (i.e. the forecast origin at time t) are available, and that forecasts for $h_{t+1}, h_{t+2}, \dots$ are desired.

Since h_{t+1} is a (deterministic) function of the variables $y_t, y_{t-1}, \dots$, it is already known at the forecast origin, and thus the first conditional variance that genuinely needs to be forecast is h_{t+2} . For simplicity, let us concentrate on the GARCH(1,1) case. Taking conditional expected values (conditional on $\{y_t, y_{t-1}, \dots\}$) of both sides of

$$h_{t+2} = \omega + \beta_1 h_{t+1} + \alpha_1 u_{t+1}^2$$

yields the optimal (in the mean squared error sense) forecast of h_{t+2} as

$$\mathbf{E}_t(h_{t+2}) = \omega + \beta_1 \mathbf{E}_t(h_{t+1}) + \alpha_1 \mathbf{E}_t(u_{t+1}^2),$$

where $\mathbf{E}_t(h_{t+1}) = h_{t+1}$ (see CEV4). Moreover, we obtain that (see CEV2 and CEV4)

$$\mathbf{E}_t(u_{t+1}^2) = \mathbf{E}_t(h_{t+1} \varepsilon_{t+1}^2) = h_{t+1} \mathbf{E}_t(\varepsilon_{t+1}^2) = h_{t+1} \mathbf{E}(\varepsilon_{t+1}^2) = h_{t+1},$$

so that

$$\mathbf{E}_t(h_{t+2}) = \omega + (\alpha_1 + \beta_1) h_{t+1}.$$

When the forecast horizon is $k \geq 3$, in a similar fashion, we obtain

$$\mathbf{E}_t(h_{t+k}) = \omega + \beta_1 \mathbf{E}_t(h_{t+k-1}) + \alpha_1 \mathbf{E}_t(u_{t+k-1}^2),$$

and finally

$$\mathbf{E}_t(h_{t+k}) = \omega \sum_{j=0}^{k-2} (\alpha_1 + \beta_1)^j + (\alpha_1 + \beta_1)^{k-1} h_{t+1}, \quad k = 2, 3, \dots,$$

where h_{t+1} is a function of variables $\{y_t, y_{t-1}, \dots\}$ known at the time when forecasts are constructed.

i Extra: Details for forecasting formulae multiple periods ahead

Consider the part

$$\mathbf{E}_t(h_{t+k}) = \omega + \beta_1 \mathbf{E}_t(h_{t+k-1}) + \alpha_1 \mathbf{E}_t(u_{t+k-1}^2),$$

Here

$$\mathbf{E}_t(u_{t+k-1}^2) = \mathbf{E}_t(h_{t+k-1} \varepsilon_{t+k-1}^2) = \mathbf{E}_t[\mathbf{E}_{t+k-2}(h_{t+k-1} \varepsilon_{t+k-1}^2)],$$

where the latter equality can be justified based on a generalization of property CEV3, that is, a generalization of the law of iterated expectations. This generalization says that $\mathbf{E}(Y|X_2) = \mathbf{E}[\mathbf{E}(Y|X_1)|X_2]$, when the components of the (potentially infinite-dimensional) random vector X_2 are a subset of the components of X_1 (or more generally, X_2 is a function of X_1).

Because

$$\mathbf{E}_{t+k-2}(h_{t+k-1} \varepsilon_{t+k-1}^2) = h_{t+k-1} \mathbf{E}_{t+k-2}(\varepsilon_{t+k-1}^2) = h_{t+k-1},$$

we obtain

$$\mathbf{E}_t(u_{t+k-1}^2) = \mathbf{E}_t(h_{t+k-1}), \quad k = 3, 4, \dots$$

To summarize, we have shown that

$$\mathbf{E}_t(h_{t+k}) = \omega + (\alpha_1 + \beta_1) \mathbf{E}_t(h_{t+k-1}), \quad k = 2, 3, \dots$$

Because $\mathbf{E}_t(h_{t+1}) = h_{t+1}$, we inductively obtain the solution

$$\mathbf{E}_t(h_{t+k}) = \omega \sum_{j=0}^{k-2} (\alpha_1 + \beta_1)^j + (\alpha_1 + \beta_1)^{k-1} h_{t+1}, \quad k = 2, 3, \dots,$$

where h_{t+1} is a function of variables $\{y_t, y_{t-1}, \dots\}$ known at the time of forecasting.

In practice,

- the unknown parameters α_1 , β_1 , and ω have to be replaced by corresponding estimates.
- Unlike in ARMA models, the quantity being predicted is now unobserved, although it can be computed using GARCH model equation for all $t \geq 1$ as long as parameter values and required initial values for h_t and y_t are available. A common choice in practice is to use the sample variance of the observed time series as the initial value h_0 . In the stationary case, the effect of the initial values diminishes as t increases.
- Forecasting with a more general GARCH(r, s) model is carried out in principle in the same way as outlined above, although the resulting forecasting formulae become more cumbersome. Deriving interval predictions is also complicated, one major reason for this being that the distribution of the conditional variance deviates heavily from a Gaussian distribution.
- Overall, **volatility forecasting** is a separate and a large area in financial econometrics that we are not considering in this course more detail.

9.6 GARCH-in-mean model

In the **GARCH-in-mean (GARCH-M) model**, the conditional variance is allowed to directly affect the conditional mean as well.

To simplify the notation, let us consider the first-order special case, i.e., the AR(1)-GARCH(1,1)-M model

$$y_t = \nu + \phi_1 y_{t-1} + \delta g(h_t) + u_t, \quad h_t = \omega + \alpha_1 u_{t-1}^2 + \beta_1 h_{t-1},$$

where, as before, $u_t = y_t - \nu - \phi_1 y_{t-1} - \delta g(h_t)$, but now the conditional variance h_t also affects the level of y_t through the function $g(h_t)$. Depending on the situation, the “in-mean effect” can be defined as

- $g(h_t) = h_t$,
- $g(h_t) = \sqrt{h_t}$, or
- $g(h_t) = \log(h_t)$.

A positive coefficient δ means that the value of y_t increases when the conditional variance increases.

- As an extension of the GARCH-M version presented above, the orders of the AR and GARCH models can naturally be greater than 1.
- In general, instead of the AR(p) model, another suitable model for the conditional mean of y_t can be used.
- Similarly, a model other than GARCH(1,1) can be chosen for the GARCH part.

The GARCH-M model is used, for example, in financial econometrics (empirical finance) to model the fundamental risk-return relation, where risk (here volatility, measured by the conditional variance or its transformation) is allowed to directly affect the expected return of a security.

Empirical example (continue). Let us consider an GARCH-in-mean extension of the AR(1)-GARCH(1,1) model (with Gaussian innovations) obtained above for the NASDAQ 100 excess stock returns. That is we consider the following model for the conditional mean

$$y_t = \nu + \phi_1 y_{t-1} + \delta \sqrt{h_t} + u_t.$$

That is, we include the conditional standard deviation in the conditional mean.

- This is one possible way to examine the fundamental risk-return relationship in (excess) stock returns (here NASDAQ 100 index). The positive risk-return relation is the cornerstone of financial economics, as postulated, e.g., by the intertemporal capital asset pricing model (ICAPM). However, the existing empirical findings have been highly ambiguous: Different GARCH-M models have led to both positive and negative relationships (see, e.g., Engle, Lilien and Robins, 1987; Glosten et al., 1993; Scruggs, 1998; Lanne and Saikkonen, 2006; Nyberg, 2012).

QMLE-based estimation result obtained with the rugarch package on AR(1)-GARCH(1,1)-M model:

```
Robust Standard Errors:
      Estimate Std. Error  t value Pr(>|t|)
mu      0.000963   0.043060  0.022373 0.982151
ar1     -0.043796   0.012596 -3.477072 0.000507
archm    0.085665   0.040331  2.124019 0.033669
omega    0.035116   0.006429  5.462160 0.000000
alpha1   0.104836   0.012296  8.526169 0.000000
beta1    0.875948   0.012702 68.958838 0.000000
```

Therefore, it appears that the estimated coefficient δ , here “archm”, is positive and also statistically significant at the 5% significance level based on the robust t -value ($2.12 > 1.96$). The statistical significance is not very strong, however, so the result gives only tentative support for a positive risk-return relation. Note also that the estimate of the constant term ν (“mu”) is no longer significant, which is typical once the in-mean term is included.

- The residual diagnostics of this GARCH-M model are essentially the same as those obtained without the in-mean effect, and hence the model seems adequate.

Chapter 10

Multivariate time series models

So far, we have considered univariate models, such as AR and ARMA models, where only a single time series is analyzed.

- This represents the “pure time series approach” as formulated at the beginning of the material.

Modelling multiple time series simultaneously is, however, of greater interest in different applications. Multivariate time series models can provide richer dynamics and predictive information, potentially improving the accuracy of empirical analyses such as forecasting.

We will next briefly introduce three important extensions to the univariate models that have been widely used in the econometric (statistics) literature:

- Linear (predictive) regressions with time series variables
- Vector autoregressive (VAR) models
- Cointegration (in connection to nonstationary time series)

10.1 Predictive regressions: Background and starting point

Let us assume, for simplicity, that throughout this section we are working with two sets of variables: y_t remains the dependent variable and either one ($M = 1$) or, more generally, multiple ($M > 1$) additional predictive variables ($x_{1t}, \dots, x_{Mt}$). We are mainly interested in a linear regression model, but now

with time series, where the lags of predictive variables are used as regressors (predictors). For example, one possible model specification is

$$y_t = \beta_0 + x_{1,t-1}\beta_1 + \cdots + x_{M,t-1}\beta_M + u_t \equiv \mathbf{x}'_t\boldsymbol{\beta} + u_t,$$

where $\mathbf{x}_t = (1, x_{1,t-1}, \dots, x_{M,t-1})'$ collects the constant and the predictors, $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_M)'$, and u_t is an iid error term or at least serially uncorrelated and uncorrelated with the regressors $\mathbf{x}_t$, with $\text{Var}(u_t) = \sigma^2$. The parameters of this type of model, contained in $\boldsymbol{\beta}$, can be estimated by (conditional) least squares in a similar fashion as discussed, e.g., in connection with the AR(p) model in Section 6. That is, by minimizing the OLS criterion (with the required initial values)

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \sum_{t=1}^T (y_t - \mathbf{x}'_t\boldsymbol{\beta})^2,$$

resulting in the OLS estimator (and estimates)

$$\hat{\boldsymbol{\beta}} = \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}'_t \right)^{-1} \sum_{t=1}^T \mathbf{x}_t y_t.$$

In contrast to the model specification above, at times we are also considering other linear regressions:

- Simultaneous values of the predictive variables—rather than their lags—can be used as regressors in certain circumstances and applications. As an example, and for simplicity, let us assume $M = 1$. In this case, the model can be specified as:

$$y_t = \beta_0 + x_{1t}\beta_1 + u_t.$$

- Including also the lags of y_t on the right hand side of the model equation. These are then autoregressive models with additional predictive variables, often denoted by ARX or ADL (Autoregressive Distributed Lag) models. One example of such a model is

$$y_t = \beta_0 + \phi_1 y_{t-1} + \beta_1 x_{1,t-1} + u_t,$$

which can be straightforwardly extended by allowing more lags of y_t and x_{1t} .

- Including an autocorrelated error term is also possible. This alternative, particularly common in the earlier econometric literature, allows us to specify, e.g.,

$$u_t = \phi u_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{iid}(0, \sigma^2)$$

(see, e.g., Verbeek, Chapters 4.6-4.7). Note that by including lags of y_t in the model, we can attempt to account for the possible autocorrelation in the error term.

Subsequent (sub)sections are organized as follows:

- We consider the case where all the (dependent and predictive/explanatory) variables are stationary variables.
- At end of this section, we also briefly introduce the case where M is high-dimensional. That is, the number of predictive variables can be very large. In contrast, throughout the rest of this section, we restrict ourselves to the case where there is only one or only a few predictive variables (i.e. M is relatively small).
- When there are also nonstationary variables present (Section 13).

10.2 Stationary variables

Let us start with the case (assumption) that all variables y_t and $x_{1t}, \dots, x_{Mt}$ are stationary, which means that their respective time series plots do not contain strong stochastic or deterministic trends. Broadly speaking then, under the following two conditions (i) and (ii), the OLS estimator $\hat{\beta}$ of the parameters of interest in a linear regression model

$$y_t = \mathbf{x}_t' \beta + u_t,$$

where $\mathbf{x}_t = (x_{1t}, \dots, x_{Mt})'$ or, for example, $\mathbf{x}_t = (x_{1,t-1}, \dots, x_{M,t-1})'$, is consistent and asymptotically normal when

- (i) The error term u_t is uncorrelated with the regressors included in $\mathbf{x}_t$, that is, $E(\mathbf{x}_t u_t) = \mathbf{0}$, and it is serially uncorrelated,
- (ii) All the regressors in $\mathbf{x}_t$ are either deterministic or stationary random variables.

If not otherwise mentioned, we assume that (i) and (ii) are valid.

- The essential condition for consistency is the orthogonality $E(\mathbf{x}_t u_t) = \mathbf{0}$ in (i). Serial uncorrelatedness of u_t is, in addition, needed for the *conventional* covariance matrix of $\hat{\beta}$ to be valid (see the discussion below).

After estimating the model, the same residual diagnostic methods used for ARMA models can be applied to assess the residuals and, consequently, the adequacy of the model. The resulting asymptotic distribution result for the OLS estimator is

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow[d]{} N(\mathbf{0}, \sigma^2 \mathbf{Q}^{-1}), \quad \mathbf{Q} = \text{plim}_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t',$$

where $\xrightarrow[d]{} \rightarrow$ denotes convergence in distribution.

- In practice the result is used in the “unscaled” form familiar from Section 6, that is, $\text{Cov}(\hat{\beta}) \approx \hat{\sigma}^2 (\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t')^{-1}$, whose diagonal elements give the (squared) approximate standard errors, and t -ratios and confidence intervals are formed exactly as before.

It turns out that, in fact, even (mild) autocorrelation of the residuals can be accepted without losing many of the useful asymptotic properties of the (conditional) OLS estimator: the estimator remains consistent and asymptotically normal, only the covariance matrix above is no longer valid. If autocorrelation occurs after the initial formation of the model, we can proceed in essentially two different ways (see detailed discussion below):

- form a (linear) model where the error term is autocorrelated.
- (preferable alternative) adjust the covariance matrix of the estimated parameters $\hat{\beta}$ (see below)
- An important exception: the above tolerance of autocorrelation requires that $\mathbf{x}_t$ does *not* contain lagged values of y_t . In an ARX or ADL model (see above), an autocorrelated u_t is correlated with the lagged y_t included in $\mathbf{x}_t$, condition (i) fails, and the OLS estimator becomes **inconsistent**. Then HAC standard errors do not help, and the model has to be respecified (e.g., by adding lags), exactly as discussed in the case of the AR(p) model in Section 6.

i Extra: Remaining autocorrelation in the linear (predictive) regression

Consider a linear regression (as above), but this time we specify an ARMA structure, such as an MA(1) structure, for the error term. This would lead to the model

$$y_t = \mathbf{x}_t' \beta + u_t, \quad u_t = a_t + \theta_1 a_{t-1},$$

where a_t is now an iid or white noise (wn) process. However, in macro and especially financial econometrics, it is typical to proceed using the latter option, which is discussed next.

As stated, and provided that $\mathbf{x}_t$ contains no lags of y_t , the OLS estimator can be shown to be still consistent and asymptotically normal even if the error term is autocorrelated and possibly also (conditionally) heteroskedastic. Only the covariance matrix of the estimator changes. The logic is the same as with the quasi-maximum likelihood estimator (QMLE): the estimates are retained, but the inference is based on a robust covariance matrix.

- Cf., for example, the section concerning the AR-GARCH model.

In such a situation, a heteroskedasticity-autocorrelation consistent (HAC) covariance matrix estimator is used, indicating that the usual standard errors of the parameter estimates it produces can be replaced with HAC

counterparts.

- Without this adjustment, the standard errors are often too small, which correspondingly increases the absolute value of the t -test statistics testing the statistical significance of individual regression coefficients (reducing the p -values).
- One formulation is based on the Newey and West (1987) estimator.

Formally, let's consider the OLS estimator and the usual covariance matrix

$$\hat{\beta} = \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right)^{-1} \sum_{t=1}^T \mathbf{x}_t y_t, \quad \text{Cov}(\hat{\beta}) = \sigma^2 \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right)^{-1},$$

where $\text{Var}(u_t) = \sigma^2$. The general HAC estimator can be written as

$$\text{Cov}(\hat{\beta})_{HAC} = \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right)^{-1} \hat{\mathbf{C}}_{HAC} \left(\sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t' \right)^{-1},$$

where different choices for the middle term $\hat{\mathbf{C}}_{HAC}$ lead to different estimators. For example, in the case of the Newey and West (1987) estimator

$$\hat{\mathbf{C}}_{HAC} = \sum_{t=1}^T \hat{u}_t^2 \mathbf{x}_t \mathbf{x}_t' + \sum_{j=1}^l w_j \sum_{s=j+1}^T (\mathbf{x}_s \hat{u}_s \hat{u}_{s-j} \mathbf{x}_{s-j}' + \mathbf{x}_{s-j} \hat{u}_{s-j} \hat{u}_s \mathbf{x}_s'),$$

where l is the so-called bandwidth parameter and w_j is an appropriate weighting function.

- For example, in the case of the so-called Bartlett kernel function

$$w_j = 1 - j/l.$$

- Newey and West recommended choosing the parameter l as the integer part of the expression $4(T/100)^{2/9}$.

Furthermore, occasionally $\hat{\mathbf{C}}_{HAC}$ may also be specified (only) allowing for heteroskedasticity (but not autocorrelation, “HC” vs. “HAC”) with the alternative (this is the so-called White estimator)

$$\hat{\mathbf{C}}_{HC} = \sum_{t=1}^T \hat{u}_t^2 \mathbf{x}_t \mathbf{x}_t'.$$

10.3 Forecasting with predictive variables

Let us consider forecast construction in more detail. When attempting to predict the value of y_{t+h} using external predictive variables, additional complications

arise especially when the forecast horizon h lengthens. As an example, consider a simple model (the main arguments generalize to more general models) containing only two lags of one predictive variable x_t

$$y_t = \beta_0 + \beta_1 x_{1,t-1} + \beta_2 x_{1,t-2} + u_t.$$

As in the ARMA case, one-step forecast is the conditional expectation given the information set at time t . That is

$$E_t(y_{t+1}) = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{1,t-1}.$$

From this, it is immediately apparent that additional challenges arise if the forecast horizon lengthens beyond one period. In other words, multi-step forecasts ($h > 1$) seem to require forecasts for x_{1t} as well.

- Vector autoregressive models provide one alternative to solve this problem by building a joint (multiple-equation) model for y_t and x_{1t} (see the next section).

An alternative approach is to use predictive models that are specific to each forecast horizon. In this case, when utilizing the predictive information available at time t , we can specify

$$y_{t+h} = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{1,t-1} + e_{t+h},$$

where e_t is a zero-mean error term and the forecast can then be constructed “directly” as

$$E_t(y_{t+h}) = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{1,t-1}.$$

Naturally the parameters $\beta = (\beta_0, \beta_1, \beta_2)'$, and finally their estimates, are then specific to the forecast horizon h . This is often called the **direct** forecasting approach, as opposed to the **iterated** one based on a one-step model.

- The properties of the error term e_{t+h} can be complicated due to the overlapping nature of the forecast horizon, from the forecast origin at time t to $t+h$. Specifically, with $h > 1$ the error term is autocorrelated by construction (an $MA(h-1)$ structure), even when the model is correctly specified. The parameters can still be estimated consistently by OLS, but the standard errors have to be computed with the HAC methods described above.
- What is introduced above for a simple model containing only one predictive variable can be generalized straightforwardly to the case of M predictors, which may also include lags of y_t .

Overall, this same principle of h -period predictive models can be viewed as a building block for various extended predictive regressions, particularly those incorporating elements of **machine learning and statistical learning**.

10.4 Extra: Regularized predictive regressions

Let us briefly focus on predicting y_{t+h} at time t (that is the forecast origin) given the collection of M predictive variables in $\mathbf{x}_t = (x_{1t}, \dots, x_{Mt})'$, which may also contain some lagged values of y_t (as discussed above).

- This is the same predictive regression we briefly considered earlier, but now extended to the case of high-dimensional predictors.

The challenge of big dependent data. The focus here is on big dependent data, meaning we're dealing with a large collection of potential predictors where the number of variables, M , can be substantial.

- In some cases, M might even be greater than the sample length T ($M > T$) used for estimating the model parameters.
- Here for “big dependent data” we mean that the number of (predictive) time series is large. “Dependent” refers to autocorrelation (serial correlation) in the data. In this section, our dependent variable y_t is still scalar-valued (i.e. only one time series) and hence we consider single-equation models.

Traditional limitation. When M is large relative to T , traditional estimation techniques, like Ordinary Least Squares (OLS), as introduced above, fail or produce highly unstable and unreliable results.

- The classic OLS estimator is not well defined if $M > T$; the conventional asymptotic results assume $M \ll T$.
- Even if M is less than T but still large, the resulting model can suffer from overfitting, where it fits the noise in the historical data too closely, leading to poor out-of-sample forecasting performance.

Need for structure: To overcome **the curse of dimensionality**, the problems that arise when the number of variables grows, we need methods that can impose structure or constraints on the predictive relationship. This is essential for selecting the most relevant predictors and stabilizing the parameter estimates in a data-rich environment.

Regularized estimation and sparsity. To enable effective predictive regressions in this high-dimensional setting, where the predictor space can be very large, a class of techniques known as regularized estimators is employed. Regularization involves adding a penalty term to the standard loss function (like the sum of squared errors) that we aim to minimize. This penalty discourages the model from assigning large values to the coefficients, effectively shrinking them towards zero. Key examples (briefly without details in this course):

- **Ridge estimator:** This technique adds a penalty proportional to the sum of the squared coefficients. It stabilizes the estimates by shrinking all coefficients but doesn't set any exactly to zero.

- **LASSO** (Least Absolute Shrinkage and Selection Operator): This method uses a penalty based on the sum of the absolute values of the coefficients. Crucially, the LASSO has the ability to perform automatic variable selection by setting the coefficients of irrelevant predictors from $\mathbf{x}_t$ exactly to zero. This produces a sparse model, meaning that only a few (the most important) predictors are ultimately used.
- **Elastic Net**: This is a hybrid that combines the penalties of both ridge and LASSO. It's often used to leverage the variable selection of LASSO while retaining the grouping effect and stability of Ridge, especially when predictors are highly correlated.

Relevance: These estimators are foundational in modern forecasting and (statistical) machine learning, particularly when integrating vast amounts of data, such as financial indicators, survey data, or text-based predictors, into an economic forecasting model. They provide a principled way to manage the trade-off between bias (from shrinkage) and variance (from the large number of predictors) to achieve superior out-of-sample forecasting accuracy.

A more detailed treatment of this topic and these estimators is reserved for the **Advanced Time Series Econometrics course**.

Chapter 11

Basics of vector autoregression

Univariate ARMA models can be generalized to the multivariate case where **multiple time series are analyzed simultaneously**. In this course, we concentrate on a brief introduction of basic ideas of a **vector autoregressive (VAR) model**. The VAR model provides a very common framework for analyzing time series dynamics, such as the effects of random shocks (e.g., changes in economic policy or technology).

- The statistical details of the VAR model are considered in more detail in the **Advanced Time Series Econometrics** course, and its very important role in applied macroeconometrics and empirical macroeconomics in the **Macroeconometrics** course.

11.1 VAR(p) process

Let us start with a bivariate model. That is, we have two stationary time series y_{1t} and y_{2t} with $E(y_{1t}) = \mu_1$ and $E(y_{2t}) = \mu_2$. A **bivariate (i.e., a two-variable) first-order VAR process** (i.e., a bivariate **VAR(1) process**) is given as

$$\begin{aligned}y_{1t} &= \nu_1 + a_{11}y_{1,t-1} + a_{12}y_{2,t-1} + u_{1t} \\y_{2t} &= \nu_2 + a_{21}y_{1,t-1} + a_{22}y_{2,t-1} + u_{2t},\end{aligned}$$

where u_{1t} and u_{2t} are two error terms (independent of the history of y_{1t} and y_{2t}) that may be correlated. In a matrix notation, the VAR(1) process can be

rewritten

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}.$$

Equivalently,

$$\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \mathbf{u}_t,$$

where now $\mathbf{y}_t = (y_{1t}, y_{2t})'$, $\boldsymbol{\nu} = (\nu_1, \nu_2)'$ and $\mathbf{u}_t = (u_{1t}, u_{2t})'$ are two 2×1 vectors.

- In this notation, likewise throughout this material, with bolded font in mathematical notations we emphasize vectors and matrices, to distinguish scalar-valued components and processes.

As an extension of two-variable and first-order case, **K -variable VAR(p) process** is

$$\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$$

where $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})'$ is a $K \times 1$ vector of variables and $\mathbf{A}_1, \dots, \mathbf{A}_p$ are $K \times K$ coefficient matrices. In other words, the VAR(p) process depends on the p lags of $y_{1t}, \dots, y_{Kt}$. The error term (“shock” or “innovation”) $\mathbf{u}_t = (u_{1t}, \dots, u_{Kt})'$ is an unobserved $K \times 1$ (vector) iid process.

- Hence $\mathbb{E}(\mathbf{u}_t) = \mathbf{0}$, $\mathbb{E}(\mathbf{u}_t \mathbf{u}_t') = \boldsymbol{\Sigma}_u$, and $\mathbb{E}(\mathbf{u}_t \mathbf{u}_s') = \mathbf{0}$, $s \neq t$.
- The bivariate and more general model show the basic idea: The idea of the VAR(p) model is to regress each component of y_t on its own lags and on the lags of the other components. The model can describe lagged or dynamic dependencies among variables. For example, one may ask how y_{2t} affects the future path of y_{1t} , and other way round.

A companion form: The VAR(p) process can be rewritten

$$\mathbf{Y}_t = \boldsymbol{\nu} + \mathbf{A} \mathbf{Y}_{t-1} + \mathbf{U}_t,$$

where $\mathbf{Y}_t = [\mathbf{y}_t' \cdots \mathbf{y}_{t-p+1}']'$ and $\boldsymbol{\nu} = [\boldsymbol{\nu}' \mathbf{0}' \cdots \mathbf{0}']'$ and $\mathbf{U}_t = [\mathbf{u}_t' \mathbf{0}' \cdots \mathbf{0}']'$ are $(Kp \times 1)$ vectors, and

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \cdots & \mathbf{A}_{p-1} & \mathbf{A}_p \\ \mathbf{I}_K & \mathbf{0}_K & \cdots & \mathbf{0}_K & \mathbf{0}_K \\ \mathbf{0}_K & \mathbf{I}_K & \cdots & \mathbf{0}_K & \mathbf{0}_K \\ \vdots & & \ddots & \vdots & \vdots \\ \mathbf{0}_K & \mathbf{0}_K & \cdots & \mathbf{I}_K & \mathbf{0}_K \end{bmatrix} \quad (Kp \times Kp)$$

In other words,

$$\begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \vdots \\ \mathbf{y}_{t-p+1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\nu} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \cdots & \mathbf{A}_{p-1} & \mathbf{A}_p \\ \mathbf{I}_K & \mathbf{0}_K & \cdots & \mathbf{0}_K & \mathbf{0}_K \\ \vdots & & \ddots & \vdots & \vdots \\ \mathbf{0}_K & \mathbf{0}_K & \cdots & \mathbf{I}_K & \mathbf{0}_K \end{bmatrix} \begin{bmatrix} \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \vdots \\ \mathbf{y}_{t-p} \end{bmatrix} + \begin{bmatrix} \mathbf{u}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}$$

where the first block row reproduces the VAR(p) equation $\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \mathbf{u}_t$ and the remaining block rows are the identities $\mathbf{y}_{t-1} = \mathbf{y}_{t-1}, \dots, \mathbf{y}_{t-p+1} = \mathbf{y}_{t-p+1}$. In other words, any VAR(p) process can be written using a Kp -dimensional VAR(1) presentation.

Lag-polynomial presentation of the VAR(p). Define again the lag-operator as $B^k \mathbf{x}_t = \mathbf{x}_{t-k}$, $k = 0, \pm 1, \pm 2, \dots$, where $\mathbf{x}_t$ might be a vector. Therefore, using the lag-operators, the VAR(p) process can be rewritten

$$\mathbf{A}(B)\mathbf{y}_t = \boldsymbol{\nu} + \mathbf{u}_t,$$

where $\mathbf{A}(B) = (\mathbf{I}_K - \mathbf{A}_1B - \cdots - \mathbf{A}_pB^p)$.

Stationarity (stability) of the VAR(p) process. A VAR(p) process is (strictly) stationary (stable) (given $\mathbf{u}_t \sim \text{iid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$) if the roots of the determinant

$$\det(\mathbf{I}_K - \mathbf{A}_1z - \cdots - \mathbf{A}_pz^p) = 0$$

are outside the unit circle in absolute value (i.e., $|z| > 1$).

Notice, however, that many statistical software packages, including the **vars** package in R, use an alternative but equivalent presentation.

- When using the companion form of the VAR(p), the stability of the VAR(p) process is determined by the eigenvalues of the companion matrix $\mathbf{A}$.
- In **vars** package” and **VAR-function**, the part “Roots of the characteristic polynomial” reported by R are precisely these eigenvalues. The key relationship is that these eigenvalues of the companion matrix are the reciprocals of the roots of the determinant polynomial. Therefore, the two conditions are equivalent:

- Roots of $\det(\mathbf{I}_K - \mathbf{A}_1z - \cdots - \mathbf{A}_pz^p) = 0$ are outside the unit circle ($|z| > 1$).
- Eigenvalues of the companion matrix $\mathbf{A}$ are inside the unit circle ($|\lambda| < 1$). That is, the eigenvalues of the matrix $\mathbf{A}$ satisfy $\det(\mathbf{I}_K\lambda^p - \mathbf{A}_1\lambda^{p-1} - \cdots - \mathbf{A}_p) = 0$.

- So, when using R's output, we should check if the moduli of the reported roots (i.e. the eigenvalues) are **all less than 1** to confirm that the VAR model is stable.

Linear process. As with the $AR(p)$ process, the stationary (stable) $VAR(p)$ process can be written

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \mathbf{u}_{t-j} = \boldsymbol{\mu} + \boldsymbol{\Psi}(B) \mathbf{u}_t,$$

where $\boldsymbol{\mu} = E(\mathbf{y}_t) = [\mathbf{A}(1)]^{-1} \boldsymbol{\nu} = (\mathbf{I}_K - \mathbf{A}_1 - \dots - \mathbf{A}_p)^{-1} \boldsymbol{\nu}$ and $\boldsymbol{\Psi}(B) = [\mathbf{A}(B)]^{-1} = \boldsymbol{\Psi}_0 + \boldsymbol{\Psi}_1 B + \boldsymbol{\Psi}_2 B^2 + \dots$ is a matrix-valued lag polynomial.

In addition to this linear process presentation, the sequence $\boldsymbol{\Psi}_0, \boldsymbol{\Psi}_1, \boldsymbol{\Psi}_2, \dots$ has another important role in macroeconometrics. It turns out that the matrix $\boldsymbol{\Psi}_s$ ($\boldsymbol{\Psi}_0 = \mathbf{I}_K$) measures the effect of one-unit increase in u_{jt} upon $y_{i,t+s}$, $i, j = 1, \dots, K$, or equivalently

$$\boldsymbol{\Psi}_s = \frac{\partial \mathbf{y}_{t+s}}{\partial \mathbf{u}'_t}.$$

The elements of the matrices $\boldsymbol{\Psi}_0, \boldsymbol{\Psi}_1, \boldsymbol{\Psi}_2, \dots$, viewed as functions of the horizon s , constitute the **(forecast error) impulse response function** (IRF) of the $VAR(p)$ process.

- In other words, the row i , column j element of $\boldsymbol{\Psi}_s$ gives the effect of a one-unit increase in the error term of the j th variable at time t on the i th variable at time $t + s$, holding all other errors constant.
- For instance, the first column of $\boldsymbol{\Psi}_1$ gives the effect in period 1 of a unit increase in the error term of the first variable (first shock) in period 0 on each variable in the system.
- Especially in (structural) macroeconometrics, the VAR model is extended by economic identification assumption, which leads to **structural VAR (SVAR) models** and identified impulse response functions. These will be considered more detail in **Macroeconometrics** course.

As in the univariate case, stability implies $\boldsymbol{\Psi}_s \rightarrow \mathbf{0}$ as $s \rightarrow \infty$, so that the effects of a shock decay over time. In empirical analysis, the unknown $\boldsymbol{\Psi}_s$ must also be replaced by their estimates, obtained from the estimation result of the VAR model.

VAR(p) model: Estimation and model selection. The stationary $VAR(p)$ model can be estimated by ordinary least squares (OLS).

- More specifically, estimation can be done equation by equation by OLS (cf. the model structure).

- Equation-by-equation OLS is efficient here, because every equation contains exactly the same regressors.
- If the distribution of $\mathbf{u}_t$ is known (assumed to be known), the model can also be estimated by the method of maximum likelihood (ML). Under normality assumption, that is $\mathbf{u}_t \sim \text{nid}(\mathbf{0}, \boldsymbol{\Sigma}_u)$, it can be shown that the conditional ML estimator is numerically equivalent to the OLS estimator.
- It can be shown that the OLS estimator is consistent and asymptotically normally distributed. Thus, standard errors, t -ratios, and confidence intervals are obtained and interpreted exactly as in the univariate case of Section 6, and (joint) restrictions can be tested with Wald or LR tests.

VAR(p): Model selection and residual diagnostics

Overall, the model selection can be based on the specification cycle presented in the case of univariate AR models but modified suitable to VARs. However, due to multiple time series and hence more complicated dynamic interrelationships between the variables, autocorrelation functions cannot be directly used to select the lag length p for the VAR(p).

- Especially information criteria, such as AIC and BIC, can be used like in the univariate case.

$$- \text{AIC}(p): \log \left(\det(\widehat{\boldsymbol{\Sigma}}_u(p)) \right) + \frac{2}{T} K(Kp + 1)$$

$$- \text{BIC}(p): \log \left(\det(\widehat{\boldsymbol{\Sigma}}_u(p)) \right) + \frac{\log(T)}{T} K(Kp + 1)$$

where $\widehat{\boldsymbol{\Sigma}}_u(p)$ is the estimated residual covariance matrix of the VAR(p) model and $K(Kp + 1)$ is the total number of estimated parameters (intercepts included). For the competing models (recall that exactly the same sample period must be used!) the one with the lowest value of the selected information criterion is the selected (preferable) model.

- Sequential testing procedure (cf. the sequential testing for univariate models) can also be used to determine the lag length p .

In practice, at times the lag length selection corresponds to the frequency of the data. That is, for the quarterly data $p = 4$ and for monthly data $p = 12$.

The adequacy of the selected and estimated VAR(p) model can be evaluated by diagnostic checks (like for ARMA models). A well-specified VAR model should capture all the systematic and linear dynamics present in the (multivariate) time series data. The primary goal of these checks is to determine if there is any significant autocorrelation left in the vector of residuals

$$\widehat{\mathbf{u}}_t = \mathbf{y}_t - \widehat{\mathbf{y}}_t,$$

where $\hat{\mathbf{y}}_t$ are the fitted values of the selected VAR model.

To formally test for remaining autocorrelation in the residuals, we use a multivariate extension of the Ljung-Box test. This is called the **Portmanteau test**. It jointly examines whether a group of residual autocovariance matrices up to lag length H are statistically different from zero. The **null hypothesis** (H_0) of the test is that there is no serial correlation in the residuals up to a specified lag H . The Portmanteau test statistic, often denoted as Q_H , is calculated as follows:

$$Q_H = T \sum_{h=1}^H \text{tr}(\hat{\mathbf{C}}'_h \hat{\mathbf{C}}_0^{-1} \hat{\mathbf{C}}_h \hat{\mathbf{C}}_0^{-1}),$$

where $\hat{\mathbf{C}}_j = \frac{1}{T} \sum_{t=j+1}^T \hat{\mathbf{u}}_t \hat{\mathbf{u}}'_{t-j}$ is the estimated residual autocovariance matrix at lag j and $\text{tr}(\cdot)$ is the trace operator, which sums the diagonal elements of a matrix.

- Under the null hypothesis of no serial correlation, the Q_H statistic approximately follows a χ^2 -distribution with $K^2(H - p)$ degrees of freedom (so that $H > p$ must be chosen) and the associated p -value can be constructed.
- The degrees of freedom are reduced by $K^2 p$ because the statistic is computed from the residuals of an estimated VAR(p) model; cf. the corresponding adjustment in the univariate case. In the **vars** package in R, the function **serial.test** reports this test, and the option **type = "PT.adjusted"** applies a small-sample correction to the statistic.
- Should the model fail in the Portmanteau test or overall in diagnostic checking, the model should be respecified (if possible), such as to increase the lag length of the model (e.g., from VAR(p) to VAR($p + 1$) or higher) and re-estimate it. The logic is that by including more lags, the model can better capture the complex temporal dependencies in the data. After respecification, the diagnostic checks must be performed again to ensure the new model is adequate.
- However, **in practice**, especially with complex multivariate data, it is often challenging to find a model that can be considered completely adequate from a diagnostic perspective. When the diagnostic checks indicate that the model is misspecified (for instance, due to some persistent residual autocorrelation), the standard assumptions for maximum likelihood estimation are violated. In such cases, the parameter estimates can be interpreted as Quasi-Maximum Likelihood Estimates (**QMLE**), a concept that was introduced on a general level in the AR-GARCH section and can be extended to the VAR models (with necessary changes in notation etc.).

VAR(p): Forecasting

Similarly as with the univariate AR(p) model, forecasts can be obtained as conditional expectations, conditional on the information at time t . The properties

CEV1–CEV4 of the conditional expectation can be extended to the vector-valued case straightforwardly (element-by-element treatment).

- One-period-ahead forecasts with the VAR(p):

$$\mathbf{E}_t(\mathbf{y}_{t+1}) = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{y}_t + \cdots + \mathbf{A}_p \mathbf{y}_{t-p+1}.$$

- Multiperiod h -step forecasts can be obtained iteratively following the recursion (cf. the AR(p) case):

$$\mathbf{E}_t(\mathbf{y}_{t+h}) = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{E}_t(\mathbf{y}_{t+h-1}) + \cdots + \mathbf{A}_p \mathbf{E}_t(\mathbf{y}_{t+h-p}),$$

where $\mathbf{E}_t(\mathbf{y}_{t+h-j}) = \mathbf{y}_{t+h-j}$ for $h \leq j$.

Similarly as in the univariate AR models, the VAR model above is assumed to include the intercept term $\boldsymbol{\nu}$, and hence a nonzero mean vector. Alternatively, one can specify the VAR model for the demeaned time series and add the mean vector $\boldsymbol{\mu}$ back to the forecasts obtained from the model estimated with the demeaned data.

Empirical example. Consider an application of a VAR model to the classic three-variable case (as described, e.g., in Stock and Watson, 2001), which is a benchmark in macroeconomic and especially monetary policy analysis.

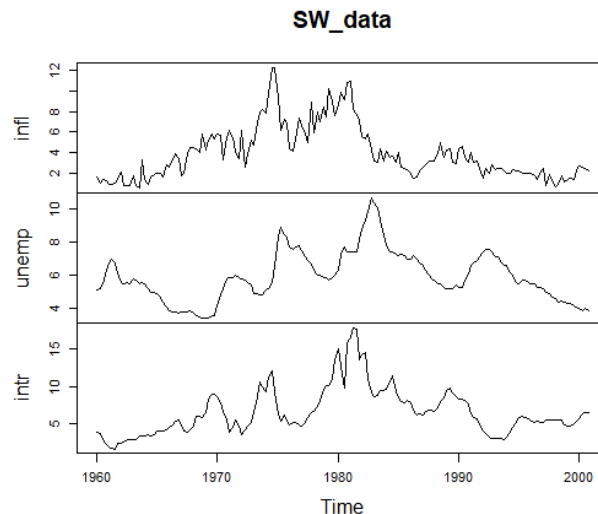
- See J.H. Stock and M.W. Watson (2001). Vector autoregressions. *Journal of Economic Perspectives*, 15(4), 101–115.

The VAR model contains the following three ($K = 3$) key quarterly U.S. macroeconomic variables (source: FRED):

- Inflation: Measured as the percent change from the preceding period in the Gross Domestic Product Implicit Price Deflator, expressed at a seasonally adjusted annual rate.
- Unemployment: The seasonally adjusted quarterly average of the civilian unemployment rate.
- Interest Rate: The quarterly average of the effective Federal Funds Rate, which is not seasonally adjusted.

These three series are modeled as an interdependent system where each variable is influenced by its own past values and the past values of the other two variables. The following analysis covers the sample period from 1960:Q1 to 2000:Q4 as in Stock and Watson (2001).

Figure: U.S. macroeconomic time series (inflation, unemployment rate and interest rate). See Stock and Watson (2001). Source of data: FRED.



Lag length selection: For this analysis and illustration, we follow Stock and Watson (2001) and specify a VAR(4) model. This choice is common in quarterly macroeconomic data as it captures potential year-over-year dynamics.

Parameter estimation: The VAR(4) model is estimated assuming the error terms are normally distributed.

- A key feature of VAR models is the large number of parameters. In this three-variable VAR(4) model, each of the three equations includes an intercept and four lags of all three variables, resulting in $3 \times (1 + 4 \times 3) = 39$ parameters to be estimated. Due to this high dimensionality, interpreting individual coefficient estimates is often impractical. Instead, analysis typically focuses on system-wide properties through Granger causality tests, as well as impulse response functions and forecast error variance decompositions (see Macroeconometrics course)

Residual diagnostics:

- The Portmanteau test for serial correlation in the residuals indicates some remaining unmodelled dynamics. The test's statistically significant result suggests that the VAR(4) specification may not fully capture all the (linear) dependencies within the data.
- While the residuals from the different variable equations appear to have low cross-correlation, an analysis of the squared residuals reveals a different pattern. The presence of autocorrelation in the squared residuals is an indicator of conditional heteroskedasticity. This means the volatility of the shocks is not constant over time. While the VAR(4) model may adequately capture the conditional mean of the variables, it fails to account for this

time-varying variance in the error terms. A more advanced model, such as a VAR-GARCH (not considered in this course), would be required to model volatility dynamics.

i Extra: Details of the estimated VAR(4) model

VAR Estimation Results:

=====

Endogenous variables: Inflation, Unemployment, Fedfunds

Deterministic variables: const

Sample size: 160

Log Likelihood: -396.638

Roots of the characteristic polynomial:

0.9696 0.9696 0.7928 0.7928 0.686 0.686 0.5674 0.5674 0.4609 0.4609 0.2053 0.2053

Call:

VAR(y = SW_data, p = 4, type = c("const"), exogen = NULL, lag.max = NULL)

Estimation results for equation Inflation:

=====

Inflation = Inflation.l1 + Unemployment.l1 + Fedfunds.l1 + Inflation.l2 + Unemployment.l2 +

	Estimate	Std. Error	t value	Pr(> t)	
Inflation.l1	0.5886863	0.0821552	7.166	3.47e-11	***
Unemployment.l1	-0.8356913	0.4006087	-2.086	0.03870	*
Fedfunds.l1	0.2370805	0.1082335	2.190	0.03007	*
Inflation.l2	0.0902554	0.0937902	0.962	0.33747	
Unemployment.l2	1.3545379	0.6841360	1.980	0.04958	*
Fedfunds.l2	-0.2168750	0.1453158	-1.492	0.13773	
Inflation.l3	0.1205897	0.0944654	1.277	0.20377	

11.2 Granger causality in VARs

Granger causality is a concept used to determine whether one time series can predict another. It doesn't imply true causality in the philosophical or strict scientific sense, but rather predictive causality based on temporal dependence and informational content.

- Granger causality can be examined for single-equation models, such as the ones considered in the previous section examining whether the additional predictor Granger causes y_t .
- Granger causality is most often studied in the context of VAR models.

Bivariate VAR model ($K = 2$): Granger-causality can be easily tested within a VAR model. Consider first, for simplicity, a bivariate VAR(p) process

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix} + \sum_{j=1}^p \begin{bmatrix} a_{11,j} & a_{12,j} \\ a_{21,j} & a_{22,j} \end{bmatrix} \begin{bmatrix} y_{1,t-j} \\ y_{2,t-j} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}.$$

If $a_{12,j} = 0$, $j = 1, 2, \dots, p$, that is the restriction

$$a_{12,1} = a_{12,2} = \dots = a_{12,p} = 0,$$

then the lags of y_{2t} do not help to forecast y_{1t} and there is no Granger-causality from y_{2t} to y_{1t} (that is, y_{2t} is not Granger-causal for y_{1t}).

- When working with an estimated VAR model, this hypothesis of no Granger causality from y_{2t} to y_{1t} **can be tested** by the Wald or likelihood ratio (LR) test, which follows the χ_p^2 -distribution under the null hypothesis (p zero restrictions).
- At times, such as in the `vars` package in R, the F -test statistic is used instead of the Wald test statistic, in which case the critical values come from the F -distribution. The two are asymptotically equivalent, but the F -version tends to behave better in small samples.

Larger VAR models ($K > 2$): When the VAR model contains more than two variables, Granger causality between two variables cannot be tested by testing zero-restrictions in a straightforward manner (as above). As an example, consider the following three-variable VAR process

$$\begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \end{bmatrix} = \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix} + \sum_{j=1}^p \begin{bmatrix} a_{11,j} & a_{12,j} & a_{13,j} \\ a_{21,j} & a_{22,j} & a_{23,j} \\ a_{31,j} & a_{32,j} & a_{33,j} \end{bmatrix} \begin{bmatrix} y_{1,t-j} \\ y_{2,t-j} \\ y_{3,t-j} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \end{bmatrix}.$$

If, for instance, $a_{12,1} = \dots = a_{12,p} = 0$, the lags of y_{2t} do not help forecasting y_{1t} one period ahead. However, because the lags of y_{2t} may affect y_{3t} , which, in turn, affects y_{1t} , the lags of y_{2t} may help forecasting $y_{1,t+1}$, $y_{1,t+2}$ etc. In other words, there may be Granger causality through indirect effects.

Generally, Granger causality from one **group of variables** to another is defined similarly. Partition $\mathbf{y}_t = [\mathbf{y}'_{1t} \ \mathbf{y}'_{2t}]'$, where $\mathbf{y}_{1t}$ is $K_1 \times 1$ and $\mathbf{y}_{2t}$ is $K_2 \times 1$, ($K_1 + K_2 = K$), and write the VAR(p) process

$$\begin{bmatrix} \mathbf{y}_{1t} \\ \mathbf{y}_{2t} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\nu}_1 \\ \boldsymbol{\nu}_2 \end{bmatrix} + \sum_{j=1}^p \begin{bmatrix} \mathbf{A}_{11,j} & \mathbf{A}_{12,j} \\ \mathbf{A}_{21,j} & \mathbf{A}_{22,j} \end{bmatrix} \begin{bmatrix} \mathbf{y}_{1,t-j} \\ \mathbf{y}_{2,t-j} \end{bmatrix} + \begin{bmatrix} \mathbf{u}_{1t} \\ \mathbf{u}_{2t} \end{bmatrix},$$

where the blocks $\mathbf{A}_{\cdot,j}$ are generally matrices (but of course may reduce to scalar-valued components in certain situations), and $\mathbf{u}_{1t}$ and $\mathbf{u}_{2t}$ are the corresponding partitions of the error vector $\mathbf{u}_t$

- If, for instance, $\mathbf{A}_{12,j} = \mathbf{0}$, $\forall j = 1, \dots, p$, then the variables in $\mathbf{y}_{2t}$ are not Granger-causal for the variables in $\mathbf{y}_{1t}$.

The hypothesis of interest

$$\mathbf{A}_{12,1} = \mathbf{A}_{12,2} = \dots = \mathbf{A}_{12,p} = \mathbf{0}$$

imposes $K_1 K_2 p$ zero restrictions and can be tested by the Wald or likelihood ratio (LR) test.

- Under the null hypothesis, the Wald and LR tests are asymptotically $\chi^2_{K_1 K_2 p}$ -distributed, and large values of the test statistics are critical for the null hypothesis.
- The above restrictions and test statistics apply also when a constant term is included in the VAR(p).
- As mentioned, at times F -test statistic is used where the number of restrictions (the first degree-of-freedom parameter in the F -distribution) is the above number of restrictions.

Example: Three-variable process/model. Consider the VAR(1) process/model:

$$\mathbf{y}_t = \begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{bmatrix} \mathbf{y}_{t-1} + \mathbf{u}_t.$$

Partitioning $\mathbf{y}_t = [y_{1t} \ \mathbf{y}'_{2t}]'$ so that $\mathbf{y}_{2t} = [y_{2t} \ y_{3t}]'$, $\mathbf{y}_{2t}$ does not Granger cause $\mathbf{y}_{1t} = y_{1t}$ because the matrix $\mathbf{A}_{12,1} = \mathbf{0}$. On the other hand, y_{1t} Granger causes $\mathbf{y}_{2t}$.

Instantaneous causality. In the bivariate VAR, there is no instantaneous causality between y_{1t} and y_{2t} if the errors u_{1t} and u_{2t} are uncorrelated. In general (not necessarily related to the VAR), a variable y_{1t} is said to be instantaneously causal for another variable y_{2t} , if knowing the value of y_{1t} in the forecast period helps to improve forecasts of y_{2t} .

It can be proved that instantaneous causality is symmetric so that if y_{1t} causes y_{2t} , then also y_{2t} causes y_{1t} . That is the concept is symmetric: This definition does not account for the “direction” of causality which must be determined from other sources.

- For instance, economic theory may suggest that the (instantaneous) causality runs only in one direction. Then significant correlation can be interpreted in favour of a causal relation.

In systems with more than two variables, there is no instantaneous causality between two groups of variables if the covariance matrix of the errors, Σ_u , is block diagonal.

- As an example, if the covariance matrix in a VAR process for $\mathbf{y}_t = (y_{1t}, y_{2t}, y_{3t})'$ is

$$\Sigma_u = \begin{bmatrix} 2.0 & 0 & 0 \\ 0 & 1.5 & 0.4 \\ 0 & 0.4 & 1.0 \end{bmatrix},$$

then there is no instantaneous causality between y_{1t} and $\mathbf{y}_{2t} = (y_{2t}, y_{3t})'$.

- Instantaneous causality can be tested by a Wald type of test of zero restrictions on the corresponding elements of the covariance matrix of the error terms of a VAR model. Under the null hypothesis the test statistic is χ^2 -distributed with $K_1 K_2$ degrees of freedom.

Empirical example: Three-variable monetary policy VAR. Continue with the three-variable VAR(4) model selected above for the Stock-Watson (2001) type of data. Consider Granger causality and instantaneous causality tests. The goal is to determine if past values of certain variables are useful in predicting others. The null hypothesis for each test is that no Granger causality/instantaneous causality exists.

The results, obtained using R, strongly indicate a highly interconnected dynamic system. We find statistically significant Granger causality in all tested directions, meaning the variables have substantial predictive power over each other. A summary of the test results is provided below:

Null hypothesis	Granger	p -value	Instantaneous	p -value
(Infl., Unemp.) $\rightarrow$ Int. rate	4.86	< 0.001	29.56	< 0.001
(Unemp., Int. rate) $\rightarrow$ Infl.	3.40	< 0.001	5.05	0.080
(Infl., Int. rate) $\rightarrow$ Unemp.	4.82	< 0.001	27.42	< 0.001

In short, the past values of each pair of variables are useful for predicting the third.

- That is, we reject the null hypotheses of no Granger causality at the conventional statistical significance levels.

Significant instantaneous (contemporaneous) relationships are also present in almost all cases. The only exception is the instantaneous link between inflation and the other two variables (unemployment rate and interest rate), which is not significant at the 5% level ($p = 0.080$). Overall, these results confirm strong dynamic interlinkages among the interest rate, inflation, and unemployment rate.

Chapter 12

Nonstationary processes

12.1 Integrated process

In some application fields, such as economics and finance, the existence of trends in the analyzed time series is common.

- Here the term “trend” is used quite generally.

Earlier in this material, it was discussed that nonstationary time series containing a trend can be investigated using methods developed for stationary time series if the trend is first removed using a suitable transformation. The most popular of such transformations is taking differences, which amounts to modelling the differenced series $\Delta y_t = y_t - y_{t-1}$

- These series might seem stationary and thus be modelled with stationary models reasonably well.
- If the differences still include some sort of trend, it is natural to extend this procedure to the difference of these differences $\Delta^2 y_t$, and so on. This, however, is very rare in practice, and typically only first differences are considered when differencing data.

Based on the aforementioned, let us introduce the following definition (here for univariate time series processes):

Integrated process. A stochastic process $\{y_t, t = 1, 2, \dots\}$ is said to be **an integrated process of order d** , or an **$I(d)$ process**, if the d times differenced process

$$\Delta^d y_t$$

is stationary (or asymptotically stationary), but the $(d - 1)$ differenced process $\Delta^{d-1} y_t$ is not.

- In some definitions, the process is first centered by subtracting its mean, that is $\Delta^d(y_t - \mathbb{E}(y_t))$, $t = 1, 2, \dots$, is stationary, while $\Delta^{d-1}(y_t - \mathbb{E}(y_t))$ is not.

If y_t is an $I(d)$ **process**, we denote

$$y_t \sim I(d) \quad (d \geq 1).$$

In this course, we will consider only $d = 1$ and $d = 0$.

Random walk. The simplest example of an $I(1)$ process is **random walk**, which was briefly covered already in connection to the AR(1) process. Here the random walk is defined with a drift term ν :

$$\Delta y_t = \nu + u_t, \quad t = 1, 2, \dots,$$

where $u_t \sim \text{iid}(0, \sigma^2)$. By recursive substitutions, we get

$$y_t = y_0 + \nu t + \sum_{j=1}^t u_j, \quad t = 1, 2, \dots$$

- Sometimes, depending on the time series and application that we are considering, it is assumed that $\nu = 0$, that is, we have a **random walk without drift**, and hence the term νt above vanishes.

Assuming the initial (starting) value y_0 as constant, for the random walk we get (details are left as an exercise)

$$\mathbb{E}(y_t) = y_0 + \nu t,$$

$$\text{Var}(y_t) = \text{Var}\left(\sum_{j=1}^t u_j\right) = \sigma^2 t,$$

$$\text{Cor}(y_t, y_{t+k}) = \frac{1}{\sqrt{1 + k/t}}, \quad k \geq 0.$$

Here we can observe that random walk is not stationary even if the value of y_0 is altered.

- As $\Delta y_t \sim \text{iid}(\nu, \sigma^2)$ is stationary, we can conclude that the random walk is an $I(1)$ process.
- From the calculations above, we can observe that for random walk $\text{Var}(y_t) \rightarrow \infty$ and $\text{Cor}(y_t, y_{t+k}) \rightarrow 1$ with any $k > 0$ as $t \rightarrow \infty$. Therefore, random walk processes are strongly autocorrelated, and due to their wandering nature, they exhibit trend-like features.

In other words, we can summarize:

- An $I(0)$ process is stationary: it fluctuates around a constant mean, has constant variance, and autocorrelations decay relatively quickly (depending on the persistence). It is often described as mean-reverting and has short memory.
- An $I(1)$ process is non-stationary: it exhibits a **stochastic trend** (see below) and **wanders** over time. Each innovation has a permanent effect, and autocorrelations decay slowly, indicating long memory.

ARIMA processes and trends in time series. Random walk is the simplest example of so-called **ARIMA processes**, as we already briefly discussed in Section 5.

- The acronym stands for AutoRegressive Integrated Moving Average, where the letter I refers to integration, the idea that the process is formed by summing (or “integrating”) past innovations, $y_t = y_0 + \sum_{j=1}^t u_j$.

Many economic and financial time series exhibit trends, and these trends are of two fundamentally different types. The distinction matters, because it determines which transformation makes the series stationary.

- A **deterministic trend** is a fixed, non-random function of time (typically linear, βt), around which the series fluctuates due to stationary noise. A process of the form

$$y_t = \alpha + \beta t + \psi(B)u_t,$$

where $\psi(B)$ is the $MA(\infty)$ representation of a stationary ARMA process (including the special case $\psi(B) = 1$), is called **trend stationary**: removing the deterministic trend $\alpha + \beta t$ leaves a stationary process, shocks have only temporary effects, and the series reverts to the trend.

- A **stochastic trend** results from the accumulation of random shocks, as in the random walk above. Such a process is **difference stationary**: stationarity is achieved by differencing, not by removing a deterministic function of time. Shocks have permanent effects, and the series wanders without reverting to any fixed path.
- The two can also appear together. In the random walk with drift, $y_t = y_0 + \nu t + \sum_{j=1}^t u_j$, the stochastic trend $\sum_{j=1}^t u_j$ is superimposed on the deterministic trend $y_0 + \nu t$.

Choosing the wrong one matters in practice: removing a deterministic trend does not make a difference-stationary series stationary, and differencing a trend-stationary series creates the artificial patterns discussed below (overdifferencing, see below). Empirical evidence suggests that stochastic trends are often more appropriate for modelling macroeconomic and financial time series.

12.2 ARIMA(p, d, q) process

If the process y_t , $t = 0, 1, \dots$, is non-stationary, but the difference Δy_t is stationary and follows an invertible ARMA(p, q) process, y_t is called an **ARIMA**($p, 1, q$) **process**.

- The order in the middle refers to the fact, that stationarity is achieved by differencing the process once.
- If the process is non-stationary after being differenced once, but the second differences $\Delta^2 y_t$ follow a stationary and invertible ARMA(p, q) process, y_t is called an ARIMA($p, 2, q$) process.

In general, y_t is called an **ARIMA**(p, d, q) **process** if it becomes a stationary and invertible ARMA(p, q) process after differencing d times.

- That is, $\Delta^d y_t \sim \text{ARMA}(p, q)$.
- In practice, $d = 1$ is by far the most common case, $d = 2$ can be encountered quite rarely, and $d > 2$ can be safely ignored.
- With $d = 0$, we obtain the special case of an ARMA(p, q) process.

The typical properties of an ARIMA($p, 1, q$) process are generally the same as for random walk. The realizations exhibit a wandering nature explained by the variance growing as a function of t and strong autocorrelation even though the autocorrelation function cannot be defined like with stationary processes.

A warning on overdifferencing. While differencing is a fundamental tool for making a nonstationary time series stationary, applying it too many times, a problem known as **overdifferencing**, can be counterproductive. It is a common pitfall to assume that if one difference helps, a second one might help more. Even more often, **you should not mechanically difference a time series that is already “stationary enough”**.

- Doing so without a strong theoretical justification (e.g., an economic model suggesting a process is integrated of order two, I(2)), is highly unlikely and can even harm your statistical/econometric analysis.

Overdifferencing does not make a series “more stationary”. Instead, it introduces new artificial patterns into the data that were not present in the original process. This complicates model building and can lead to poor analyses and forecasts.

- The most significant problem is that overdifferencing creates a specific, predictable correlation structure.

Illustration: Overdifferencing white noise. Let us see what happens when we difference a series that is already stationary: a white noise process, $y_t = u_t$, where $u_t \sim \text{wn}(0, \sigma^2)$. This series has by definition zero autocorrelation at all lags. If we mistakenly difference it, we create a new series, $z_t = \Delta y_t =$

$y_t - y_{t-1} = u_t - u_{t-1}$. This new series, z_t , is no longer white noise. Let's examine its properties:

- $\text{Var}(z_t) = \text{Var}(u_t - u_{t-1}) \stackrel{wn}{=} \text{Var}(u_t) + \text{Var}(u_{t-1}) = 2\sigma^2$. We have doubled the variance!
- Autocorrelation: The autocovariance at lag 1 is

$$\text{Cov}(z_t, z_{t-1}) = \text{Cov}(u_t - u_{t-1}, u_{t-1} - u_{t-2}) = -\text{Var}(u_{t-1}) = -\sigma^2.$$

The autocorrelation at lag 1 is therefore:

$$\rho_1 = \frac{\text{Cov}(z_t, z_{t-1})}{\sqrt{\text{Var}(z_t)\text{Var}(z_{t-1})}} = \frac{-\sigma^2}{\sqrt{2\sigma^2 \cdot 2\sigma^2}} = \frac{-\sigma^2}{2\sigma^2} = -0.5.$$

By differencing a series with zero autocorrelation, we created a new series with a perfectly defined negative autocorrelation of -0.5 at lag 1. In fact, $z_t = u_t - u_{t-1}$ is an MA(1) process with $\theta_1 = -1$, that is, a **non-invertible** MA process (a unit root in the MA polynomial). Such a process cannot be approximated with a finite-order AR model, which is a further reason why overdifferencing complicates the subsequent model building.

As we will discuss later on, always check for stationarity using tools like the ACF plot and unit root tests (e.g., ADF test, see the coming section), and visual inspection, before applying another round of differencing. If the series already appears stationary enough, stop!

12.3 Unit root process

Let's consider the (univariate) AR(p) process

$$a(B)z_t = u_t,$$

where $a(B) = 1 - a_1B - \dots - a_pB^p$.

- Here notation z_t , as in Section 2, will be used when the unit root process is not assumed to have a constant or other deterministic trends.

It can be shown (see the Extra tag below) that the polynomial $a(B)$ can be rewritten as

$$a(B) = \Delta - \phi B - \phi_1 \Delta B - \dots - \phi_{p-1} \Delta B^{p-1}, \quad \phi = -a(1),$$

so the process z_t can be represented as

$$\Delta z_t = \phi z_{t-1} + \phi_1 \Delta z_{t-1} + \dots + \phi_{p-1} \Delta z_{t-p+1} + u_t, \quad t = 1, 2, \dots,$$

where $u_t \sim \text{iid}(0, \sigma^2)$ (or that u_t is white noise).

Let's assume that for $a(z)$ the following holds:

$$\text{If } a(z) = 0, \text{ then } |z| > 1 \text{ or } |z| = 1.$$

In other words, the roots of the polynomial $a(z)$ lie outside or at the unit circle on the complex plane.

- If all the roots lie outside the unit circle, z_t is (at least) asymptotically stationary (as meeting the stationarity condition of the $\text{AR}(p)$ process).

Let us now assume there is exactly one **unit root**. From the known properties of polynomials, it follows that

$$a(z) = (1 - z)b(z),$$

where $b(z)$ is a polynomial of the degree (at most) $p - 1$ with its roots inevitably lying outside the unit circle (see the Extra tag below).

- It is clear that this is equivalent to $\phi = 0$ above. When $\phi = 0$, the above-mentioned polynomial becomes $b(z) = 1 - \phi_1 z - \dots - \phi_{p-1} z^{p-1} := \phi(z)$.

Thus, Δz_t is $I(0)$ and as the roots of the polynomial $\phi(z)$ lie outside the unit circle, such initial values $\Delta z_{t-1}, \dots, \Delta z_{t-p+1}$ can be found that Δz_t is stationary.

- Therefore, the process Δz_t can be written as $\Delta z_t = \phi(B)^{-1} u_t$, which is a special case of the random walk we looked at earlier but with the stationary $\text{AR}(p - 1)$ process $\phi(B)^{-1} u_t$ replacing the innovation u_t .

i Extra: Polynomials and power series

Consider polynomial function of power m

$$p(z) = \sum_{k=0}^m a_k z^k,$$

where a_k and z are real or complex numbers and $a_m \neq 0$. In the case of complex numbers $z \in \mathbb{C}$, by the *fundamental theorem of algebra* the polynomial $p(z)$ can be given the form

$$p(z) = a_m (z - \zeta_1) \cdots (z - \zeta_m),$$

where $\zeta_1, \dots, \zeta_m$ are the roots of $p(z)$ and thus $p(\zeta_k) = 0$. The case $\zeta_k = \zeta_l$ ($k \neq l$) is possible and in the case of a real coefficient $a_k \in \mathbb{R}$ the roots occur as complex conjugates for all k ; that is if $\zeta_k = x_k + iy_k$ ($i^2 = -1$), for some $l \neq k$ it holds $\zeta_l = x_k - iy_k =: \bar{\zeta}_k$.

Let's return to a normal polynomial $p(z) = \sum_{k=0}^m a_k z^k$. It is easy to conclude that $p(z)$ can be written as

$$p(z) = p(1) + (1 - z)q(z),$$

where

$$q(z) = \sum_{k=0}^{m-1} b_k z^k, \quad b_k = - \sum_{j=k+1}^m a_j, \quad k = 0, 1, \dots, m-1.$$

A corresponding alternative way to present this would be

$$p(z) = p(1)z + (1-z)r(z),$$

where

$$r(z) = \sum_{k=0}^{m-1} c_k z^k, \quad c_0 = a_0 \quad \text{and} \quad c_k = - \sum_{j=k+1}^m a_j, \quad k = 1, \dots, m-1$$

These can be generalized to the power series, that is to say to the case

$$p(z) = \sum_{k=0}^{\infty} a_k z^k \quad (|z| \leq 1).$$

If we assume

$$\sum_{k=1}^{\infty} k |a_k| < \infty,$$

it can be easily concluded that the above equations hold when $|z| \leq 1$ and m are replaced with ∞ . In addition $\sum_{k=1}^{\infty} |b_k| < \infty$ and $\sum_{k=1}^{\infty} |c_k| < \infty$.

As discussed above in connection to integrated and ARIMA processes, the presence of a unit root is a particularly interesting question from an economic point of view. In unit root processes, shocks (such as policy or technological interventions and disruptions) have persistent effects that last forever, whereas with stationary cases such shocks have only a temporary effect. Therefore, it is of particular interest to test the unit root hypothesis in y_t .

12.4 ADF unit root test

Let us continue to work with unit root processes and examine how the unit root can be tested. Among various alternatives, we focus on the Augmented Dickey-Fuller (ADF) test, which is the most commonly used unit root test.

Additional background and motivation for unit root testing are provided below in the Extra tab before proceeding to the ADF test.

i Extra: Background and details on the unit root test statistics

Let us continue with the autoregressive unit root process from the previous section

$$\Delta z_t = \phi z_{t-1} + u_t, \quad t = 1, 2, \dots,$$

where for the sake of simplicity $z_0 = 0$ and $u_t \sim \text{iid}(0, \sigma^2)$. The question of interest is to **test the unit root hypothesis**

$$H_0 : \phi = 0$$

against the stationary (or stable) alternative $\phi < 0$. Recall that $\phi = -a(1)$, where $a(B)$ is the autoregressive polynomial, so that $\phi = 0$ corresponds to a unit root ($a(1) = 0$) and $\phi < 0$ is a necessary (though in the general $\text{AR}(p)$ case not sufficient) condition for stability.

It seems natural to base the test on the least squares estimator of ϕ :

$$\hat{\phi} = \left(\sum_{t=1}^T z_{t-1}^2 \right)^{-1} \sum_{t=1}^T z_{t-1} \Delta z_t \stackrel{H_0}{=} \left(\sum_{t=1}^T z_{t-1}^2 \right)^{-1} \sum_{t=1}^T z_{t-1} u_t,$$

where T is the number of observations. The unit root test can be based directly on the estimator $\hat{\phi}$, but it is more common to use the “t-ratio” printed by different statistical programs

$$\tau := \frac{\hat{\phi}}{\text{s.e.}(\hat{\phi})} = \frac{\sqrt{\sum_{t=1}^T z_{t-1}^2}}{\hat{\sigma}} \hat{\phi},$$

which follows, under the null (unit root) hypothesis, a **nonstandard** asymptotic distribution.

- The limiting distribution is free of the unknown parameters (in particular of σ^2) and can therefore be tabulated; tables are presented in books and computer programs.
- The percentage points of the limiting distributions have no closed form and are obtained by numerical integration or simulation. Both the (suitably normalized) estimator and the “t-ratio” τ have a non-Gaussian, left-skewed asymptotic distribution. For example, for the τ statistic the asymptotic 5% percentage point is about -1.94 , while for a $N(0, 1)$ distribution it would be -1.645 . Since the rejection region is of the form $\tau < \text{critical value}$, using standard normal critical values would reject the unit root hypothesis far too easily.
- Note that $\hat{\phi}$ is *superconsistent* under H_0 : it converges at the rate T rather than the usual $\sqrt{T}$, so it is $T\hat{\phi}$, and not $\sqrt{T}\hat{\phi}$, that has a nondegenerate limiting distribution.

The above unit root test can be generalized to a case with a deterministic constant and trend (if appropriate due to the shape of the time series or

background information on it). A common model for this is

$$y_t = \mu_t + z_t, \quad t = 1, 2, \dots,$$

where z_t is as determined above and the deterministic term is

$$\mu_t = \mu \quad \text{or} \quad \mu_t = \mu_1 + \mu_2 t.$$

Indicator variables can also be used to account for seasonal variation if necessary.

- The limiting distribution of the “t-ratio” can again be tabulated. It deviates from the one with no constant and is located further to the left (the asymptotic 5% percentage point is now about -2.86).
- The limiting distribution shifts still further to the left when the “t-ratio”-based test is generalized to the linear trend case $\mu_t = \mu_1 + \mu_2 t$ (the asymptotic 5% percentage point is then about -3.41 ; further details are skipped here).
- The critical values above are asymptotic. In finite samples the appropriate percentage points are somewhat further to the left, and modern software (and the tables of MacKinnon) reports critical values or p -values interpolated as a function of the sample size.

The asymptotic distributions. Under the null hypothesis $z_t = \sum_{j=1}^t u_j$ is a random walk, so the usual limit theorems for stationary processes do not apply and the analysis instead relies on functional central limit theory. It can be shown that

$$T\hat{\phi} \xrightarrow{d} \left(\int_0^1 W(u)^2 du \right)^{-1} \int_0^1 W(u) dW(u)$$

and that the “t-ratio” follows the (non-standard) asymptotic distribution

$$\tau := \frac{\sqrt{\sum_{t=1}^T z_{t-1}^2} \hat{\phi}}{\hat{\sigma}} \xrightarrow{d} \left(\int_0^1 W(u)^2 du \right)^{-1/2} \int_0^1 W(u) dW(u),$$

where $\hat{\sigma}^2 = (T-1)^{-1} \sum_{t=1}^T (\Delta z_t - \hat{\phi} z_{t-1})^2$ is the usual regression estimator for the error variance σ^2 (T can also be used as the denominator) and $W(u)$ is standard Brownian motion. Note that normality of u_t is *not* needed for these results; the Brownian motion arises from a functional central limit theorem.

In the case with a constant term, applying the operator $\Delta - \phi B$ to both sides of the equation $y_t = \mu + z_t$ (and using $(\Delta - \phi B)z_t = u_t$ and $(\Delta - \phi B)\mu = -\phi\mu$), we get

$$\Delta y_t = \nu + \phi y_{t-1} + u_t, \quad t = 1, 2, \dots,$$

where $u_t \sim \text{iid}(0, \sigma^2)$ and $\nu = -\phi\mu$. When the unit root hypothesis holds, for the “t-ratio” τ_μ of the parameter ϕ it holds that

$$\tau_\mu \xrightarrow{d} \left(\int_0^1 \bar{W}(u)^2 du \right)^{-1/2} \int_0^1 \bar{W}(u) dW(u),$$

where $\bar{W}(u) = W(u) - \int_0^1 W(s) ds$ is the centered (demeaned) version of the standard Brownian motion. In the linear trend case the corresponding result holds with $\bar{W}$ replaced by the residual of the projection of W on a constant and a linear trend; this is what shifts the distribution still further to the left.

An important practical implication of the parameterization $\nu = -\phi\mu$ is that **under H_0 the intercept vanishes**, $\nu = 0$: the constant-only test regression corresponds to a random walk *without drift* under the null. This is why software also reports joint tests of ϕ and the deterministic parameters (see below).

Augmented Dickey-Fuller (ADF) test. Based on the discussion in the previous section, assume (for simplicity) that $y_t = \mu + z_t$, where $z_t \sim \text{AR}(p)$, and hence y_t also follows an $\text{AR}(p)$ process. To test for a unit root in y_t , we use the following test regression model

$$\Delta y_t = \nu + \phi y_{t-1} + \sum_{j=1}^{p-1} \phi_j \Delta y_{t-j} + u_t, \quad u_t \sim \text{iid}(0, \sigma^2), \quad t = 1, 2, \dots$$

This is simply a reparameterization of the $\text{AR}(p)$ model, obtained from the autoregressive polynomial $a(B)$ with $\phi = -a(1)$; the lagged differences Δy_{t-j} (“augmentation”) are what allow the errors u_t to be serially uncorrelated even though y_t is autocorrelated.

The null hypothesis is $H_0 : \phi = 0$, which implies that y_t has a unit root and is non-stationary. Under the alternative hypothesis $H_1 : \phi < 0$, the process is stationary around a deterministic mean.

- The interpretation of H_0 as “a unit root” rests on a **maintained assumption**: the remaining roots of $a(B)$ are assumed to lie outside the unit circle, i.e. y_t has at most one unit root and no explosive or seasonal unit roots. If y_t were, say, $I(2)$, the test would be applied to Δy_t first (testing “from the top down”).
- Here in the ADF test the intercept term ν is related to the mean of the process via $\nu = -\phi\mu$. Consequently $\nu = 0$ under H_0 , so that the constant-only specification treats the null as a driftless random walk. In practice ν is nevertheless left unrestricted in the test regression, which is what makes the test statistic invariant to the value of μ .

- The test statistic is the **t-ratio** of the estimated coefficient $\hat{\phi}$, often denoted $\tau_{\phi} = \frac{\hat{\phi}}{\text{s.e.}(\hat{\phi})}$. Its asymptotic distribution under the null is non-standard but known, and critical values are tabulated in the literature. The relevant table depends on which deterministic terms are included in the test regression, not on the true data generating process. Note that the test is one-sided: the null hypothesis is rejected only for sufficiently large *negative* values of τ_{ϕ} .
- Importantly, the asymptotic properties of the test hold even without assuming normality of the errors; it is sufficient that u_t are independent and identically distributed with finite variance (a martingale difference sequence with finite variance would in fact suffice).

If the null hypothesis of a unit root ($H_0 : \phi = 0$) cannot be rejected, according to the above equation an $AR(p-1)$ model can be built on the differences Δy_t .

- In other words, this implies the need to take first differences to get a stationary time series.
- Note also that failing to reject H_0 is *not* evidence in favour of a unit root. Unit root tests have low power against stationary alternatives with a root close to unity, especially in short samples, and the conclusion should be supported by graphical inspection, economic reasoning, and possibly a stationarity test such as KPSS, in which stationarity is the null hypothesis.

Determining the lag length p and deterministic terms in the ADF test. Executing the ADF test regression requires specifying the lag length p , which can be selected using information criteria (such as AIC or BIC), sequential testing (covered in the next section), or based on common fixed choices related to the data frequency.

- The choice involves a trade-off: too few lags leave the errors autocorrelated, which invalidates the asymptotic distribution and distorts the size of the test, whereas too many lags reduce the power of the test.
- Whichever rule is used, the residuals of the chosen test regression should be checked for remaining autocorrelation.
- It is generally advisable to try a few different values of p and compare the resulting test outcomes for robustness.
- Be careful with the terminology used by software: in the R function `ur.df`, the argument `lags` is the number of **lagged differences** Δy_{t-j} in the test regression, that is, $p-1$ in the notation above. The default is `lags = 1` together with `selectlags = "Fixed"`; setting `selectlags = "AIC"` or `"BIC"` selects the number of lagged differences automatically up to the value given in `lags`. The default `type = "none"` includes no deterministic terms at all, which is rarely what one wants with economic data.

The ADF test regression can also be extended to include a trend (together with the constant). In this case, the test regression includes the term $\delta_0 + \delta_1 t$, allowing for a deterministic trend in the data:

$$\Delta y_t = \delta_0 + \delta_1 t + \phi y_{t-1} + \sum_{j=1}^{p-1} \phi_j \Delta y_{t-j} + u_t.$$

The null hypothesis remains $H_0 : \phi = 0$, indicating a unit root.

- As in the constant-only case, the coefficients of the test regression are linked to the deterministic terms of the process. Writing $y_t = \mu_1 + \mu_2 t + z_t$ gives $\delta_1 = -\phi\mu_2$ and $\delta_0 = \mu_2 - \phi\mu_1 + \phi\mu_2$. Hence under H_0 we have $\delta_1 = 0$ and $\delta_0 = \mu_2$: the null model is a random walk **with drift**, and the trend term drops out. Including t in the regression nonetheless makes the test invariant to the values of μ_1 and μ_2 .
- In `ur.df` (with `type = "drift"` or `"trend"`) the summary also reports joint F-type statistics `phi1`, `phi2` and `phi3` for hypotheses that restrict ϕ and the deterministic parameters simultaneously (e.g. `phi1`: $\nu = \phi = 0$; `phi3`: $\delta_1 = \phi = 0$). These have their own non-standard distributions and their own tables — the usual F critical values do not apply.

Deciding which deterministic terms to include is a crucial step, and the decision matters in both directions.

- A constant term is typically included by default, while the inclusion of a trend term depends on whether a trending pattern is evident in the data. This decision can be guided by graphical inspection or theoretical considerations relevant to the application.
- Omitting a needed trend makes the test badly biased towards non-rejection (the test is then inconsistent against trend-stationary alternatives), whereas including an unnecessary trend shifts the critical values to the left and costs power.
- A practical rule of thumb: include the deterministic terms that would be needed to describe the series under the *alternative* hypothesis.
- Finally, note that a structural break in the level or slope of a stationary series makes the ADF test strongly favour the unit root hypothesis. If a break is suspected, tests that explicitly allow for it (e.g. Zivot-Andrews) should be used instead.

i Extra: Some other unit root tests

Other widely used unit root tests include the Phillips-Perron test, which handles serial correlation non-parametrically instead of by augmentation,

and the DF-GLS (Elliott-Rothenberg-Stock) test, which detrends the series by GLS and has better power than the ADF test. They are not discussed further here.

Empirical examples. Let us consider unit root testing for the few time series considered above in different parts. As discussed, the first thing in the ADF test is to determine which deterministic terms to include in the test regression. Basically you should always include a constant term, but whether the deterministic trend component should be included as well should be determined based on visual inspection and background knowledge on the time series at hand.

In the following illustrations, for simplicity, we fix the number of lagged differences in the ADF test regression to 6, 8 or 10. The lag length could and should be varied to examine whether the testing results are intact for this selection.

- (i) Consider **log** of the CPI time series (Introduction): In this case, there is clearly an upward trend present. It seems reasonable to argue that due to different (economic) reasons we can think that price level is steadily going up and a deterministic trend seems reasonable to be included in the ADF test.

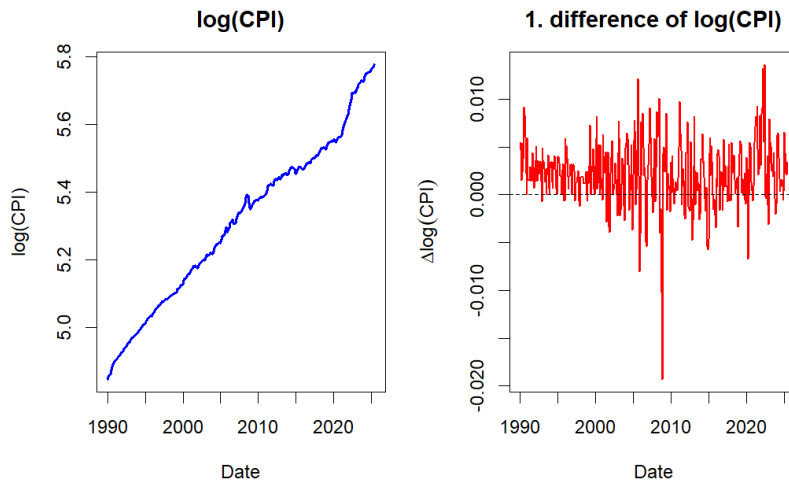


Figure: Monthly $\log(\text{CPI})$ time series (left) and its first difference $\Delta \log(\text{CPI}_t)$ (right), 1990:1–2025:6.

The first difference is shown next to the level series for comparison: the level series trends upwards throughout the sample, whereas the differenced series fluctuates around a constant (positive) level without any trend. This is exactly

the visual signature of an $I(1)$ series, and it anticipates the test results reported in part (iii) below.

```

ADF-test (deterministic terms: constant and trend)
p=8: Test statistic: -1.113, p-value: 0.925
p=6: Test statistic: -1.517, p-value: 0.823
p=10: Test statistic: -1.451, p-value: 0.845

```

- (ii) Log of the S&P 500 index. Rising stock prices illustrate a possible stochastic trend very well, but it is difficult to argue that there is a deterministic upward trend in stock prices. Therefore, including a constant in the ADF test is surely a good choice, but for robustness we can also examine the unit root hypothesis when a deterministic trend is included as well (together with the constant) in the ADF regression.

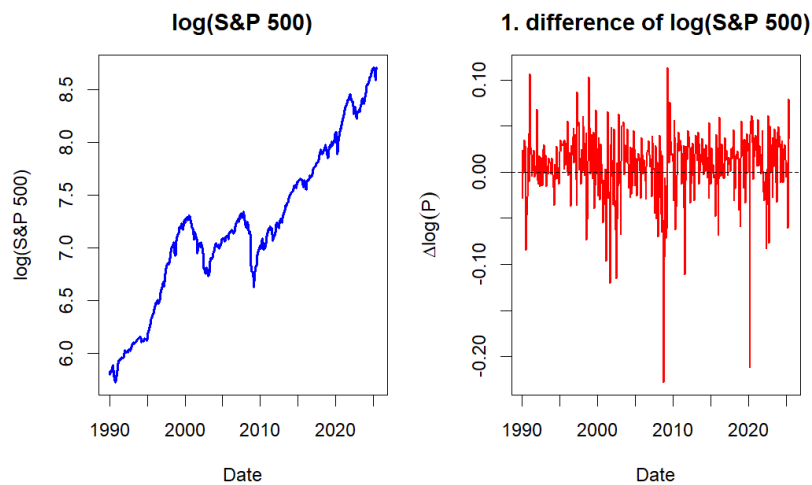


Figure: Monthly $\log(\text{S\&P } 500)$ time series (left) and its first difference $\Delta \log(P_t)$ (right), 1990:1–2025:6.

Again the differenced series is plotted next to the level series. The first difference, that is, the monthly log-return, shows no trend and fluctuates around a constant level, although the size of the fluctuations clearly varies over time (recall the volatility clustering discussed in the AR-GARCH section).

```

ADF-test (deterministic terms: constant)
p=8: Test statistic: -0.769, p-value: 0.826
p=6: Test statistic: -0.302, p-value: 0.922
p=10: Test statistic: -0.772, p-value: 0.825

ADF-test (deterministic terms: constant and trend)
p=8: Test statistic: -2.193, p-value: 0.492

```

- (iii) The log-differences of the CPI and S&P 500 indices are the series plotted on the right-hand side of the two figures above (and, up to scaling, the series previously shown in the Introduction). A visual inspection alone suggests that the unit root hypothesis is no longer reasonable for these differenced series. To confirm this, let's examine the Augmented Dickey-Fuller (ADF) test results below. The test includes a constant, as a deterministic trend is clearly no longer present in the data.

These conclusions are reinforced by the test results for their log-differences, which appear to be stationary (i.e., $I(0)$ variables). This provides a consistent picture: the original series are non-stationary $I(1)$, but their first differences are stationary ($I(0)$).

```

ADF-tests (deterministic terms: constant)

logdiff(CPI):
p=8: Test Statistic: -7.744 , p-value: 1.248e-11
p=6: Test Statistic: -4.764, p-value: 7.568e-05
p=10: Test Statistic: -7.886, p-value: 5.331e-12

logdiff(S&P 500)
p=8: Test statistic: -6.095, p-value: 1.303e-07
p=6: Test statistic: -7.011, p-value: 8.741e-10
p=10: Test statistic: -5.451, p-value: 3.299e-06

```

Putting the unit root testing results together, the results show that the ADF tests fail to reject the unit root hypothesis for the original level series of both the CPI and the S&P 500.

- For the CPI levels, even after including a constant and a linear trend in the test regression, we still cannot reject the null hypothesis of a unit root. This strongly suggests that the CPI can be treated as an $I(1)$ process (integrated of order one).
- A similar conclusion holds for the S&P 500 index.

A critical limitation: The low power of the ADF test. An important limitation of the ADF test is its low statistical power. In practical terms, this means: **The ADF test often fails to reject the null hypothesis of a unit root, even when the time series process is actually stationary.**

- In other words, the ADF (or other) unit root test is not always very good at correctly identifying a stationary process, especially if the process is close to non-stationary (e.g., an AR(1) process where the autoregressive coefficient ϕ_1 is close to 1).
- If we are unable to reject the presence of a unit root, it does not necessarily mean that it is true and the process is necessarily $I(1)$. It could just be (but of course not always!) that there is not sufficient amount of information in the data to reject the unit root.
- The practical takeaway is that it's common for a truly stationary time series to produce a p-value from the ADF test that is too high to be considered statistically significant. You can observe this phenomenon by running the following code, which simulates AR(1) processes and computes the ADF test statistic, with a varying value of the AR(1) coefficient.

From an empirical point of view, the choice whether the process is a unit root process (nonstationary) or a “near” unit root process (stationary) is interesting but at times complicated.

- One practical remedy is to complement the ADF test with a test that reverses the roles of the hypotheses, such as the KPSS test, in which stationarity is the null hypothesis. Agreement between the two tests strengthens the conclusion; disagreement is a sign that the data are not informative enough to settle the question.
- Particularly ambiguous are, for example, interest rates. Interest rates are highly persistent and the unit root hypothesis cannot often be rejected, although nonstationary such as random walk interest rates do not seem to be very plausible from an economic point of view.

```
# Simulating AR(1) process with different values of phi_1
# - Change n_obs and phi_1

# Load the libraries
library(urca)
library(tseries)

n_obs = 200      # number of observations (do not name this T: T is TRUE in R)
phi_1 = 0.98
adf_lags = 8     # number of lagged differences in the ADF test regression

epsilon = rnorm(n_obs) # random draws from nid(0,1)
```

```
y = epsilon
y[1] = 0 # for simplicity and  $E(y_t) = 0$ 
for(i in 2:n_obs) y[i] = phi_1*y[i-1]+epsilon[i]
plot(y, type="l", main="Simulated realization")

# The simulated process has zero mean, so the test regression needs no
# deterministic terms: type = "none" and, correspondingly, trend = "nc".
test_ur_none <- ur.df(y, type = "none", lags = adf_lags)
stat_none <- test_ur_none@teststat[1, 1]
p_val_none <- punitroot(q = stat_none, N = length(test_ur_none$res), trend = "nc")
cat("Test statistic:", stat_none, "\n")
cat("P-value:", p_val_none, "\n")
```


Chapter 13

Linear regressions with $I(1)$ variables

Instead of stationary variables, as considered earlier in Section 10, let us now assume that there are two variables, y_t and x_t , and either of them or both are nonstationary and specifically $I(1)$ variables.

- y_t is again the dependent variable and x_t is the predictive/explanatory variable.

Before continuing to examine linear (predictive) regressions between y_t and x_t in different cases, let us introduce an important concept of cointegration.

13.1 Basics of cointegration

An important special case of the linear regression between y_t and x_t , where y_t and x_t are nonstationary $I(1)$ variables, arises when there is a common stochastic trend in both series.

In other words, suppose that there is a linear relationship between variables so that there exists some value δ such that $y_t - \delta x_t$ is stationary ($I(0)$) although y_t and x_t are nonstationary ($I(1)$).

- In such case, it is said that y_t and x_t are **cointegrated** and they share the common trend.

In principle, an alternative way to construct a linear regression model, when y_t and x_t are $I(1)$, is based on the differences Δy_t and Δx_t which are then stationary ($I(0)$).

- This is not an optimal strategy if there is indeed a cointegration relationship between the variables. Cf. the discussion on overdifferencing in Section 12.

Let us examine cointegration more detail. **If y_t and x_t are $I(1)$ and cointegrated**, then

$$z_t = y_t - \delta x_t = [y_t \ x_t][1 \ -\delta]' \sim I(0).$$

The vector $[1 \ -\delta]'$ is called the **cointegration vector**.

- Cointegration is often interpreted as a long-run relationship between the variables. Assume that the equilibrium of the variables y_t and x_t is defined by the relationship $y_t = \tilde{\delta}x_t$ for some fixed $\tilde{\delta}$. Then $\hat{z}_t = y_t - \hat{\delta}x_t$ is the “equilibrium error” which measures the extent y_t deviates from its “equilibrium value”.
- As z_t is $I(0)$, the equilibrium error is stationary and fluctuating around zero. In other words, on average, the system is in equilibrium.

Notice that we can also use a slightly modified and general definition of cointegration for the above, especially when there will be more than two variables involved.

- We will consider this definition, and cointegration in connection to the VAR model more detail in the **Advanced Time Series Econometrics course**.
- **General definition of cointegration:** The situation described above can be formalized for $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})'$. If $\mathbf{y}_t \sim I(1)$, but there is a linear combination $\beta'\mathbf{y}_t$, which is $I(0)$, then an $I(1)$ process $\{\mathbf{y}_t, t = 1, 2, \dots\}$ ($K \times 1$) is said to be cointegrated if there exists a nonzero parameter vector β ($K \times 1$) such that:

$$\beta'\mathbf{y}_t \sim I(0).$$

The parameter β is called the cointegrating (cointegration) vector. This definition allows for trivial cases (like the overdifferencing example above) where $\beta = (0, \dots, 1, \dots, 0)$, meaning one variable is already stationary.

13.2 Testing cointegration between two variables

From the discussion above, it is obvious that it is important to distinguish whether there is a cointegration relationship between the variables.

Consider a “cointegration regression” between two $I(1)$ variables y_t and x_t :

$$y_t = \alpha + \delta x_t + u_t.$$

If y_t and x_t are cointegrated, then u_t is $I(0)$. If not, u_t will be $I(1)$. Therefore, after the model is estimated by OLS, the presence of a possible cointegration relationship can be evaluated by testing for a unit root in the residuals $\hat{u}_t$.

Testing for cointegration with a known coefficient. In some applications, the cointegration coefficient δ (and the intercept α) is known in advance due to economic theory or other application-specific knowledge. When the coefficient is known, the cointegration relationship can be tested as follows. For the sake of simplicity, we will assume $\alpha = 0$ in the procedure described below

- Construct the series $z_t = y_t - \delta x_t$.
- Use the ADF test for the null hypothesis of unit root, which implies $z_t \sim I(1)$. If H_0 is rejected, then there is a cointegration relationship.

Testing for cointegration with unknown coefficients. If δ (and α) are unknown, then we can proceed as follows (Engle-Granger ADF-test):

- Estimate the following model by using OLS

$$y_t = \alpha + \delta x_t + u_t.$$

- Construct the series $\hat{u}_t = y_t - \hat{\alpha} - \hat{\delta}x_t$, which is the residual of the regression.
- Test the null hypothesis implying $\hat{u}_t \sim I(1)$.
- Critical values are now different than in the above case, and in the ADF test, as unit root testing is based on the (estimated) residuals.

Notice that if y_t and x_t are indeed cointegrated, OLS yields a consistent estimator for the cointegration coefficient δ . However, the OLS estimator of δ has a non-normal asymptotic distribution, and the inferences based on the standard t -test statistic can be misleading.

Empirical example: Consumption and income. Let us consider an important empirical question derived from Hall's (1978) Permanent Income Hypothesis (PIH) in the context of cointegration.

- R. E. Hall (1978). Stochastic implications of the life cycle-permanent income hypothesis: Theory and evidence. *Journal of Political Economy*, 86(6), 971–987.
- Source of data: FRED.

The testable hypothesis is that there is a stable long-run relationship exist between real consumption and real income (that is for inflation adjusted consumption and income), such that the difference between them (the cointegration residual) is stationary.

- In other words, the cointegration theory predicts that since consumption c_t is proportional to permanent income y_t so that $c_t = a y_t$, $0 \leq a \leq 1$ (the present value of expected future income), and both are driven by a common stochastic trend (permanent income shocks), they should be cointegrated. That is there is a long-term equilibrium between c_t and y_t , and if c_t and y_t are now $I(1)$ processes, then there should be a cointegration relationship between the variables.

So, let us test the prediction of the PIH that real per capita consumption ($\log(c_t)$) and real per capita disposable income ($\log(y_t)$) share a single cointegrating relationship.

- Hypothesis: The two variables ($\log(c_t)$ and $\log(y_t)$) are $I(1)$ and cointegrated.
- Long-run equation: $\log(c_t) = \alpha + \delta \log(y_t) + u_t$
- PIH implication: The cointegrating vector should be $(1, -\delta)$, where δ is theoretically close to 1 (or exactly 1 in the simplest PIH model) because, in the long run, consumption should grow proportionally with permanent income. The error term u_t and the resulting residuals $\hat{u}_t$ must be $I(0)$ (stationary).

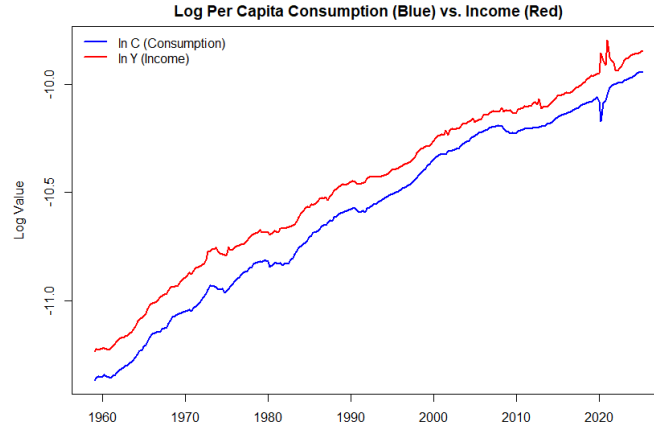


Figure: Monthly U.S. data (sample period 1959:1–2025:8) on (log) consumption and income. (Source: FRED)

Check first the unit root hypothesis for the two time series. In these ADF tests, a constant and linear trend component is included in the test regression and the number lagged differences p is 4:

```
ADF tests (deterministic terms: constant and trend)
```

```
log(c_t):
p=4: Test Statistic: -2.005, p-value: 0.595
p=6: Test Statistic: -2.160, p-value: 0.510
```

```
log(y_t)
p=4: Test Statistic: -2.583, p-value: 0.289
p=6: Test Statistic: -2.473, p-value: 0.341
```

Estimation result of the cointegration regression:

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.383122   0.044448    8.62 6.29e-16 ***
ln_Y         1.048293   0.004244   246.99 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

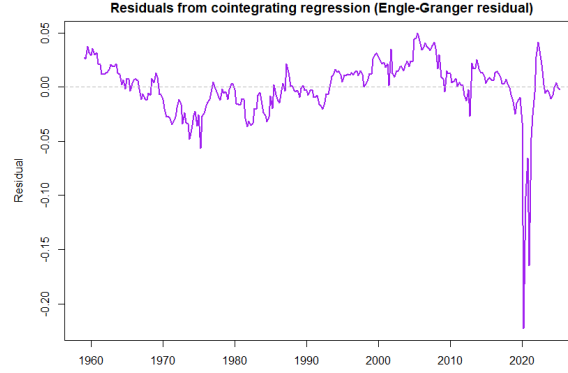
Residual standard error: 0.02692 on 264 degrees of freedom
Multiple R-squared:  0.9957,    Adjusted R-squared:  0.9957
F-statistic: 6.1e+04 on 1 and 264 DF,  p-value: < 2.2e-16
```

Based on the Engle-Granger two-step procedure, the null hypothesis of no cointegration can be rejected at 5% level (see testing result below). Despite one large outlier related to the Covid-10, the cointegration residual series seems to fluctuate as stationary $I(0)$ variable.

- In the ADF test below, we are using the critical values of the ADF test even though, as discussed above, that is not completely correct.

Figure: PIH residual time series (from the estimated cointegrated model).

```
Test Statistic: -4.561388
> cat("P-value:", p_val_drift, "\n")
P-value: 0.000197051
```



13.3 Linear regressions containing $I(1)$ variables

Recall the linear regression

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + u_t,$$

and the two assumptions (i)–(ii) set in Section 10:

- (i) The error term u_t is serially uncorrelated and uncorrelated with the regressors included in $\mathbf{x}_t$.
- (ii) All the regressors in $\mathbf{x}_t$ are either deterministic or stationary random variables.

Assume now that only (i) holds, that is the error term u_t is serially uncorrelated and uncorrelated with the regressors included in $\mathbf{x}_t$.

- Even if the model cannot be written in this way, the OLS estimator of the coefficients of the $I(1)$ regressors $\mathbf{x}_t$ is consistent. However, its asymptotic distribution is, in general, nonstandard such that usual inference does not apply.
- If the model can be written such that all the parameters of interest are coefficients of mean zero stationary variables, their OLS estimator is consistent and asymptotically normal.

To examine these points more detail, let us consider again the following relatively simple regression model

$$y_t = \beta_0 + \beta_1 x_{t-1} + \beta_2 x_{t-2} + u_t,$$

where $x_t \sim I(1)$. The model can be rewritten as

$$y_t = \beta_0 + (\beta_1 + \beta_2)x_{t-1} - \beta_2(x_{t-1} - x_{t-2}) + u_t,$$

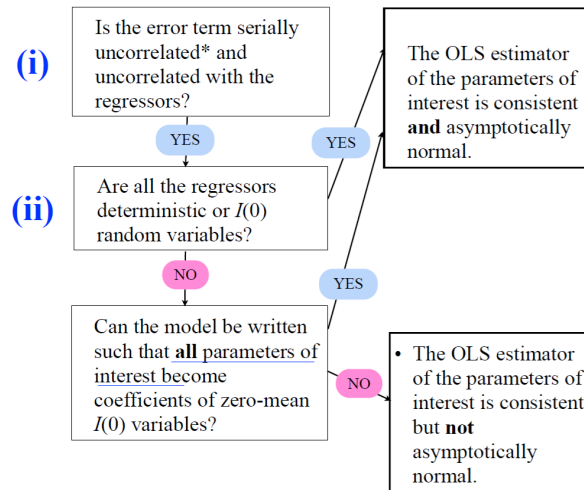
or

$$y_t = \beta_0 + \beta_1(x_{t-1} - x_{t-2}) + (\beta_1 + \beta_2)x_{t-2} + u_t.$$

As $(x_{t-1} - x_{t-2}) \sim I(0)$, and hence, standard inference on β_2 (or β_1) holds. Therefore, both β_1 and β_2 cannot simultaneously be written as coefficients of $I(0)$ variables (unless higher lags are included in the model).

- The OLS estimator of β_1 and β_2 is not, in general, jointly asymptotically normal.
- The test statistic on a hypothesis concerning both coefficients (for instance $H_0 : \beta_1 = \beta_2$), in general, does not have the usual asymptotic χ^2 distribution.

These remarks are compiled to the next figure.



* Not autocorrelated after $H-1$ lags.

Figure: Cheat sheet on inference in OLS regression models with $I(1)$ variables (References: Stock and Watson (1988) and lecture notes by Markku Lanne).

Let us continue with linear regression where now also the assumption (i) does not hold and the dependent variable is $I(1)$.

- If the dependent variable is not cointegrated with any of the regressors, the OLS estimator of the coefficients of the $I(1)$ regressors is inconsistent.
- If the dependent variable is cointegrated with at least one of the regressors, the OLS estimator of the parameters of interest
 - that can be written as coefficients of stationary variables, is inconsistent.

- that cannot be written as coefficients of stationary variables, is consistent, but not asymptotically normal.

For instance, consider the following regression model:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + u_t,$$

where all variables are $I(1)$, and y_t and x_{2t} are cointegrated and assumption (i) does not hold.

- If x_{1t} and x_{2t} are cointegrated such that $(x_{1t} - \gamma x_{2t}) \sim I(0)$, the model can be written as

$$y_t = \beta_0 + \beta_1(x_{1t} - \gamma x_{2t}) + (\beta_1\gamma + \beta_2)x_{2t} + u_t,$$

and the OLS estimator of β_1 is inconsistent.

- If x_{1t} is not cointegrated with x_{2t} , β_1 cannot be written as a coefficient of an $I(0)$ variable, and hence, its OLS estimator is consistent, but not asymptotically normal.

The complete *cheat sheet* on linear regression with $I(0)$ and $I(1)$ variables is summarized hereby:

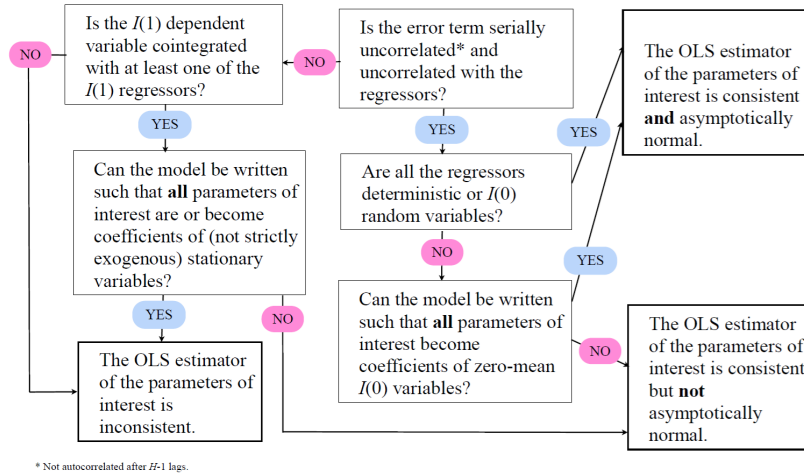


Figure: Complete cheat sheet on inference in OLS regression models with $I(1)$ variables.

A spurious regression emerges when a statistically significant relationship between two $I(1)$ variables y_t and x_t is found although those are completely unrelated. Assume that a researcher estimates a linear regression

$$y_t = \beta_0 + \beta_1 x_t + u_t,$$

but assuming wrongly that the variables y_t and x_t are stationary. Instead, they are generated by two independent random walks (i.e., y_t and x_t are $I(1)$ variables)

$$\begin{aligned} y_t &= y_{t-1} + u_{1t}, & u_{1t} &\sim \text{iid}(0, \sigma_1^2) \\ x_t &= x_{t-1} + u_{2t}, & u_{2t} &\sim \text{iid}(0, \sigma_2^2). \end{aligned}$$

This typically leads to the estimation result of the above linear regression characterized by a fairly high R^2 (the coefficient of determination), highly autocorrelated residuals $\hat{u}_t$ and a (highly) statistically significant estimate of β_1 . This result is clearly spurious given that the variables are completely independent!

- Estimation results like this should not be taken seriously.
- The reason is that with y_t and x_t being $I(1)$ variables, the error term u_t will also be $I(1)$, not stationary ($I(0)$).

Solution to this problem is to include the lags of y_t and x_t as predictors in the model. In other words, we end up a model (cf. dynamic regression models in Section 10)

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 y_{t-1} + \beta_3 x_{t-1} + u_t.$$

In this case the OLS estimator is consistent. Thus, in general, including lagged values in the regression is sufficient to solve many of the problems associated with possibly spurious regressions.

An example of a spurious regression situation can be replicated with the following program code in R lab.

